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Condominium Booking and Operations Management System

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

The integration of information technology into business operations became a critical factor in improving organizational efficiency, data accuracy, and overall service quality across industries. As digital transformation accelerated in every sector of the global economy, businesses increasingly adopted computerized systems to streamline daily operations, minimize human error, and establish more reliable data management infrastructures. Organizations that failed to embrace technological advancement found themselves unable to compete with digitally equipped counterparts, particularly as client expectations continued to rise and regulatory requirements for accurate record-keeping intensified.

In the real estate and property management industry, digital transformation played an especially vital role. Managing bookings, processing financial transactions, maintaining tenant records, and scheduling property maintenance were inherently complex activities that became unmanageable at scale without the support of automated systems. Property managers overseeing multiple units across different buildings or locations faced compounding administrative burdens when relying on manual processes, resulting in delayed responses to tenant concerns, inaccurate financial reporting, and missed business opportunities. The adoption of integrated property management platforms therefore shifted from a competitive advantage to an operational necessity.

The growing demand for condominium rentals in urban areas, driven by rapid urbanization, increased migration of students and young professionals to city centers, and shifting preferences toward high-density, amenity-rich residential environments, amplified the need for efficient property management solutions. Condominium rentals offered distinct advantages over traditional house rentals, including enhanced security features, proximity to commercial and educational institutions, shared facility access, and predictable utility arrangements. These qualities made condominium units the preferred housing choice for a growing demographic of urban residents, placing greater pressure on property management businesses to maintain seamless operations and high standards of tenant service.

However, managing multiple rental units simultaneously introduced considerable administrative complexity. Tracking tenant inquiries across different communication channels, maintaining accurate booking schedules to prevent conflicts, monitoring payment statuses for numerous tenants with varying payment cycles, and responding promptly to maintenance requests required systems capable of processing, organizing, and retrieving large volumes of data in real time. Without such systems, administrative staff were compelled to dedicate disproportionate amounts of time to clerical tasks, reducing their capacity to engage in higher-value activities that directly improved tenant satisfaction and business performance.

Gales Urban Haven, operating under the registered business name One Spatial Iloilo and located in Iloilo City, Western Visayas, Philippines, provided condominium rental and

property management services to a target market composed primarily of young professionals, university students, and small family units seeking affordable, accessible, and conveniently located residential accommodations. The business occupied a strategically significant position in Iloilo City's emerging urban rental market, which had experienced consistent growth in recent years as the city's academic institutions, business process outsourcing sector, and commercial developments drew an increasing number of residents from surrounding provinces.

Despite its growing clientele and expanding portfolio of rental units, Gales Urban Haven relied predominantly on manual processes for the management of its core operational functions. Booking records were maintained through handwritten logs or basic spreadsheet entries, with no automated mechanism to verify unit availability or flag scheduling conflicts. Payment tracking required staff to manually reconcile tenant payment receipts against ledger records, a process that became increasingly error-prone as the volume of transactions grew. Revenue computation depended on spreadsheet-based calculations that were vulnerable to formula errors, data entry inconsistencies, and version control issues arising from multiple users accessing shared files. Maintenance activities were recorded informally through verbal communications, handwritten notes, or disconnected digital files, with no centralized system to assign tasks, monitor progress, or verify resolution.

Manual management of booking and operational records presented several significant risks to the business. Data redundancy emerged as information was duplicated across

multiple records without synchronization, leading to conflicting versions of critical data. Record misplacement occurred as paper-based documents were filed inconsistently or lost entirely, disrupting administrative continuity and creating gaps in the historical record. Delayed financial reporting resulted from the time-intensive nature of manual reconciliation, preventing management from accessing timely revenue data for informed decision-making. Inaccurate revenue computation exposed the business to financial discrepancies that could go undetected for extended periods, potentially affecting tax compliance and investor relations. Perhaps most consequential, double bookings arose when the same unit was inadvertently reserved for multiple clients during overlapping periods, damaging the business's reputation and eroding tenant trust.

The use of paper-based or basic spreadsheet-driven systems also severely limited data accessibility. Management personnel were unable to retrieve operational information quickly when needed, particularly during high-demand periods when rapid decision-making was essential. The absence of real-time dashboards or automated reporting tools forced administrators to compile reports manually, consuming valuable time and introducing additional opportunities for data transcription errors. Furthermore, without a centralized and searchable database, historical records were difficult to query, making it challenging to analyze booking trends, identify peak occupancy periods, or assess the financial performance of individual units over time.

The absence of an integrated calendar and maintenance tracking system further limited the organization's ability to effectively monitor unit availability, coordinate booking schedules, and address property maintenance concerns in a timely manner. Scheduling conflicts arose when staff members operating with incomplete or outdated availability information inadvertently committed the same unit to multiple prospective tenants. Maintenance concerns, if not systematically recorded and tracked, were deprioritized, deferred, or forgotten entirely, accelerating unit deterioration and diminishing the living experience of current tenants. These operational deficiencies collectively undermined both the internal productivity of the business and its ability to deliver the standard of service that contemporary tenants expected.

An automated Condominium Booking and Operations Management System comprehensively addressed these challenges by consolidating all operational data into a single, centralized, database-driven platform. Such a system eliminated the fragmentation and redundancy inherent in manual record-keeping by serving as the authoritative source of truth for all booking, payment, scheduling, and maintenance information. Automation replaced error-prone manual calculations and verification processes, reducing the likelihood of financial discrepancies and operational conflicts. Real-time reporting capabilities provided management with actionable insights drawn from current, accurate data, enabling more informed and timely strategic decisions. Integration of a dynamic calendar scheduling module ensured that unit availability was accurately reflected at all times, preventing double bookings and

enabling staff to manage reservations with confidence. A dedicated maintenance monitoring component allowed property concerns to be formally recorded, assigned to appropriate personnel, tracked through to resolution, and reviewed for quality assurance purposes.

This study therefore proposed the development of a Condominium Booking and Operations Management System specifically tailored to the operational requirements of Gales Urban Haven (One Spatial Iloilo). The proposed system automated booking record management, payment tracking and revenue monitoring, calendar-based scheduling, and maintenance activity tracking through a unified, user-friendly, and scalable platform. By eliminating manual inefficiencies and consolidating operational data within a single integrated system, the proposed solution aimed to substantially improve organizational efficiency, ensure consistent data accuracy, reduce the administrative workload on staff, and equip management with the decision-support tools necessary to guide the business toward sustained growth and enhanced tenant satisfaction.

1.2 Statement of the Problem

Gales Urban Haven, operating under the registered business name One Spatial Iloilo, managed its condominium booking and operational processes through predominantly manual methods. These processes encompassed a broad range of administrative functions, including recording and updating tenant booking details, tracking payment receipts and outstanding balances, computing and reporting revenue

figures, scheduling unit reservations through a non-integrated calendar, and monitoring property maintenance activities from initial request through to verified resolution. While such manual approaches proved nominally manageable during periods of limited occupancy and low transaction volume, the anticipated and ongoing growth in demand for condominium rentals in Iloilo City rendered these practices increasingly inefficient, time-consuming, and susceptible to consequential human error.

The absence of an automated information system gave rise to a constellation of significant and interrelated operational challenges. Data redundancy emerged as a persistent issue, with identical or overlapping information recorded across multiple paper-based forms, handwritten logs, and disconnected spreadsheet files without any mechanism for automatic synchronization or conflict resolution. When the same data point was maintained in multiple locations by different staff members, discrepancies arose, and determining which record represented the authoritative version required time-consuming manual review. Over extended periods, these redundancies accumulated into a fragmented data landscape that severely complicated administrative continuity and undermined organizational trust in the accuracy of its own records.

Inaccurate record-keeping constituted a further systemic vulnerability. Manual data entry processes were inherently prone to transcription errors, particularly when staff

members worked under time pressure, managed multiple inquiries simultaneously, or transferred data between different record formats. Inconsistent data entry practices across different personnel produced records that were difficult to standardize, audit, or consolidate. Over time, these inaccuracies propagated through related records, creating compounding discrepancies that distorted the organization's understanding of its own operational state. The consequences of inaccurate record-keeping extended beyond administrative inconvenience, potentially affecting tenant billing accuracy, contractual compliance, and the reliability of financial reports presented to business owners or external stakeholders.

Payment verification experienced significant delays as administrative staff manually cross-referenced tenant payment receipts against booking records and financial ledgers. This reconciliation process, which required staff to locate, retrieve, and compare multiple physical or digital documents for each transaction, became increasingly burdensome as the tenant population grew and transaction volume increased. Revenue miscalculations arose from formula errors in spreadsheet-based accounting tools, transactions that were inadvertently omitted from revenue summaries, or timing discrepancies between when payments were received and when they were recorded. These financial inaccuracies impeded management's ability to assess the true profitability of the business, plan for operational expenses, and fulfill tax and regulatory reporting obligations accurately.

Perhaps the most operationally damaging consequence of the manual system was the occurrence of double bookings—situations in which the same condominium unit was inadvertently reserved for two or more clients during overlapping periods. Double bookings arose when staff members, lacking access to a centralized and continuously updated availability record, committed a unit to a new booking without verifying that the unit had not already been reserved during the same period. The repercussions of double bookings were severe: affected tenants experienced significant inconvenience and frustration, the business incurred reputational damage that was difficult to reverse, and the administrative resources required to resolve resulting disputes and arrange alternative accommodations placed additional strain on already burdened staff.

The lack of an integrated scheduling and maintenance monitoring tool further compounded these operational difficulties. Without a centralized, real-time calendar system, staff struggled to maintain accurate and up-to-date visibility into unit availability across all properties. Scheduling conflicts arose not only from double bookings but also from situations where units undergoing maintenance or cleaning were inadvertently committed to incoming tenants before they were ready for occupancy. The absence of a formal check-in and check-out tracking mechanism also made it difficult to plan unit turnaround timelines and coordinate cleaning and inspection activities efficiently.

Maintenance requests were tracked through informal and unreliable channels, including handwritten logs maintained by individual staff members, verbal communications between tenants and property managers, or disconnected digital files that were not accessible to all relevant parties. This fragmented approach to maintenance tracking made it difficult to prioritize repairs based on urgency or impact, assign maintenance tasks to appropriate personnel with the necessary skills and availability, monitor the progress of ongoing repairs, and verify that completed work met quality standards before the unit was returned to service. Deferred or overlooked maintenance accelerated the deterioration of unit fixtures and common areas, increased long-term capital expenditure requirements, and diminished the living experience of current tenants, which in turn elevated tenant turnover rates.

These cumulative inefficiencies reduced operational productivity across all levels of the organization. Administrative staff expended excessive time and cognitive resources on repetitive data entry, manual reconciliation, and error correction tasks that a well-designed automated system could have handled instantaneously and with far greater accuracy. The opportunity cost of this allocated labor was substantial: hours dedicated to manual clerical work represented hours not invested in tenant relationship management, proactive property maintenance, marketing initiatives, or strategic business development activities. At the management level, the absence of real-time operational data and automated reporting tools forced decision-makers to rely on outdated or incomplete

information when formulating business strategies, allocating resources, or responding to emerging operational challenges. Without access to timely revenue reports, occupancy rate analyses, or maintenance cost summaries, management was ill-equipped to identify performance trends, forecast demand fluctuations, or evaluate the return on investment of individual units. This information deficit hindered the organization's capacity to adapt to changing market conditions and position itself competitively within Iloilo City's growing urban rental sector.

From the perspective of tenant experience, the cumulative effects of manual operational inefficiencies manifested as slower response times to inquiries and maintenance requests, a higher incidence of billing errors, and a greater likelihood of booking conflicts. As prospective and current tenants increasingly compared service quality across competing rental providers, these shortcomings placed Gales Urban Haven at a disadvantage relative to competitors who had adopted more sophisticated property management solutions. Declining tenant satisfaction translated into reduced occupancy rates, higher turnover costs, and diminished word-of-mouth referrals, all of which negatively affected the long-term financial viability of the business.

This study aimed to develop a Condominium Booking and Operations Management System to comprehensively address the identified operational concerns of Gales Urban Haven (One Spatial Iloilo). The proposed system centralized all

booking, payment, scheduling, and maintenance data within a single integrated platform, eliminating data redundancy and ensuring consistency across all operational records. Automation replaced manual calculations, verification processes, and scheduling coordination, substantially reducing the frequency and severity of errors while freeing staff to concentrate on higher-value activities. Real-time reporting capabilities empowered management with actionable, accurate, and timely insights necessary for sound strategic decision-making. The system was designed with scalability in mind, ensuring that it could continue to support the organization's operations effectively as its tenant population and property portfolio expanded in the future.

Specifically, this study sought to answer the following questions:

1. How would booking records be organized and managed efficiently within a centralized system to prevent data loss, minimize redundancy, and eliminate the occurrence of double bookings?

2. How would payment tracking and revenue monitoring be automated to reduce computational errors, accelerate financial reconciliation, and ensure the timely and accurate generation of financial reports?

3. How would an integrated calendar scheduling system improve the management of unit availability, booking coordination, and occupancy planning to prevent scheduling conflicts and maximize rental efficiency?

4. How would a formalized maintenance monitoring module systematically record, assign, track, and verify property

maintenance activities to prevent operational conflicts, reduce response times, and preserve unit condition?

5. How would the implementation of a centralized Condominium Booking and Operations Management System enhance overall organizational efficiency, data accuracy, staff productivity, and management decision-making for Gales Urban Haven (One Spatial Iloilo)?

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study was to design, develop, and implement a centralized Condominium Booking and Operations Management System for Gales Urban Haven, operating under the business name One Spatial Iloilo. The proposed system was intended to serve as a comprehensive digital solution that would automate and integrate the core operational workflows of the business, including booking management, payment monitoring, revenue tracking, calendar-based scheduling, and maintenance recording.The system will be engineered to replace the current manual and fragmented processes with a secure, reliable, and intelligent web-based platform that will significantly improve operational efficiency, minimize human errors, enhance data accessibility, and support faster and more accurate business decision-making. By centralizing all operational data into a single unified platform, the system will empower administrators and property managers to manage

day-to-day operations with greater confidence, control, and precision.

Ultimately, the system will not only address the present operational challenges of Gales Urban Haven but will also establish a scalable digital infrastructure that will adapt to the growing demands of the property management business, enabling the organization to sustain competitive service quality, optimize resource utilization, and make informed strategic decisions well into the future.

1.3.2 Specific Objectives

Specifically, this study aimed to:

1. Analyze the existing booking and operational procedures of Gales Urban Haven to systematically identify inefficiencies, recurring operational challenges, workflow bottlenecks, and system requirements necessary for automation and process improvement. This analysis involved a thorough examination of how reservations were recorded, how payments were tracked, how maintenance requests were managed, and how scheduling conflicts were handled. The findings derived from this phase served as the foundational basis for the design and development of a system that was precisely aligned with the real operational needs of the business, ensuring that the proposed solution was both practical and impactful. The analysis included direct consultations with business owners and administrative staff, whose firsthand experiences and operational insights provided the researchers with a comprehensive and accurate understanding of the existing challenges and inefficiencies that the proposed system needed to address. The researchers

also examined current documentation practices, record-keeping methods, and communication workflows to identify specific areas where automation and digital integration could yield significant improvements in operational efficiency and accuracy. Additionally, the analysis assessed the volume and frequency of bookings, payment transactions, and maintenance requests to determine the scale and complexity of data management requirements that the proposed system needed to accommodate. The findings of this analysis were also used to establish clear and measurable system requirements, ensuring that every feature and functionality incorporated into the proposed system directly responded to a specific and validated operational need of Gales Urban Haven. Moreover, the insights gathered during this phase informed the prioritization of system features during the development process, allowing the researchers to focus first on the most critical functionalities that delivered the greatest immediate impact on the business's operational performance. Ultimately, a thorough and systematic analysis of the existing booking and operational procedures of Gales Urban Haven ensured that the proposed Condominium Booking and Operations Management System was developed on a solid and well-informed foundation, maximizing its relevance, usability, and long-term value to the business and all of its stakeholders.

2. Design a secure, accessible, and user-friendly web-based system that provided administrators and property managers with a well-organized, intuitive, and reliable platform for managing all aspects of condominium operations. The system

design incorporated clearly defined user roles and access controls to ensure data privacy and operational security. The interface was structured to minimize the learning curve for end users, allowing staff to navigate and operate the system efficiently without extensive technical training. The design phase also considered scalability and adaptability so that the system remained functional and relevant as the business continued to grow and evolve. The system design followed established principles of user-centered design, ensuring that every interface element, navigation structure, and functional layout was developed with the specific needs, preferences, and technical capabilities of the end users in mind. The design team also conducted a thorough review of existing property management system interfaces and best practices in web-based application design, drawing upon proven design patterns and usability standards to inform the development of a visually clear and operationally effective user interface. Additionally, the system was designed with a responsive layout that ensured consistent functionality and visual clarity across a variety of devices and screen sizes, allowing administrators and property managers to access and operate the platform conveniently from desktop computers, laptops, and tablets. The design phase also incorporated robust data validation mechanisms and error-handling protocols that guided users in entering accurate information and prevented the submission of incomplete or incorrect data that could compromise the integrity of the system's records. Moreover, the security architecture of the system was carefully designed to incorporate industry-standard encryption protocols, secure authentication mechanisms, and role-based access controls

that collectively ensured the confidentiality, integrity, and availability of all sensitive operational and financial data stored within the platform. The design also prioritized system performance and loading efficiency, ensuring that all pages, modules, and data retrieval functions operated smoothly and responsively even during periods of high usage or large data volumes. Ultimately, the design of a secure, accessible, and user-friendly web-based system established a strong and reliable technological foundation for the proposed Condominium Booking and Operations Management System, ensuring that it effectively served the operational needs of Gales Urban Haven while delivering a consistently positive and productive experience for all authorized users.

3. Develop an automated booking management module that allows administrators to efficiently record and process reservation requests, check real-time unit availability, and prevent scheduling conflicts and double bookings. The module maintained a comprehensive and up-to-date repository of all active, pending, and completed reservations, enabling staff to retrieve booking information instantly and manage the reservation pipeline with greater accuracy. By automating the booking process, the system reduced reliance on manual entries, eliminated redundant record-keeping, and ensured that unit availability data remained current and reliable. The system also generated automated notifications and confirmation alerts that kept administrators and clients informed throughout every stage of the booking process. Additionally, the module produced detailed booking reports and analytics that enabled

management to monitor occupancy trends, forecast demand, and make data-driven decisions. The platform integrated with existing property management tools, ensuring that booking data was synchronized across departments. As the system evolved, it supported scalable functionalities that accommodated a growing volume of reservations without compromising speed, accuracy, or overall system performance. The module further incorporated a centralized dashboard that provided administrators with a clear and organized overview of reservation activities, allowing for faster response times and more efficient allocation of available units.

4. Implement a payment tracking system capable of recording and monitoring tenant down payments, outstanding balances, payment history, and cumulative revenue in a structured and auditable manner. The system generated clear and accurate financial records for each tenant transaction, providing administrators with a real-time view of the payment status of every unit. This module significantly improved financial accuracy by reducing discrepancies arising from manual computation and providing a transparent and verifiable trail of monetary transactions. The payment tracking system also included automated reminders and alerts that helped administrators proactively follow up on pending or overdue payments.

5. Create a revenue reporting module that automatically generated comprehensive financial summaries, itemized payment reports, and revenue analytics on demand and at

scheduled intervals. The module produced visual and tabular representations of income data, enabling management to quickly assess the financial performance of the property across different periods. By automating the generation of financial reports, the system substantially reduced the time and effort devoted to manual financial reporting, minimized the risk of computational errors, and ensured that decision-makers had access to accurate and timely financial information for strategic planning and business forecasting.

6. Integrate a calendar-based scheduling system that enabled administrators to monitor and manage reservations, tenant check-ins and check-outs, unit availability windows, and scheduled maintenance activities through a unified and interactive calendar interface. The scheduling module provided a clear visual overview of time-sensitive events and activities, allowing the property management team to coordinate operations more effectively and avoid scheduling conflicts. The calendar system was synchronized with the booking and maintenance modules, ensuring that changes in reservation status or maintenance scheduling were reflected across relevant parts of the system. The calendar-based scheduling system was equipped with color-coded event indicators and category filters that allowed administrators to quickly distinguish between different types of operational activities, such as new bookings, ongoing reservations, pending check-outs, and scheduled maintenance tasks, enabling faster and more efficient event management. The interactive calendar interface also supported multiple viewing modes, including daily, weekly, and monthly

perspectives, providing administrators with flexibility in monitoring operational schedules at varying levels of detail. Additionally, the scheduling module generated automated alerts and notifications for upcoming events and deadlines, ensuring that administrators and property managers were informed of critical activities in advance and could make the necessary preparations. The calendar system also maintained a complete and accurately timestamped record of scheduled and completed activities, providing administrators with a reliable historical reference that supported operational review, performance evaluation, and long-term planning. Moreover, integration with the centralized database ensured that all event data was securely stored, consistently updated, and readily accessible to authorized users, eliminating the need for separate scheduling tools or manual calendar management processes. The scheduling module was also designed with a user-friendly interface that allowed administrators to add, modify, and remove calendar entries with minimal effort. Ultimately, the integration of a robust and synchronized calendar-based scheduling system enhanced the operational visibility and coordination capabilities of the proposed Condominium Booking and Operations Management System, enabling the property management team of Gales Urban Haven to provide a well-organized and professionally managed rental experience.

7. Develop a maintenance monitoring module for systematically recording, tracking, and managing repair requests, preventive maintenance schedules, and service completion updates to ensure operational continuity and

improved property upkeep. The module allowed administrators to log maintenance requests with detailed descriptions, assign tasks to responsible personnel, set priority levels, and track the progress of each maintenance activity from initiation through completion. By providing a structured and auditable record of maintenance activities, the system helped management identify recurring issues, allocate maintenance resources more efficiently, and uphold high standards of property condition and tenant satisfaction. The maintenance monitoring module was integrated with the calendar-based scheduling system, ensuring that scheduled maintenance activities were automatically reflected in the operational calendar and that administrators could view and manage maintenance tasks alongside other time-sensitive operational events within a unified interface. The module also generated automated reminders and notifications for upcoming preventive maintenance schedules and overdue service requests, ensuring that maintenance activities were not neglected or delayed due to oversight or miscommunication. Additionally, the maintenance monitoring module maintained a comprehensive and timestamped history of maintenance activities, providing management with a reliable record that supported property condition assessments, warranty tracking, and long-term maintenance planning. The module also supported the categorization and filtering of maintenance records by unit, maintenance type, priority level, and completion status, allowing administrators to quickly locate and review specific maintenance activities and generate accurate maintenance reports. Moreover, the structured data captured by the maintenance monitoring module enabled management to identify patterns and trends in property maintenance needs,

providing insights that supported proactive maintenance planning and the prevention of costly repairs. The module also facilitated better communication and accountability among maintenance personnel by providing a clear and transparent record of task assignments, progress updates, and service completions accessible to authorized users. Ultimately, the development of a comprehensive and integrated maintenance monitoring module enhanced the property management capabilities of the proposed Condominium Booking and Operations Management System, enabling Gales Urban Haven to maintain high standards of property condition, operational continuity, and tenant satisfaction.

8. Establish a centralized database management system that securely stored, organized, and managed all operational data, including booking records, payment transactions, tenant profiles, maintenance logs, and revenue reports, within a single and cohesive data environment. The database would be designed with data integrity, consistency, and security as primary considerations, incorporating appropriate access controls, backup mechanisms, and data validation protocols. By consolidating all records into one centralized repository, the system will drastically reduce the time required for information retrieval, eliminate data silos, and provide administrators with a single source of truth for all operational and financial data. The centralized database management system will be structured using a well-defined relational database schema that will establish clear and logical relationships between all data entities, ensuring that information stored across different

system modules would remain consistently connected, accurately referenced, and efficiently retrievable at all times. The database will also be equipped with automated backup procedures that will regularly create secure copies of all stored data, protecting the business from the risk of data loss due to system failures, accidental deletions, or unforeseen technical disruptions. Additionally, the centralized database will support concurrent multi-user access, allowing multiple authorized administrators and staff members to simultaneously retrieve and update records without compromising data consistency or system performance, ensuring smooth and uninterrupted operational workflows across all departments. The data validation protocols embedded within the database management system will also enforce strict input rules and constraints that will prevent the entry of incomplete, inaccurate, or duplicate records, maintaining the overall quality and reliability of all data stored within the platform. Moreover, the centralized database will serve as the unified data source for all system-generated reports and analytics, ensuring that every financial summary, booking record, and maintenance log presented to administrators and business owners will be drawn from a single, consistently maintained, and accurately updated repository. The database architecture was also designed with scalability in mind, ensuring that the system was capable of accommodating a growing volume of records, transactions, and user accounts as the operations of Gales Urban Haven continued to expand over time. Ultimately, the establishment of a robust and secure centralized database management system served as the backbone of the proposed Condominium Booking and Operations Management System, providing a stable, reliable, and

well-organized data environment that supported all operational modules and contributed to the sustained efficiency, accuracy, and long-term success of Gales Urban Haven under One Spatial Iloilo.

9. The functionality, usability, and efficiency of the developed system were evaluated through structured user acceptance testing, system performance testing, and comprehensive feedback collection from end users and stakeholders. The evaluation phase assessed the degree to which the system fulfilled its intended objectives, responded to the operational needs of Gales Urban Haven, and performed reliably under real-world usage conditions. The results of this evaluation were used to identify areas for refinement and improvement, ensuring that the final system was fully validated as an effective, dependable, and sustainable solution for the operational challenges that the business was currently facing and anticipated encountering in the future.

The structured user acceptance testing involved direct participation from administrators, business owners, and other authorized end users of Gales Urban Haven, whose firsthand interactions with the system provided the researchers with valuable and practical insights into the overall usability, functionality, and relevance of the developed platform in the context of real operational scenarios. The system performance testing also assessed critical technical parameters such as response time, data processing speed, and system stability under varying levels

of user activity and data volume, ensuring that the proposed system consistently delivered a smooth and reliable user experience even during periods of peak operational demand.

Additionally, the comprehensive feedback collection process employed structured evaluation instruments, including questionnaires and rating scales, that allowed end users and stakeholders to provide detailed and measurable assessments of the system's features, interface design, and overall operational impact, generating data that directly informed the refinement and finalization of the platform. The evaluation phase also examined the accuracy and consistency of all system-generated reports, records, and data outputs, verifying that the centralized database and all integrated modules produced reliable and precise information that administrators and business owners could confidently utilize for operational and decision-making purposes.

Moreover, any issues, deficiencies, or areas of improvement identified during the evaluation phase were systematically documented and addressed through targeted refinements and system updates, ensuring that the final delivered system fully met the quality standards, functional requirements, and usability expectations established at the outset of the development process. The evaluation results were also analyzed and interpreted within the context of the study's research objectives, providing a comprehensive and evidence-based assessment of the extent to which the

proposed Condominium Booking and Operations Management System successfully addressed the identified operational inefficiencies of Gales Urban Haven.

1.4 Scope and Delimitation

This chapter defined the boundaries and parameters of the proposed study, establishing a clear understanding of what the research covered, what it accomplished, and what it deliberately excluded from its scope. The precise delineation of scope and delimitation was critical in this academic research endeavor as it provided clarity of purpose, prevented scope creep, and ensured that the resulting system met the expectations of its intended users within a realistically achievable framework.

The following sections discussed in detail the focus areas of the study, the specific modules and functionalities that were developed, the limitations that were set, and the rationale behind each of these decisions. Collectively, these elements formed the structural and operational boundaries within which the proposed system was built, tested, and evaluated. The clearly defined scope and delimitation of this study served as a guiding framework throughout the entire research and development process, ensuring that all decisions regarding system design, feature development, and implementation strategies remained consistently aligned with the established boundaries and objectives of the proposed project.

The scope of the study also provided a transparent and well-documented reference point for all stakeholders involved in the development and evaluation of the proposed system, ensuring that business owners, administrative staff, academic advisers, and evaluators shared a common and accurate understanding of the system's intended capabilities and limitations. Additionally, the delimitation of the study reflected the researchers' commitment to delivering a focused and high-quality system within the constraints of the available time, resources, and technical environment, prioritizing the development of core functionalities that delivered the most immediate and meaningful operational benefits to Gales Urban Haven.

The boundaries established in this chapter also protected the academic integrity and feasibility of the research by preventing the inclusion of features or functionalities that fell outside the realistic capacity of the development team, ensuring that the proposed system was completed, tested, and evaluated within the defined timeframe of the study. Moreover, the scope and delimitation outlined in this chapter provided future researchers with a clear and comprehensive understanding of the extent and boundaries of this study, enabling them to identify specific areas for expansion, enhancement, and further academic inquiry in subsequent research projects related to property management systems and web-based business solutions.

The structured presentation of scope and delimitation also demonstrated the methodological rigor and careful planning that underpinned the entire research process, reinforcing the credibility and academic soundness of the proposed Condominium Booking and Operations Management System as a well-conceived and purposefully developed solution for the operational challenges of Gales Urban Haven. Ultimately, the precise and transparent definition of scope and delimitation presented in this chapter ensured that the proposed system was developed, evaluated, and delivered as a focused, feasible, and academically sound solution that effectively addressed the core operational needs of Gales Urban Haven under One Spatial Iloilo, while also contributing meaningfully to the broader academic discourse on the application of information systems in property management and hospitality operations.

1.4.1 Focus of the Study

This study focused on the design, development, and evaluation of a web-based Condominium Booking and Operations Management System intended for use by Gales Urban Haven, operating under the registered business name One Spatial Iloilo. The business was a condominium property management enterprise that previously relied on manual or fragmented methods to manage its day-to-day rental operations. In its present state at the time, the organization faced recurring inefficiencies in tracking unit availability, recording payments, monitoring maintenance tasks, and generating financial reports —

challenges that directly impacted operational productivity and customer satisfaction.

The proposed system was developed to address these operational gaps by automating and centralizing the major business processes related to condominium rental and property management. Specifically, the system served as a unified digital platform that allowed administrators to manage bookings, monitor financial transactions, track maintenance activities, and view real-time availability data from a single, accessible web interface.

The development of this system followed a structured software development approach, incorporating requirements analysis, system design, coding, testing, and evaluation phases. The final product was assessed based on its alignment with the operational needs of One Spatial Iloilo and its capacity to improve the overall efficiency and accuracy of the organization's property management processes. Furthermore, the system was evaluated using established software quality standards to ensure that it met the usability, reliability, and functionality requirements expected of a modern property management solution.

The findings of the evaluation served as a basis for identifying areas of improvement and contributed to the overall body of knowledge in the field of web-based information systems for real estate and property

management. It was anticipated that the successful implementation of the system would not only streamline internal operations but would also enhance the experience of tenants and prospective clients by providing a more transparent, responsive, and organized booking and management process.

Ultimately, this study aimed to demonstrate that a purpose-built digital system could serve as a viable and scalable solution to the operational challenges faced by One Spatial Iloilo, laying the groundwork for future system expansions and potential adoption by similar property management enterprises.

1.4.2 Core System Modules and Functionalities

The proposed system included seven (7) core modules, each designed to address a specific operational domain within the condominium management workflow. Each module interacted with the others to form a cohesive, integrated system. The following subsections described each module in detail, including its specific functions, operational logic, and the particular problem it sought to resolve.

The seven core modules were developed in alignment with the identified operational needs and workflow requirements of Gales Urban Haven, ensuring that each module directly contributed to the resolution of a specific and validated

operational challenge faced by the business in its previous manual and fragmented management processes. The integration of all seven modules within a single, unified platform eliminated the need for separate tools or disconnected systems, providing administrators and business owners with a comprehensive and centralized solution that addressed the full spectrum of condominium management operations through one cohesive and consistently maintained system.

Additionally, each module was designed with a clear and intuitive interface that allowed authorized users to navigate and operate its features with minimal technical difficulty, ensuring that the functionality of every module was readily accessible and practically useful to all categories of end users at Gales Urban Haven. The modular architecture of the system also provided a flexible and scalable framework that allowed individual modules to be updated, enhanced, or expanded independently without disrupting the overall functionality of the platform, ensuring that the system remained adaptable and relevant as the operational needs of the business continued to evolve over time.

Moreover, the seamless interaction and data sharing between all seven core modules ensured that information entered or updated in any part of the system was immediately and accurately reflected across all related modules, maintaining the consistency and integrity of all operational data stored within the centralized database at all times. The development of each module was also guided

by the findings of the operational analysis conducted during the initial phase of the research, ensuring that every feature and functionality incorporated into the proposed system was purposefully designed to deliver measurable and meaningful improvements to the specific operational processes it was intended to support.

Ultimately, the seven core modules of the Condominium Booking and Operations Management System collectively formed a powerful, integrated, and operationally comprehensive platform that transformed the management workflows of Gales Urban Haven, enabling the business to operate with greater efficiency, accuracy, and confidence as it continued to grow and serve its tenants and clients under One Spatial Iloilo.

Module 1: User Authentication Module

The User Authentication Module served as the gateway to the entire system, ensuring that only authorized personnel were granted access to its features and data. This module was foundational to the system's security architecture and implemented industry-standard login mechanisms to protect sensitive information from unauthorized access.

The module included the following functionalities:

- Secure administrator login using username and password credentials
- Session management to maintain authentication state across pages

- Role-based access control (RBAC) to regulate which features were accessible to different types of users
- Password management features, including the ability to change credentials
- Automatic session expiration after a defined period of inactivity .

This module prevented unauthorized individuals from accessing financial records, booking data, and other sensitive operational information. It also provided a foundation for future expansion of user roles should the organization grow and require differentiated access levels among its staff members.

Module 2: Booking Management Module

The Booking Management Module automated and centralized the process of recording and managing tenant reservations. At the time, many property management offices relied on manual registers or spreadsheets to track bookings, which were prone to data loss, duplication, and human error. This module replaced those processes with a structured, database-driven booking system.

The module covered the following core functions:

- Creation and recording of new reservations, including tenant personal information, preferred unit, and booking dates.
- Retrieval and display of complete booking history for both individual tenants and specific units.

- Editing and updating of existing reservation details as needed .
- Cancellation and archiving of past and canceled bookings.
- Generation of booking summaries and reservation confirmation records.

All booking records were stored in a centralized database, enabling administrators to retrieve, review, and manage reservations with speed and accuracy. The module also served as the primary data source for several other modules, including the Availability Checking Module and the Payment Recording Module.

Module 3: Availability Checking Module

The Availability Checking Module provided real-time visibility into the occupancy status of each condominium unit managed by One Spatial Iloilo. This module was designed to address the problem of double bookings and scheduling conflicts, which were common in manually operated rental systems.

The module included the following capabilities:

- Real-time monitoring of unit occupancy status, distinguishing between available, occupied, reserved, and under-maintenance units
- Prevention of double bookings through automated conflict detection during the reservation process

- Display of unit availability* within specified date ranges
- Visual dashboard representation of unit availability for quick reference by administrators
- Integration with the Booking Management and Calendar Integration modules to maintain synchronized data across all related functions

By providing instant and accurate availability information, this module significantly reduced the risk of operational errors and improved the responsiveness of the booking process. Administrators no longer needed to manually cross-reference multiple records to determine which units were free, as this information was automatically computed and displayed by the system.

Module 4: Payment Recording Module

The Payment Recording Module managed all financial transactions related to tenant bookings, providing a structured and transparent record of all payments received and obligations outstanding. This module directly addressed the challenge of maintaining accurate and up-to-date financial records, which was critical for the operational sustainability of any rental business.

The module included the following functionalities:

- Recording of initial down payments at the time of booking
- Documentation of subsequent balance payments and installment transactions.

- Maintenance of a complete and chronological payment history for each tenant account .
- Tracking of outstanding balances and overdue accounts.
- Generation of payment receipts and acknowledgment records.
- Linking of payment records to specific bookings and tenant profiles for traceability.

All financial data entered into this module was securely stored and fed directly into the Revenue Monitoring and Financial Reporting Module, enabling the automatic computation of revenue figures without the need for manual data re-entry. This integration reduced the administrative workload and the risk of computational errors in financial reporting.

Module 5: Revenue Monitoring and Financial Reporting Module

The Revenue Monitoring and Financial Reporting Module provided management-level visibility into the financial performance of One Spatial Iloilo's condominium operations. This module aggregated payment data from the Payment Recording Module and generated structured financial reports that supported informed decision-making.

The module delivered the following features:

- Automated computation of total revenue within user-specified periods (daily, weekly, monthly, annual) .

- Generation of payment summaries categorized by unit, tenant, and payment type.
- Production of financial reports* suitable for administrative review and record-keeping .
- Display of revenue trends and payment status overviews through summary dashboards.
- Exportable reports in printable format for documentation and compliance purposes .

This module eliminated the need for manual computation of revenue figures, a process that was both time-consuming and susceptible to arithmetic errors. By providing on-demand financial reports, the module enabled administrators to assess the financial health of the business at any point in time.

Module 6: Calendar Integration Module

The Calendar Integration Module provided a visual and chronological interface for managing tenant reservations, check-in and check-out schedules, and unit availability timelines. This module was designed to improve the clarity and organization of scheduling activities, which were central to the day-to-day operations of a condominium rental business.

The module included the following features:

- An interactive calendar view displaying all active and upcoming reservations.

- Visual markers for tenant check-in and check-out dates, color-coded by unit or status.
- Calendar-based navigation for reviewing past, current, and future bookings.
- Synchronization with the Booking Management and Availability Checking modules to reflect real-time data.
- Date-range filtering to support planning and scheduling activities.

The calendar interface allowed administrators to quickly identify scheduling patterns, detect potential conflicts, and plan unit turnover activities. It served as an intuitive visual tool that complemented the more detailed data management functions of other modules.

Module 7: Maintenance Tracking Module

The Maintenance Tracking Module managed the recording, scheduling, and monitoring of all maintenance and repair activities related to the condominium units. Proper maintenance management was essential to preserving the quality and safety of rental properties, and this module provided the infrastructure needed to keep maintenance operations organized and accountable.

- The module included the following functionalities:
- Recording of maintenance requests submitted for specific units.
- Scheduling and assignment of maintenance tasks* to responsible personnel.

- Tracking of task status, including pending, in-progress, and completed.
- Maintenance history logs for each unit to monitor recurring issues.
- Notifications and alerts for upcoming or overdue maintenance activities.

All maintenance records were linked to specific units and reflected in the Availability Checking Module, ensuring that units under maintenance were not made available for booking. This module will help management ensure that maintenance issues are addressed promptly and systematically, preventing minor problems from escalating into larger and more costly repairs. It will also contribute to the overall documentation of unit conditions, which may be useful in future property assessments or tenant disputes.

Module Summary

The table below provides a consolidated overview of all seven core modules, their key functionalities, and the expected operational outcomes.

Module	**Key Functionalities**	**Expected Outcome**
User Authentication	Login, session management, role-based access control	Secure, authorized system access

Booking Management	Reservation creation, editing, history tracking, cancellation	Centralized, error-free booking records
Availability Checking	Unit status monitoring, double-booking prevention, real-time display	Conflict-free scheduling
Payment Recording	Down payments, balance tracking, payment history, receipts	Accurate financial transaction records
Revenue Monitoring & Reporting	Revenue computation, financial reports, payment summaries	Data-driven financial oversight
Calendar Integration	Reservation calendar, check-in/out scheduling, date filtering	Organized visual scheduling interface
Maintenance Tracking	Request recording, repair scheduling, status monitoring, history logs	Systematic property maintenance management

1.4.3 Academic Context and Development Framework

The study was limited to the development, implementation, testing, and evaluation of the proposed system within an academic setting. The project was treated as a capstone research undertaking, adhering to the academic standards and methodological requirements set by the institution.

The research followed a structured systems development lifecycle that encompassed requirements gathering, system analysis and design, development and coding, testing, and final evaluation. The system was developed as a functional web-based prototype, meaning it was fully operational and capable of performing all the functions described within its scope, but it was not configured or optimized for large-scale commercial deployment.

As a prototype, the system demonstrated the viability of the proposed solution and served as a working model that the client organization could choose to adopt, further develop, or enhance following the completion of the academic study.

Prototype vs. Commercial Deployment

The distinction between a functional prototype and a commercially deployed system is important. A functional prototype operates correctly within a controlled or limited environment and demonstrates the full capabilities of its

intended design. Commercial deployment, by contrast, involves additional considerations such as scalability, load balancing, enterprise security architecture, and regulatory compliance — aspects that fall outside the boundaries of this academic study.

The development of the system was conducted using web technologies appropriate for the academic environment and consistent with the technical competencies developed throughout the researchers' program of study. The final evaluation of the system was conducted with selected evaluators whose feedback was gathered using structured assessment instruments aligned with recognized software quality standards.

1.4.4 Delimitations of the Study

While the proposed system covered a wide range of operational functions essential to the management of One Spatial Iloilo condominium operations, certain features and capabilities were deliberately excluded from the scope of the study. These exclusions were referred to as delimitations and represented conscious decisions made by the researchers to maintain the focus, feasibility, and academic appropriateness of the project. Each delimitation was described below, along with the rationale for its exclusion.

The delimitations of this study were clearly defined and transparently communicated to ensure that all stakeholders, evaluators, and future researchers had a precise and accurate understanding of the boundaries within which the proposed system was developed and assessed. The identification of these delimitations also served to protect the integrity and credibility of the study, as clearly established boundaries prevented misinterpretations regarding the intended scope, capabilities, and applicability of the proposed Condominium Booking and Operations Management System.

Additionally, the delimitations reflected the practical constraints of the research environment, including limitations in time, resources, and technical infrastructure, that influenced the researchers' decisions regarding which features and functionalities were prioritized within the development scope of the proposed system. The conscious exclusion of certain features also allowed the research team to allocate their efforts and resources more effectively toward the development and refinement of the core system functionalities that delivered the most immediate and impactful operational benefits to Gales Urban Haven.

Moreover, the delimitations outlined in this section provided a valuable reference point for future researchers who sought to expand upon or enhance the proposed system, as the clearly documented exclusions highlighted specific areas and functionalities that represented natural

opportunities for further development, innovation, and academic inquiry. Ultimately, the transparent and well-reasoned presentation of the study's delimitations reinforced the academic rigor and methodological soundness of the research, demonstrating that the scope of the proposed system was carefully and deliberately defined to ensure the delivery of a focused, feasible, and high-quality solution that effectively addressed the core operational needs of Gales Urban Haven under One Spatial Iloilo.

A. Integration with External Online Payment Gateways or Banking Systems

The system did not integrate with third-party online payment gateways such as PayMaya, GCash, PayPal, or any banking application programming interfaces (APIs). While the integration of digital payment processing would have significantly enhanced the convenience and functionality of the system from a commercial standpoint, such integrations required compliance with financial regulations, merchant account agreements, secure communication protocols, and ongoing API maintenance — all of which were beyond the scope of an academic capstone project.

Instead, the Payment Recording Module focused on the manual documentation of payments that had already been received, providing a structured and accurate record-keeping system rather than a payment processing platform. This approach was consistent with the current operational practice of the client and remained fully relevant to the organization's

immediate needs. The groundwork established by this module served as a solid foundation upon which future developers could build more advanced payment functionalities.

B. Mobile Application Development for Android or iOS Platforms

The study did not include the development of a native mobile application for Android or iOS devices. Mobile application development constituted a distinct technical discipline with its own set of tools, frameworks, and platform-specific requirements. Developing a mobile companion application in parallel with the primary web-based system would have significantly expanded the scope of the project beyond what was achievable within the time and resource constraints of the study.

The proposed web-based system was, however, designed with a responsive user interface where practicable, allowing it to be accessed from modern mobile browsers without a dedicated native application. This approach ensured basic mobile accessibility while keeping the development scope manageable and focused. By prioritizing the web-based platform as the primary mode of access, the study was able to direct its full attention and resources toward delivering a robust, fully functional, and well-evaluated system that addressed the core operational needs of One Spatial Iloilo.

C. SMS or Email Notification Services

Automated notification services, including SMS alerts and email confirmations sent directly to tenants or administrators, were not incorporated into the system. Such features required integration with third-party messaging platforms such as Twilio, SendGrid, or local telecommunications services, and involved subscription costs, API access agreements, and data privacy considerations governed by applicable telecommunications regulations.

The system handled all notification-related information internally, providing administrators with the data they needed to communicate with tenants through their preferred external communication channels. This limitation did not impede the core operational functions of the system and was considered for inclusion in a future version of the software.

D. Multi-Property or Enterprise-Level Management Features

The system was designed specifically for use by Gales Urban Haven operating as One Spatial Iloilo, a single-property condominium management operation. As such, it did not include multi-property management features, enterprise-level dashboards, or tools designed to support the simultaneous administration of multiple buildings or branch locations. Enterprise-level systems typically required complex multi-tenant database architectures, hierarchical administrative structures, and extensive

customization capabilities that were beyond the scope of this study.

The proposed system was tailored to the specific operational structure and requirements of a single property, ensuring that its design remained practical, relevant, and deliverable within the academic timeline. The modular architecture established during the current phase ensured that the addition of enterprise-level features in subsequent versions would be technically feasible and structurally supported by the existing system framework.

E. Cloud-Based Hosting and Advanced Cybersecurity Implementations

The study did not cover cloud-based hosting configurations, content delivery networks (CDNs), server load balancing, or advanced cybersecurity implementations beyond standard authentication mechanisms and basic database security practices. While these elements were important considerations for any production-level web application, their implementation required specialized infrastructure knowledge and resources that were not within the scope of an academic capstone study.

The system's security architecture included essential measures such as hashed password storage, session management, and basic input validation to protect against common vulnerabilities. These measures were appropriate for

the academic and prototype context in which the system was deployed and evaluated.

Summary of Delimitations

Excluded Features	Rationale for Exclusion
External Payment Gateway Integration	Requires financial regulatory compliance, merchant agreements, and ongoing API maintenance beyond academic scope.
Mobile Application (Android/iOS)	Constitutes a separate technical discipline; inclusion would exceed the project's achievable scope.
SMS/Email Notification Services	Requires third-party API integration and may involve data privacy regulations.
Multi-Property Management Features	Enterprise-level architecture is not aligned with the single-property context of One Spatial Iloilo.
Cloud Hosting & Advanced Cybersecurity	Requires specialized infrastructure and resources outside the constraints of an academic prototype.

1.4.5 Evaluation Framework

The evaluation of the proposed system was conducted through a structured assessment process involving selected evaluators who interacted with the fully functional prototype under controlled conditions. The evaluation measured the quality and effectiveness of the system across five (5) dimensions, each of which corresponded to a recognized criterion in software quality assurance.

Evaluation Dimensions

Functionality • Usability • Efficiency • Reliability • Accuracy

Functionality

Functionality referred to the degree to which the system performed all of its specified features correctly and completely. Evaluators assessed whether each of the seven core modules operated as intended, whether all defined features were present and accessible, and whether the system correctly responded to user inputs and commands. A functionally complete system demonstrated that all described capabilities had been successfully implemented and were working as designed.

Usability

Usability referred to the ease with which users could learn, navigate, and operate the system to accomplish their intended tasks. Evaluators assessed the clarity of the

system's interface, the intuitiveness of its navigation structure, the consistency of its design elements, and the degree to which it reduced cognitive effort on the part of the user. A highly usable system required minimal training and allowed administrators to perform their tasks efficiently from the outset. The usability evaluation also examined the responsiveness and visual clarity of the system's interface across different devices and screen sizes, ensuring that administrators and authorized users were able to access and operate the platform conveniently and effectively regardless of the device they used. The evaluation also assessed the degree to which the system's workflow design aligned with the natural operational routines and task sequences of the end users, ensuring that the navigation structure and feature layout felt logical, familiar, and efficient to administrators performing their daily responsibilities. Additionally, usability testing involved direct observation of end users interacting with the system in realistic operational scenarios, allowing evaluators to identify specific interface elements or workflow sequences that caused confusion, inefficiency, or unnecessary difficulty during actual system operation. The feedback gathered from end users during the usability evaluation was systematically analyzed to identify patterns and recurring concerns that informed targeted interface refinements and design improvements aimed at enhancing the overall user experience of the proposed system. Moreover, the usability of the system was assessed against established usability standards and best practices in web-based application design, ensuring that the proposed platform met the quality benchmarks expected of a professional and operationally effective information

system. The results of the usability evaluation were also used to measure the degree to which the system aligned with the principles of the Technology Acceptance Model, particularly in terms of perceived ease of use, which was a critical determinant of the system's overall adoption and sustained utilization by all categories of users at Gales Urban Haven. Ultimately, a thorough and structured assessment of the system's usability ensured that the proposed Condominium Booking and Operations Management System was delivered as an intuitive, accessible, and user-centered platform that empowered administrators and property managers to perform their operational responsibilities with confidence, efficiency, and minimal technical difficulty.

Efficiency

Efficiency refers to the system's ability to perform its functions within an acceptable time frame and with minimal use of system resources. Evaluators assessed page load times, the speed of data retrieval and processing, and the system's responsiveness under normal usage conditions. An efficient system demonstrated that it could support day-to-day operations without unnecessary delays that could disrupt the workflow of the user.

Reliability

Reliability referred to the consistency and dependability of the system's performance over time and across different usage scenarios. Evaluators assessed whether the system

behaved predictably, handled errors gracefully, and maintained data integrity under various conditions. A reliable system demonstrated that it could sustain consistent performance without unexpected failures or data loss.

Accuracy

Accuracy refers to the correctness of the data produced, processed, and displayed by the system. Evaluators verified that financial computations, booking records, availability statuses, and maintenance logs were correctly generated and faithfully reflected the information entered by the user. An accurate system demonstrated that it could be trusted as a reliable source of operational data for administrative and decision-making purposes.

The evaluators were selected based on their relevance to the system's intended use and included information technology professionals, property management practitioners, academic faculty members, and end users from the client organization. Their assessments were gathered through structured questionnaires based on recognized software evaluation standards, and the results were analyzed to determine the overall quality and readiness of the proposed system. The accuracy evaluation also examined the consistency of data across all integrated system modules, ensuring that any information entered or updated in one part of the system was correctly and immediately reflected in all related modules and records without discrepancies or data mismatches. Evaluators also conducted systematic testing of the system's financial computation

functions, including payment calculations, revenue summaries, and outstanding balance monitoring, to verify that all mathematical operations produced correct and reliable results that administrators and business owners could confidently rely upon for financial reporting and decision-making. Additionally, the accuracy assessment included a thorough review of the system's data validation mechanisms, ensuring that all input fields, forms, and data entry processes were equipped with appropriate constraints and error-checking protocols that effectively prevented the submission of incorrect, incomplete, or inconsistent information. The evaluation also tested the accuracy of the system's real-time data synchronization functions, verifying that booking statuses, unit availability windows, and maintenance schedules were correctly and promptly updated across all relevant modules whenever changes were made by authorized users. Moreover, the accuracy assessment included a review of the system's generated reports and data summaries, which were cross-referenced against manually verified records to confirm that all system outputs faithfully and precisely represented the underlying operational data stored within the centralized database. The results of the accuracy evaluation were also used to identify computational errors, data inconsistencies, or system vulnerabilities that could compromise the reliability of the platform, ensuring that all identified issues were thoroughly resolved before the final deployment of the proposed system. Ultimately, a rigorous and comprehensive assessment of the system's accuracy ensured that the proposed Condominium Booking and Operations Management System functioned as a trustworthy, precise, and dependable operational tool that provided Gales Urban Haven

with the accurate and reliable data it needed to manage its condominium rental business effectively and with full confidence in the integrity of its information.

1.4.6 Significance of the Defined Scope

The careful delineation of scope and delimitation serves multiple important purposes in the context of this capstone research. By clearly defining what the system will and will not include, the researchers are able to direct their efforts toward achievable, high-impact deliverables that will meaningfully address the operational needs of One Spatial Iloilo.

From a practical standpoint, the defined scope ensures that the seven core modules collectively provide a comprehensive solution to the most pressing operational challenges faced by the client, including fragmented booking records, manual payment tracking, uncoordinated maintenance management, and the absence of real-time availability data. The system, as scoped, will introduce a level of operational efficiency and data accuracy that is currently unavailable to the organization.

From an academic standpoint, the defined scope and delimitations ensure that the study remains executable within the academic timeline and resource constraints of the researchers, while still producing a system of sufficient complexity and practical value to satisfy the requirements of a capstone research project. The exclusion

of features beyond the defined scope is not a reflection of their lack of importance, but rather a deliberate and strategic decision to maintain the integrity and manageability of the study.

It is also worth noting that the features identified as delimitations in this study — including online payment integration, mobile application development, notification services, and cloud-based deployment — represent natural directions for the future development and enhancement of the system. As such, this capstone project will not only deliver immediate operational value to the client but will also lay the groundwork for a more expansive and commercially viable system in the future.

1.5 Significance of the Study

The proposed Condominium Booking and Operations Management System was expected to provide benefits to various stakeholders by improving business processes and operational management.

• **Business Owner** – The study helped business owners improve operational efficiency through automated booking management, accurate payment monitoring, revenue tracking, and centralized record management. It also supported better decision-making through readily available reports and organized business data. Furthermore, business owners were able to monitor real-time updates on reservations and

cancellations, allowing them to respond promptly to changes in demand. The system also provided detailed financial summaries that enabled owners to identify trends, forecast future revenue, and allocate resources more effectively. In addition, business owners benefited from a reduced manual workload, as the automated processes minimized human errors and saved valuable time. The centralized platform also allowed owners to access and review all business records from a single location, which significantly improved overall business management and long-term planning.

• **Administrative Staff** – The system simplified daily operational tasks such as booking management, payment recording, schedule monitoring, and maintenance tracking. This reduced manual workload, minimized errors, and improved productivity. Additionally, the system provided administrative staff with a more organized and efficient workflow, allowing them to focus on higher-priority tasks rather than repetitive manual processes. Staff members were also able to retrieve and update records quickly, which eliminated unnecessary delays and ensured that all information remained accurate and up to date. The system further assisted administrative staff in generating reports and summaries, reducing the time spent on paperwork and manual data compilation. Moreover, real-time notifications and alerts kept the staff well-informed about upcoming bookings, pending payments, and scheduled maintenance, ensuring that operational concerns were addressed in a timely manner. As a result, administrative staff experienced a more manageable and less stressful work environment, ultimately contributing to a higher level of service quality and overall operational success.

• **Tenants and Clients** – Clients benefited from improved booking organization, reduced scheduling conflicts, and more efficient handling of reservations and maintenance concerns, resulting in better service quality and customer satisfaction. Furthermore, clients experienced a smoother and more convenient booking process, as the system allowed them to make and confirm reservations with greater ease and accuracy. They also received timely updates and notifications regarding their bookings, payments, and any changes to their reservations, ensuring that they remained well-informed. In addition, clients benefited from faster response times to maintenance concerns and service requests, as the system enabled staff to track and address issues more efficiently. The organized record management also ensured that client information and transaction history were stored securely and accurately, reducing the likelihood of disputes or miscommunication. Ultimately, the improved system fostered a more trustworthy and professional relationship between the business and its clients, leading to higher levels of loyalty and long-term customer satisfaction.

• **BSIT Students** – This study served as a practical application of concepts in web development, database management, software engineering, and systems analysis and design. It also provided students with experience in solving real-world business problems using information technology. Moreover, BSIT students gained hands-on experience in developing and implementing a fully functional system, which strengthened their technical skills and deepened their understanding of the software development lifecycle. The study also exposed students to

the challenges and complexities of building real-world applications, preparing them for future professional roles in the field of information technology. Additionally, students developed critical thinking and problem-solving abilities as they designed and refined system features to meet the specific needs of the business. The project also enhanced their ability to work collaboratively, manage project timelines, and communicate technical solutions effectively, which were valuable skills for their future careers. Furthermore, this study served as a significant addition to their academic portfolio, demonstrating their capability to create practical and impactful technology solutions that addressed genuine business needs.

• **Future Researchers** – The study served as a reference for future research related to property management systems, booking automation, revenue monitoring, and web-based business solutions. Researchers could use this study as a foundation for enhancing similar systems or exploring additional features and technologies. Furthermore, future researchers were able to build upon the findings and methodologies of this study to develop more advanced and comprehensive property management solutions. The documented processes, system design, and implementation strategies presented in this study served as a valuable guide for researchers who undertook similar projects in the future. Additionally, future researchers could use this study to identify the limitations and challenges encountered during development, which helped them formulate better approaches and more effective solutions. The study also opened opportunities for further exploration of emerging technologies such as artificial intelligence, machine

learning, and cloud-based systems that could be integrated into property management platforms. Moreover, the data and insights gathered from this study contributed to the growing body of knowledge in the field of information systems and web-based business solutions, supporting the advancement of future academic and professional research.

• **Academic Institution** – The institution benefited from this study as an additional academic resource that demonstrated the practical use of information systems in solving business operational challenges and promoting innovation among students. Furthermore, the study served as a testament to the institution's commitment to producing competent and skilled graduates who were capable of developing practical technology solutions for real-world business needs. The completed system and its documentation were made available as reference materials that future students and faculty members could utilize for academic purposes and further research. Additionally, the institution was able to showcase this study as an example of student achievement that reflected the quality and effectiveness of its information technology curriculum. The study also contributed to the institution's academic resources by providing a well-documented case of applied systems development, which supported instructors in teaching relevant concepts in web development, database management, and systems analysis. Moreover, the success of this study encouraged more students to engage in research and development projects, fostering a culture of innovation

and academic excellence that strengthened the institution's reputation in the field of information technology.

1.6 Definition of Terms

The following terms were operationally defined to provide a clearer and more comprehensive understanding of the key concepts, components, and processes central to this study. These definitions were contextualized within the scope of the Condominium Booking and Operations Management System developed for Gales Urban Haven, operating under the business name One Spatial Iloilo, and were intended to guide readers in interpreting the discussions, analyses, and findings presented throughout this research.

Automation – Referred to the application of technology and software logic to execute repetitive, rule-based operational processes with minimal or no manual human intervention. In the broader context of information systems and business operations, automation was widely recognized as a critical driver of productivity, operational consistency, and error reduction. In this study, automation was applied across multiple functional areas of the proposed system, including the processing of booking requests, the computation and recording of payments, the generation of financial reports, and the scheduling of maintenance activities. By automating these processes, the system reduced the administrative burden on staff, minimized the occurrence of human errors inherent in manual operations, and ensured that critical business workflows

were carried out consistently and reliably. The automation features integrated into the system enabled administrators and staff members to dedicate more of their time and attention to higher-value operational tasks that required human judgment and decision-making, rather than expending their efforts on repetitive and time-consuming manual processes that could be more efficiently handled by the system. The automated booking processing functions also ensured that reservation confirmations, availability updates, and scheduling adjustments were executed promptly and accurately without requiring manual intervention, significantly reducing the likelihood of double bookings, scheduling conflicts, and reservation errors that had previously affected the operational efficiency of Gales Urban Haven. Additionally, the automation of payment computation and recording eliminated the need for manual financial calculations and data entry, ensuring that all payment transactions were processed with a high degree of accuracy and that all financial records were consistently and immediately updated within the centralized database upon the completion of each transaction.

Booking Management System – Referred to a computerized system designed to record, organize, process, and monitor reservation transactions, encompassing booking details, tenant information, unit availability tracking, and comprehensive reservation history. In the context of condominium property management, a booking management system served as the primary operational tool for handling all stages of the reservation lifecycle, from initial inquiry and unit assignment to check-in confirmation and booking completion. In this study, it specifically referred

to the automated module embedded within the proposed system that enabled One Spatial Iloilo to manage condominium bookings efficiently, accurately, and in real time, replacing the cumbersome and error-prone manual booking processes previously practiced by the business. As the system was utilized and refined, the Booking Management System served as the central backbone of the organization's operational workflow, ensuring that all reservation-related transactions were handled with greater speed, consistency, and reliability. The module included advanced filtering and search functionalities that allowed administrators to retrieve booking records more efficiently and generate comprehensive reports based on specific time periods, unit types, or tenant profiles. The system was also positioned to accommodate an increasing volume of booking transactions as One Spatial Iloilo expanded its property portfolio and grew its tenant base over time. During the development phase, the Booking Management System was capable of integrating with other modules such as payment recording and tenant management, creating a fully interconnected platform that provided administrators with a holistic view of all ongoing and completed reservations. Furthermore, the data accumulated through system usage served as a valuable source of insights that developers and management personnel could analyze to identify booking trends, optimize unit availability, and improve overall reservation strategies. The transition from manual to automated booking management streamlined day-to-day administrative tasks and positioned One Spatial Iloilo to operate with a higher level of professionalism and data-driven decision-making.

Broadcast Console – Referred to a dedicated communication management interface integrated within the proposed system that enabled authorized administrators of Gales Urban Haven to compose,schedule,and disseminate official announcements, notices, billing advisories, maintenance alerts, and other community-relevant messages to residents, tenants, or targeted groups in a structured and auditable manner. In the broader context of property management and community operations, a broadcast console served as the centralized communication hub that bridged management and residents, ensuring that all stakeholders received timely, accurate, and consistent information without relying on fragmented and informal communication channels such as verbal relays or untracked messaging applications.

In this study,the broadcast console provided administrators with the capability to direct messages to individual units, specific floors or blocks, or the entire resident population simultaneously, with each transmission automatically logged in the system audit trail to create a verifiable and retrievable record of all official communications issued by the management of One Spatial Iloilo. The console also supported delivery status tracking, enabling administrators to confirm message receipt and maintain transparency in all community-wide and unit-specific communications. By centralizing and formalizing the communication process, the broadcast console reduced miscommunication, eliminated information gaps among residents, and ensured that operational and financial notices were delivered promptly, professionally, and in accordance with the communication standards of the business.

Calendar Integration – Referred to a scheduling feature that visually organized and presented booking schedules, tenant check-ins and check-outs, unit availability windows, and maintenance activities within an interactive calendar interface. Calendar integration was a widely used tool in property management and hospitality systems, providing administrators with a consolidated and time-oriented view of all scheduled events and operational activities. In this study, calendar integration referred to the scheduling module incorporated into the proposed system that allowed administrators of Gales Urban Haven to view, monitor, and coordinate all time-sensitive operational activities in a single, centralized calendar view. This feature significantly improved coordination among staff, reduced scheduling conflicts, and enabled proactive planning for upcoming reservations and maintenance requirements. Furthermore, the calendar integration feature provided administrators with a visual overview of unit availability, making it easier to identify open slots and accommodate new bookings without the risk of double reservations or overlapping schedules. The interactive calendar interface also allowed staff to filter and sort scheduled activities by date, unit, or event type, enabling them to quickly locate and manage specific operational tasks with greater efficiency. Additionally, the system utilized the calendar module to generate automatic reminders and alerts for upcoming check-ins, check-outs, and maintenance schedules, ensuring that no critical activity was overlooked or

missed. The calendar integration also served as a communication tool among staff members, as it provided a shared and consistently updated view of all operational events, promoting better teamwork and coordination across departments. Ultimately, this feature contributed to a more organized and well-managed operational environment, allowing Gales Urban Haven to deliver a higher standard of service to its tenants and clients.

Cash Ledger – Referred to a comprehensive and systematically organized financial record maintained within the proposed system that documented all monetary transactions associated with each resident, tenant, and unit under the management of Gales Urban Haven, including but not limited to assessment fees, down payments, installment payments, penalty charges, credit adjustments, debit memos, and outstanding balances. In the broader discipline of accounting and financial management, a cash ledger served as the authoritative book of financial accounts, providing a chronological and itemized record of all inflows and outflows attributed to a specific account or entity, thereby forming the evidentiary foundation for financial reporting, collections management, and audit verification. In this study, the cash ledger module was embedded within the proposed system as a dynamic and real-time financial tracking tool that automatically recorded each transaction at the moment it was posted, computed running balances per account, flagged overdue or delinquent obligations, and generated individual and consolidated financial statements on demand. The cash ledger also served as the primary data source for the revenue monitoring and financial analytics functions of the

system, enabling administrators and finance officers of One Spatial Iloilo to maintain full visibility over the financial standing of every resident account, identify collection gaps, and produce accurate and audit-ready financial reports at any point in the operational cycle. By replacing manual ledger books and fragmented spreadsheet records, the cash ledger module eliminated transcription errors, prevented unauthorized alterations to financial entries, and ensured that all monetary data was preserved with integrity, consistency, and traceability throughout the financial lifecycle of the business.

Centralized Database – Referred to a structured and unified digital repository used to securely store, organize, and manage all system-generated and system-dependent data, including booking records, tenant profiles, payment transactions, financial summaries, and maintenance logs, within a single, integrated platform. In information systems management, a centralized database eliminates data silos by consolidating disparate data sources into one coherent and consistently maintained environment, enabling authorized users to retrieve accurate and up-to-date information rapidly. In this study, the centralized database served as the backbone of the proposed system, housing all operational data for Gales Urban Haven in a structured manner that ensured data integrity, consistency, and security while supporting fast and reliable information retrieval for administrative and decision-making purposes. Furthermore, the centralized database enabled administrators to perform comprehensive data queries and generate accurate reports, providing business owners with the reliable information they needed to make well-informed

operational and financial decisions. The database was also designed with proper access controls and security protocols, ensuring that only authorized personnel were able to view, modify, or delete sensitive business and tenant information. Additionally, the centralized structure of the database eliminated the risks associated with data redundancy and inconsistency, as all system modules drew from and wrote to a single, unified data source. The database also supported the scalability of the system, allowing Gales Urban Haven to accommodate a growing volume of records and transactions without compromising system performance or data reliability. Moreover, the centralized database served as a long-term digital archive for all operational records, ensuring that historical data remained accessible and well-preserved for future reference, auditing, and business analysis purposes. Ultimately, the implementation of a centralized database laid a strong and stable foundation for the overall functionality and efficiency of the proposed system, supporting all operational modules and contributing to the sustained growth and success of Gales Urban Haven.

Condominium Booking and Operations Management System – Referred to a web-based information system developed to automate, integrate, and streamline the core operational processes of a condominium rental business, including booking management, payment monitoring, revenue tracking, calendar-based scheduling, and maintenance management. As the proposed system in this study, it represented a purpose-built digital solution designed specifically to address the operational inefficiencies of Gales Urban Haven under One Spatial Iloilo. The system consolidated all key

operational functions into a single platform, enabling administrators to perform their duties more efficiently, reduce manual workloads, and access accurate and comprehensive business information. The system served not only as an immediate operational solution but also as scalable infrastructure that supported the long-term growth and digital transformation of the business. Furthermore, the system provided business owners and administrative staff with real-time access to operational data, enabling them to monitor the status of bookings, payments, and maintenance activities without the need for manual data gathering or cross-referencing multiple records. The web-based nature of the system also ensured that authorized users were able to access the platform from any device with an internet connection, providing greater flexibility and convenience in managing day-to-day business operations. Additionally, the system was equipped with a user-friendly interface that minimized the learning curve for staff members, allowing them to navigate and utilize all system features with ease and efficiency. The integration of multiple operational modules within a single platform eliminated the need for separate tools or manual processes, reducing the likelihood of errors and ensuring that all business data remained consistent and accurately reflected across all system functions. Moreover, the system served as a reliable digital record-keeping solution, ensuring that all historical transactions, booking records, and maintenance logs were securely stored and readily accessible for future reference, auditing, and business reporting purposes. Ultimately, the Condominium Booking and Operations Management System represented a significant step forward in the modernization of Gales Urban Haven's

operations, empowering the business to deliver a higher standard of service while building a stronger and more efficient foundation for its continued growth and success.

Data Accuracy – Referred to the degree to which recorded information within a system was correct, reliable, consistent, and free from errors or discrepancies. In the domain of information systems, data accuracy was one of the most critical dimensions of data quality, as inaccurate records could lead to flawed decision-making, financial misreporting, and operational disruptions. In manual systems, data accuracy was frequently compromised by transcription errors, illegible handwriting, duplicate entries, and outdated records. In this study, data accuracy referred to the correctness and reliability of all information stored and managed by the proposed system, including booking details, payment figures, tenant records, and maintenance logs. The system incorporated input validation, automated computation, and structured data entry mechanisms to ensure that all recorded data remained accurate and trustworthy throughout the operational lifecycle of the business. As the system became fully operational, the data accuracy mechanisms embedded within its architecture served as a continuous safeguard against the entry and propagation of erroneous information across all modules, ensuring that administrators had access to reliable and up-to-date records upon which sound operational and financial decisions could be based. The structured data entry interfaces and automated validation protocols incorporated into the system significantly reduced dependence on manual data handling, thereby minimizing the risk of human error that had historically

undermined the integrity of records maintained through traditional paper-based and spreadsheet-driven management approaches. The system also incorporated data validation mechanisms, including error detection, duplicate record identification, and data reconciliation functionalities, which further strengthened the overall accuracy and consistency of the information maintained within the platform. The high standard of data accuracy established by the system also contributed to the organization's ability to produce reliable financial reports, conduct meaningful performance assessments, and respond confidently to inquiries or audits from regulatory authorities and other relevant stakeholders. The data accuracy framework of the system also included comprehensive data audit trails that allowed administrators to track the history of changes made to critical records, providing an additional layer of accountability and transparency that further reinforced the trustworthiness of the system's data environment. The consistent maintenance of accurate records across all operational modules also supported the analytical capabilities of the organization, as administrators and developers were able to draw upon a well-maintained historical database to generate meaningful insights, identify operational trends, and inform strategic planning decisions.

Maintenance Tracking – Referred to a system feature used to record, schedule, monitor, and update maintenance requests, repair activities, preventive maintenance schedules, and service completion statuses related to condominium units and property facilities. Effective maintenance tracking was a fundamental component of property management operations,

as it ensured that all repair and upkeep needs were addressed promptly and that the physical condition of the property was consistently maintained to the standards expected by tenants and management. In this study, maintenance tracking referred to the dedicated module within the proposed system that provided administrators of Gales Urban Haven with a structured platform for logging maintenance requests, assigning responsibilities, monitoring progress, and maintaining a complete historical record of all maintenance activities conducted across the property. Once the system was fully deployed and integrated into the daily operations of Gales Urban Haven, the Maintenance Tracking module became an important tool for ensuring that all property-related concerns were resolved in a timely, organized, and well-documented manner. The module also incorporated priority-based categorization of maintenance requests, allowing administrators to distinguish between urgent repair needs and routine upkeep tasks, thereby enabling more efficient allocation of maintenance resources and personnel.

The system also generated detailed maintenance reports that provided management with a comprehensive overview of recurring issues, completed repairs, and pending service requests, supporting more informed and proactive decision-making regarding property upkeep. The Maintenance Tracking module was also capable of sending automated reminders and status updates to relevant personnel, ensuring that no maintenance request was overlooked or left unresolved beyond an acceptable timeframe.

Furthermore, the historical maintenance data accumulated within the system served as a valuable reference that administrators and developers could analyze to identify patterns of property deterioration, forecast maintenance costs, and develop more effective preventive maintenance strategies. The continuous use and refinement of this module contributed significantly to the preservation of property value, the satisfaction of tenants, and the overall operational excellence of Gales Urban Haven as a professionally managed condominium facility.

Efficiency – Referred to the capacity of an organization to execute its business processes accurately, swiftly, and with minimal resource wastage through the adoption of optimized workflows, technological tools, and systematic procedures. Operational efficiency was a key performance indicator in business management and was directly linked to the profitability, service quality, and competitiveness of an organization. In the context of property management, operational efficiency encompassed the speed and accuracy of booking processes, the reliability of financial tracking, the responsiveness of maintenance management, and the overall productivity of administrative staff. In this study, operational efficiency referred to the measurable improvement in the speed, accuracy, and reliability of the operational processes of Gales Urban Haven that was achieved through the deployment of the proposed Condominium Booking and Operations Management System.

Payment Tracking – Referred to a system feature responsible for monitoring, recording, and managing all financial transactions made by tenants, including initial down

payments, scheduled installment payments, outstanding balances, and complete payment histories. In property rental and condominium management, effective payment tracking was essential for maintaining accurate financial records, ensuring that all receivables were properly accounted for, and providing management with a real-time view of the financial standing of each tenant account. In this study, payment tracking referred to the financial monitoring module of the proposed system that enabled administrators of One Spatial Iloilo to record and access all payment-related data efficiently, generate payment status reports, and maintain a transparent and auditable record of all financial transactions processed by the business. As the system was fully implemented and integrated into the daily financial operations of One Spatial Iloilo, the Payment Tracking module became a cornerstone of the organization's financial management framework, providing administrators with a reliable, centralized, and consistently updated platform for overseeing all tenant-related financial activities. The structured and comprehensive financial records maintained by this module enabled management to conduct periodic financial reviews with greater accuracy and confidence, supporting more informed budgeting decisions and more effective long-term financial planning for the organization. The Payment Tracking module included automated payment reminder notifications, overdue account flagging, and penalty fee computation functionalities that further streamlined the financial management processes of One Spatial Iloilo and reduced the likelihood of uncollected or misrecorded payments. The system also generated tenant-specific payment summaries and detailed

transaction histories that served as critical documentation in cases of financial disputes, lease renewals, or compliance audits conducted by relevant regulatory bodies. The Payment Tracking module was also capable of integrating with other system components such as the Booking Management and Tenant Information modules, creating a fully interconnected financial and operational data environment that eliminated redundancy and ensured consistency across all records maintained within the platform. The accumulation of historical payment data over time also provided management with a valuable analytical resource that administrators and developers could leverage to identify payment trends, assess the financial reliability of tenant accounts, and develop more proactive and data-driven revenue collection strategies. Furthermore, the Payment Tracking module was positioned to evolve in alignment with emerging technologies and industry practices, ensuring that One Spatial Iloilo remained competitive, compliant, and financially well-managed. Ultimately, the Payment Tracking module developed into a more robust and feature-rich financial management tool that continued to serve the operational and reporting needs of One Spatial Iloilo.

Prototype – Referred to a functional and preliminary version of the proposed system developed for the purposes of testing, evaluation, and demonstration prior to full-scale implementation and deployment. In systems development, a prototype served as a working model that allowed developers and stakeholders to assess the design, functionality, and usability of the system in a controlled environment before the final version was released.

Prototyping was a critical phase in the systems development life cycle, as it enabled the identification of design flaws, functional gaps, and usability issues at an early stage, reducing the cost and risk associated with post-deployment modifications. In this study, the prototype referred to the initial operational version of the Condominium Booking and Operations Management System that was presented to end users and evaluators for testing and feedback collection.

Resident Directory – Referred to a centralized and dynamically maintained digital registry embedded within the proposed system that served as the authoritative and comprehensive record of all individuals and households registered as residents or active tenants of Gales Urban Haven under the management of One Spatial Iloilo. In the broader context of property and community management information systems, a resident directory functioned as the foundational data layer upon which all other operational modules—including financial tracking, scheduling, communication, and audit functions—depended for accurate identification, association, and retrieval of person-specific and unit-specific records. In this study, the resident directory maintained a structured profile for each registered occupant, capturing essential biographical and contact information, unit assignment details, tenancy classification, move-in and move-out dates, verification status, and linked financial accounts. This consolidated registry enabled administrators to rapidly retrieve complete resident information, enforce access controls based on residency status, and ensure that all transactional, communicative, and operational records

generated by the system were accurately attributed to the correct individual and unit. The resident directory also served as the primary reference for the cash ledger module, the broadcast console, and the system audit trail, ensuring that data consistency and relational integrity were maintained across all modules. By replacing fragmented paper-based or spreadsheet-driven resident records, the directory eliminated redundancy, prevented the coexistence of conflicting resident profiles, and provided management with a single, reliable, and continuously updated source of information for all resident-related records throughout the operational lifecycle of the condominium property.

Revenue Monitoring – Referred to the automated process of recording, tracking, aggregating, and computing the total income generated by the business from all booking transactions, including payments received, down payments collected, and outstanding balances pending collection. Revenue monitoring was a critical financial management function in a property rental business, as it provided management with an accurate and timely picture of the financial performance of the operation. Without a reliable revenue monitoring mechanism, businesses were susceptible to income leakage, undetected discrepancies, and delayed financial reporting. In this study, revenue monitoring referred to the financial analytics component of the proposed system that automatically captured and processed all income-related data for Gales Urban Haven, enabling administrators to generate revenue reports, track financial trends, and make data-driven business decisions with confidence and precision.

System Audit Trail – Referred to a comprehensive, tamper-evident, and chronologically ordered electronic record automatically generated and maintained by the proposed system that documented every action, transaction, modification, and access event performed by any user within all modules of the Condominium Booking and Operations Management System. In the discipline of information systems security and governance, an audit trail served as the foundational accountability mechanism that ensured complete traceability of all system activities, providing organizations with the evidentiary basis necessary for compliance verification, forensic investigation, dispute resolution, and performance auditing. In this study, the system audit trail captured and permanently preserved detailed records of every login and logout event, every booking creation, modification, or cancellation, every financial transaction posted to the cash ledger, every broadcast message composed and transmitted through the broadcast console, every update made to the resident directory, and every administrative override or access privilege change executed within the system.

Each audit entry recorded the identity of the acting user, the nature and scope of the action performed, the specific data fields affected, the precise timestamp of the event, and the resulting state of the system following the action. The audit trail was accessible exclusively to users holding audit or administrative privileges and was protected against deletion, alteration, or unauthorized access through role-based security controls. By maintaining a continuous and unalterable record of all system activities, the system audit trail strengthened the operational

accountability of Gales Urban Haven, deterred unauthorized or fraudulent actions, supported the timely detection and resolution of discrepancies, and ensured that administrators and authorized reviewers were able to reconstruct the complete operational history of any transaction, communication, or data modification at any point in time, thereby upholding the standards of data governance and institutional integrity required in the management of a residential condominium property.

User Authentication – Referred to a security mechanism that verified and confirmed the identity of individuals attempting to access a system by requiring the presentation of valid login credentials, such as usernames and passwords, before granting access to system features and data. User authentication was a foundational element of information security in any digital system, ensuring that sensitive operational data and administrative functions were accessible only to authorized personnel. In systems handling financial records, tenant information, and business-critical data, robust user authentication was indispensable for protecting the integrity and confidentiality of system data. In this study, user authentication referred to the access control mechanism integrated into the proposed system that restricted system access to verified administrators and authorized users of Gales Urban Haven, preventing unauthorized access and safeguarding the integrity of all stored operational and financial data. Furthermore, the user authentication feature supported role-based access control, ensuring that different levels of system users were granted access only to the specific features and data relevant to their

designated responsibilities and authority within the organization. The system also implemented secure password protocols and encryption measures, ensuring that login credentials were stored and transmitted in a protected manner that minimized the risk of unauthorized access or data breaches. Additionally, the authentication mechanism served as the first line of defense against potential security threats, providing a reliable barrier that prevented unauthorized individuals from accessing, modifying, or deleting sensitive business and tenant information. The system also maintained an access log that recorded all login activities, including successful and failed authentication attempts, allowing administrators to monitor system access and detect suspicious or unauthorized activity in a timely manner. Moreover, the user authentication feature contributed to the overall trustworthiness and reliability of the system, assuring business owners, administrative staff, and tenants that their personal and financial data were handled with a high level of security and confidentiality. Ultimately, the implementation of a robust user authentication mechanism ensured that the system operated in a secure and controlled environment, protecting the integrity of business data and supporting the safe and efficient management of Gales Urban Haven's operations.

Web-Based System – Referred to a software application that was deployed on a web server and accessed through a standard web browser using an internet connection or a local area network, without requiring the installation of dedicated software on the user's device. Web-based systems had become the dominant architecture for modern business

applications due to their platform independence, ease of access, centralized data management, and simplified maintenance and update procedures. They allowed multiple authorized users to access and interact with the system simultaneously from different locations and devices. In this study, web-based systems referred to the architectural approach adopted in the development of the Condominium Booking and Operations Management System, which was accessible to administrators and authorized personnel of One Spatial Iloilo through a web browser, enabling flexible, convenient, and real-time access to system features and operational data. Furthermore, the web-based nature of the system eliminated the need for costly hardware installations or device-specific software configurations, making it a practical and cost-efficient solution for the operational needs of Gales Urban Haven. The system also benefited from the scalability that web-based architecture provided, allowing additional features, modules, and user accounts to be incorporated into the platform as the business grew and evolved. Additionally, the web-based system enabled administrators to perform updates, modifications, and maintenance procedures on the platform without disrupting overall system operations, ensuring continuous availability and minimal downtime for users. The centralized nature of the web-based architecture also ensured that all data entered and retrieved by authorized users were consistently synchronized and accurately reflected across all system modules in real time, eliminating discrepancies and data inconsistencies.

Moreover, the web-based system provided a responsive and accessible interface that was compatible with various devices, including desktop computers, laptops, and tablets, ensuring that administrators and authorized personnel were able to manage operational tasks conveniently from different locations and at different times. Ultimately, the adoption of a web-based system architecture positioned the Condominium Booking and Operations Management System as a modern, flexible, and practical solution that effectively supported the digital transformation and operational management of Gales Urban Haven under One Spatial Iloilo.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

According to Imkumnert (2002), the proposed system handles most information management in this condominium such as customer information, room information, service information room renting, bills and receipts operation, and reports for the management team. The Rental Condominium business is now becoming popular in the lodging market. Formerly the nature of the real-estate business, focused on the condominium market, was the construction methodology for selling. Halkiopoulos et al. (2020), the study focuses on exploration of knowledge for online booking systems and on the views of local students-users concerning the booking rate based on these online systems. Another perspective of this project is to investigate the decision-making process (emotion-focused) that they follow in order to choose a tourist destination via online booking systems.

Varutbangkul (2000), in Assumption University: Condominium management information system, this project is implemented to improve the efficiency of the existing system in Fair Tower, a condominium that belongs to GOS Limited Partnership, by designing a new computerized system to replace the old manual system in front office. Therefore, this project is aimed at providing a computerized system for the staff at the front desk in order to carry out their

work more efficiently and accurately. The proposed system handles most information management in this condominium such as customer information, room information, service information room sale, room renting, bills and receipts operations, and reports for the general manager.

According to Kuo et al. (2012), in Science Direct: Enhancement of condominium management based on the effect of quality attributes on satisfaction improvement (Expert Systems with Applications 39 (5), 5418-5425), condominium management (CM) businesses rely on continuous improvements to service quality (SQ) to increase consumer satisfaction (CS) and thereby improve operations. Lacking knowledge of SQ and CS results in poor efficiency. To develop cost-effective strategies for improvements, the relationship between SQ and CS must be investigated. This work proposes novel procedures to explore the relationship between identified SQ attributes and CS in the CM business. First, we conduct exploratory factor analysis to cluster intrinsically similar attributes into three constructs and identify significant attributes for subsequent assessment. Furthermore, resident satisfaction scores (RSSs) are computed to assess the relative importance rate (RIR) of investigated attributes by applying an adapted multivariate method to regress the RSSs on the performance of SQ attributes. To facilitate the decision-making process, the service-quality model evolved from the integration of an improved Kano's approach and importance–performance analysis (IPA) is utilized to develop a prioritized strategy for satisfaction improvements in the CM business.

2.2 Local Studies

According to Bajado (2025), this research investigated the impact of hotel business tools and technologies on guest satisfaction in Tagaytay City, Philippines, a rapidly growing tourist destination. Hotels globally are adopting technology to enhance guest experiences and satisfaction. Gosela et al. (2024), it is a unified application aiming to address existing challenges by incorporating user-centric features. Through advanced location-based services and streamlined booking processes, the application intends to enhance efficiency, accessibility, and sustainability in the tourism sector by providing real-time tracking, efficient transit solutions, and a comprehensive directory of establishments.

Gallenero et al. (2022), IEEE 14th International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management (HNICEM), 1-5: Occupy-Real-Estate Management System with Sales Analysis and Customer Segmentation, the increase in population equates to the increasing demand of the real estate. Managing real estate manually takes a lot of time and manpower, which is prone to damage and misplacement of documents and client’s data. Customer support and community were also considered. Real-Estate Management System automates complex tasks and duties. Sales analysis through customer segmentation was implemented in the system to come up with better sales and understand the current, possible, and future customers of the real-estate company. Agile

Methodology was applied in developing the system, in which different users – who has different roles and administrator rights, are needed to access, and manage the system. Testing, verification, validation, usability testing, and acceptance testing were done to ensure that the requirements needed were met, having both alpha and beta testing for the software.

2.3 Related Literature

This chapter presents a comprehensive review of related literature that serves as the theoretical and empirical foundation for the development of the Condominium Booking and Operations Management System proposed in this study. The literature reviewed covers a broad range of thematic areas directly relevant to the proposed system, including property management systems, automated booking and reservation technologies, payment tracking and revenue monitoring, maintenance management, centralized database systems, web-based application development, and the application of the Agile development methodology in information systems projects. The insights and findings drawn from the reviewed literature collectively inform the design decisions, functional specifications, and architectural choices that underpin the proposed system for Gales Urban Haven, operating under the business name One Spatial Iloilo.

Property Management Systems

Property management systems (PMS) have been the subject of extensive scholarly and professional inquiry over the past two decades, owing to the growing recognition that the efficient management of residential, commercial, and mixed-use properties requires the support of robust, integrated digital tools. The literature broadly defines a property management system as a software platform that consolidates the core operational functions of a property management business, including tenant record management, lease administration, financial tracking, maintenance scheduling, and reporting, into a single, centralized, and accessible interface.

Bardi (2011) affirmed that property management systems have fundamentally transformed the hospitality and real estate industries by replacing fragmented, paper-based administrative processes with automated, data-driven workflows that significantly reduce operational errors and improve service delivery standards. The author noted that the integration of booking, billing, and reporting functions within a unified platform enables property managers to maintain a comprehensive and current view of all operational activities, thereby improving their capacity for informed decision-making and proactive problem resolution.

Tesone (2012) similarly argued that modern property management systems represent an indispensable operational infrastructure for any property management enterprise operating in a competitive market environment. His research

emphasized that the ability to retrieve accurate and up-to-date information across all operational domains, from reservation status to financial performance, at any point in time and from a single platform, constitutes a decisive competitive advantage for property management organizations that have adopted such systems over those that continue to rely on manual methods.

In the Philippine context, the Department of Information and Communications Technology (DICT, 2020) has acknowledged the growing importance of information and communications technology adoption among small and medium enterprises (SMEs) in the service sector, noting that businesses in property management, hospitality, and rental services that adopt digital management systems report significantly higher levels of operational efficiency, financial accuracy, and customer satisfaction compared to their non-digitalized counterparts. This observation is particularly relevant to the present study, given that Gales Urban Haven represents a small-scale condominium management operation whose administrative processes remain largely manual and therefore vulnerable to the inefficiencies that digital systems are specifically designed to address.

Cobanoglu et al. (2013) further highlighted that the effectiveness of a property management system is not determined solely by its technological sophistication but also by the degree to which its design and functionality align with the specific operational workflows and data

management needs of its target users. This finding underscores the importance of conducting a thorough requirements analysis prior to system development and of involving end users in the design and evaluation process, both of which are key activities in the Agile development approach adopted in the present study.

Automated Booking and Reservation Management

The automation of booking and reservation processes has been extensively studied in both the hospitality and real estate management literature, with a consistent body of evidence demonstrating that automated reservation systems substantially outperform manual methods in terms of speed, accuracy, reliability, and scalability. The literature identifies scheduling conflicts, double bookings, and incomplete reservation records as the most common and consequential problems arising from manual booking systems, and collectively affirms that these problems are effectively mitigated through the implementation of automated booking management platforms.

O'Connor and Murphy (2008) conducted a landmark study on the adoption of online reservation systems in the hospitality sector and found that properties utilizing automated booking platforms reported a significant reduction in reservation errors, an increase in booking processing speed, and a marked improvement in the accuracy of unit availability information available to both administrators and clients. The authors concluded that the real-time availability checking functionality embedded in automated booking systems is among the most operationally

impactful features such systems provide, as it eliminates the information lag that is inherent in manual availability verification processes.

Siguaw and Enz (1999) similarly documented that the implementation of computerized reservation systems in lodging establishments resulted in measurable improvements in front-desk operational efficiency, with staff able to process reservations in a fraction of the time previously required under manual systems. Their findings further indicated that automated reservation systems enabled managers to access historical booking data more readily, facilitating trend analysis and demand forecasting activities that were previously impractical under manual record-keeping regimes.

Kasavana and Brooks (2009) noted that the fundamental advantage of an automated booking system lies in its capacity to maintain a continuously updated, centrally accessible, and conflict-free reservation schedule, ensuring that no two reservations can be recorded for the same unit at the same time without the system generating an alert or blocking the entry. This conflict-prevention mechanism, which is a standard feature of well-designed booking management systems, directly addresses one of the most critical operational risks associated with manual reservation processes: the inadvertent recording of overlapping bookings that results in tenant disputes and administrative complications.

In the context of small-scale condominium and apartment rental management, Pizam and Ellis (1999) emphasized that the operational benefits of booking automation are proportionally more significant for smaller properties than for large hotel chains, as smaller operations typically have fewer administrative staff and therefore less capacity to absorb the time and error costs associated with manual booking management.

This observation is directly applicable to the context of Gales Urban Haven, where the administrative team is small and the operational demands of managing multiple condominium units simultaneously require a system capable of handling reservation data with speed, accuracy, and minimal staff intervention.

Payment Tracking and Financial Record Management

The management of tenant payments, including the recording of initial deposits, monitoring of outstanding balances, and maintenance of complete payment histories, is one of the most critical and error-sensitive functions in any property rental operation. The literature on financial record management in property contexts consistently identifies the absence of a systematic and automated payment tracking mechanism as a primary source of financial discrepancies, revenue leakage, and administrative disputes between property managers and tenants.

Warren and Jones (2010) conducted a study on financial management practices among small residential rental

operations and found that businesses relying on manual payment recording methods experienced significantly higher rates of payment recording errors, missed payment notifications, and balance computation discrepancies compared to those using automated financial tracking tools. The authors concluded that even relatively simple automated payment tracking systems produce substantial improvements in financial accuracy and accountability, reducing the administrative burden of manual reconciliation and improving the reliability of financial records available to management.

Clauretie and Sirmans (2010) further argued that accurate and timely payment tracking is not merely an administrative convenience but a strategic financial management imperative for property rental businesses, as the accuracy of payment records directly determines the reliability of cash flow projections, revenue reports, and business planning decisions. Their research demonstrated that property managers who maintained comprehensive and up-to-date payment records were significantly better positioned to make informed decisions regarding pricing, capacity expansion, and operational investment than those whose financial records were incomplete or delayed.

Turner (2012) examined the integration of payment tracking with broader financial management systems in property management operations and found that the consolidation of payment data, revenue figures, and financial reporting functions within a single platform eliminated the data reconciliation delays that commonly arise when these

functions are managed through separate, disconnected tools. This finding is directly relevant to the design of the proposed system for Gales Urban Haven, in which the payment tracking module will be fully integrated with the revenue reporting module to ensure that all financial data is processed, stored, and reported within a single, cohesive system environment.

Revenue Monitoring and Financial Reporting

Revenue monitoring and financial reporting are foundational functions of any business operation, and their importance in property management contexts has been the subject of considerable scholarly attention. The literature consistently emphasizes that accurate, timely, and accessible financial reporting is essential for enabling property management organizations to assess their financial performance, identify revenue trends, manage cash flow effectively, and make evidence-based strategic decisions.

Brigham and Houston (2019) described revenue monitoring as the systematic and continuous process of tracking all income streams, computing aggregate revenue figures across defined time periods, and comparing actual revenue performance against projected targets. Their work emphasized that automated revenue monitoring systems, which perform these functions without requiring manual data compilation, enable management to access accurate financial performance data in real time, dramatically reducing the lag between financial activity and financial awareness that characterizes manual reporting processes.

Horngren, Datar, and Rajan (2015) further noted that the quality of management decision-making is directly and substantially influenced by the timeliness and accuracy of the financial information available to decision-makers. Their research demonstrated that organizations with access to real-time financial reporting tools consistently outperformed those relying on periodic manual reports in terms of their ability to identify and respond to financial challenges before they escalate into serious operational problems. This finding strongly supports the inclusion of an automated revenue reporting module in the proposed system for Gales Urban Haven, where the current absence of such a tool has been identified as a significant impediment to effective financial management.

Kimmel, Weygandt, and Kieso (2016) similarly highlighted the importance of financial report accuracy in small business contexts, noting that errors in manually prepared financial reports can have disproportionately serious consequences for small enterprises, given their typically limited financial reserves and narrower margins for error. The authors advocated for the adoption of automated financial reporting tools by small business operators as a critical risk management strategy, a recommendation that directly reinforces the rationale for incorporating revenue reporting automation into the proposed system for One Spatial Iloilo.

Maintenance Management in Property Operations

Maintenance management is widely recognized in the property management literature as a critical determinant of property value retention, tenant satisfaction, and operational continuity. The systematic recording, scheduling, monitoring, and resolution of maintenance activities is essential for ensuring that condominium units and shared property facilities remain in optimal condition throughout their operational lifecycle.

Dhillon (2006) provided a comprehensive framework for maintenance management in built environments, arguing that the effectiveness of a maintenance program is directly determined by the quality of the systems and processes used to track and manage maintenance activities. His research demonstrated that organizations employing structured, digitally supported maintenance management systems consistently achieved higher rates of timely maintenance resolution, lower incidences of repeated maintenance failures, and higher levels of tenant satisfaction compared to those relying on informal or paper-based maintenance tracking methods.

Mobley (2002) similarly emphasized the strategic importance of preventive maintenance scheduling in property management, noting that a systematic approach to maintenance planning, supported by accurate historical maintenance records and a reliable scheduling system, enables property managers to address potential issues before they develop into costly emergency repairs. His work

highlighted the critical role of digital maintenance logs in enabling managers to identify patterns of recurring maintenance problems and make data-driven decisions regarding maintenance resource allocation and preventive intervention priorities.

Wireman (2004) further documented that the absence of a systematic maintenance tracking system in property management operations frequently leads to the loss of maintenance history records, making it difficult for managers to assess the long-term performance of maintenance interventions, track the lifecycle of property assets, and demonstrate maintenance accountability to property owners. These findings provide strong empirical support for the inclusion of a dedicated maintenance monitoring module within the proposed system for Gales Urban Haven, ensuring that all maintenance activities are systematically recorded, tracked, and accessible for review by administrators at any time.

Calendar-Based Scheduling in Operations Management

The use of calendar-based scheduling tools in operations management has been widely studied across multiple industries, including hospitality, healthcare, education, and property management. The literature consistently affirms that visual scheduling interfaces provide administrators with a significant operational advantage by consolidating time-sensitive operational data into a

single, chronologically organized display that enables rapid situational awareness and more effective operational planning.

Gaither and Frazier (2002) noted that scheduling is among the most cognitively demanding and error-prone functions in operations management when performed manually, as it requires administrators to simultaneously track and reconcile multiple overlapping activities, availability windows, and time constraints without the support of a visual organizational framework. Their research demonstrated that digital scheduling tools reduce the cognitive load associated with scheduling tasks and substantially decrease the frequency of scheduling errors and conflicts in complex operational environments.

Pinedo (2012) further documented that integrated scheduling systems, which synchronize calendar data with other operational modules such as reservations, availability tracking, and maintenance records, provide the most significant operational benefits compared to standalone scheduling tools. The synchronization of scheduling data with real-time operational information ensures that the calendar view presented to administrators always accurately reflects the current state of operations, enabling them to make scheduling decisions based on reliable and up-to-date information. This principle directly informed the design of the calendar scheduling module in the proposed system, which will be fully integrated with the booking and maintenance modules to ensure real-time data consistency across all scheduling displays.

Centralized Database Systems in Information Management

The design and implementation of centralized database systems has been a central area of inquiry in information systems research, with the literature consistently affirming that centralized relational databases provide a superior data management foundation for multi-functional information systems compared to fragmented or distributed data storage approaches. The literature emphasizes that centralized databases enable organizations to eliminate data redundancy, enforce data consistency and integrity, and provide all authorized users with access to the same, current version of organizational data, regardless of their location or the device they are using.

Date (2004) provided a seminal treatment of relational database theory, arguing that the relational model, implemented through systems such as MySQL, provides the most robust and flexible framework for organizing complex, interrelated data sets in organizational contexts. His work emphasized that the use of structured query language (SQL) in relational database systems enables powerful and precise data retrieval operations that would be impractical or impossible in non-relational data storage environments, including the spreadsheet-based and paper-based systems commonly used by small property management operations.

Elmasri and Navathe (2016) further documented the critical importance of database normalization in ensuring the integrity and consistency of data stored in relational databases, noting that a properly normalized database schema eliminates data redundancy, prevents update anomalies, and ensures that all data within the system

remains internally consistent. Their work also highlighted the importance of implementing appropriate indexing strategies within the database to optimize query performance and ensure that the system remains responsive even as the volume of stored data grows over time, a consideration that is particularly relevant to the proposed system for Gales Urban Haven, which will accumulate increasingly large volumes of booking, payment, and maintenance records over the course of its operational life.

Silberschatz, Korth, and Sudarshan (2019) addressed the security dimensions of centralized database management, emphasizing that robust access control mechanisms, parameterized query execution, and regular data backup protocols are essential components of any database system that handles sensitive organizational or financial data. Their recommendations directly inform the security specifications incorporated into the database design of the proposed system, which will implement role-based access controls, prepared statement query mechanisms, and scheduled backup procedures to protect all operational data stored for One Spatial Iloilo from unauthorized access, data corruption, and accidental loss.

Web-Based Information Systems

Web-based information systems have become the dominant architectural paradigm for organizational information management across virtually all industry sectors, owing to their platform independence, accessibility, scalability,

and ease of maintenance. The literature on web-based systems consistently documents their advantages over locally installed desktop applications, particularly in organizational contexts where multiple users need to access and interact with shared data simultaneously from different workstations or locations.

Laudon and Laudon (2020) provided a comprehensive treatment of management information systems in organizational contexts, arguing that web-based systems represent the most effective and adaptable information management infrastructure available to modern organizations. Their work highlighted that web-based systems enable organizations to deploy and update their information systems more efficiently than locally installed alternatives, as modifications to the application need only be applied at the server level and are immediately available to all users without requiring software updates on individual client devices.

O'Brien and Marakas (2011) similarly emphasized that web-based systems offer a critical advantage in data consistency, as all users access the same centrally hosted application and data repository, eliminating the data synchronization problems that arise when multiple users maintain separate local copies of organizational data. This advantage is particularly significant in the context of property management operations, where multiple staff members may need to access and update shared booking, payment, and maintenance records simultaneously, and where

data inconsistencies arising from unsynchronized local copies can lead to serious operational errors.

Turban, Volonino, and Wood (2015) further noted that the accessibility of web-based systems through standard browsers without requiring specialized software installation significantly lowers the barrier to adoption for small and medium enterprises, as it eliminates the hardware and software procurement costs commonly associated with proprietary desktop application deployments. This observation is particularly relevant to the context of One Spatial Iloilo, where the adoption of a web-based system will enable the organization to implement a comprehensive digital management platform without incurring the substantial upfront costs associated with traditional enterprise software installations.

Agile Software Development Methodology

The Agile software development methodology has been extensively studied and widely adopted in the information systems development community since the publication of the Agile Manifesto in 2001, which articulated a set of values and principles prioritizing iterative development, customer collaboration, responsiveness to change, and the continuous delivery of working software over the rigid adherence to comprehensive upfront planning that characterizes traditional development models.

Pressman and Maxim (2020) provided a thorough examination of Agile development practices in their seminal software engineering textbook, documenting that Agile methodologies consistently produce higher-quality software products than traditional Waterfall-based approaches when applied to projects with evolving or initially unclear requirements. Their analysis demonstrated that the iterative structure of Agile development enables development teams to incorporate user feedback and stakeholder input at each stage of the development process, resulting in systems that more closely align with the actual needs and expectations of their intended users at the point of delivery.

Sommerville (2016) similarly noted that Agile development is particularly well-suited to projects in which the full scope of system requirements cannot be completely determined at the outset, as its iterative cycles allow requirements to be progressively clarified, refined, and expanded as development proceeds and as users develop a clearer understanding of the system's capabilities and their own operational needs. This characteristic makes Agile an especially appropriate methodology for the development of the proposed system for Gales Urban Haven, where the complexity of integrating multiple operational modules and the need to align system functionality with the nuanced and evolving operational workflows of the property management team make iterative development and continuous feedback essential to the success of the project.

Schwaber and Sutherland (2017), the co-creators of the Scrum framework, one of the most widely used

implementations of the Agile methodology, documented that organizations applying Agile practices in information systems development projects report higher rates of project completion within budget and schedule, higher levels of end-user satisfaction with the delivered system, and greater organizational capacity to adapt to changing requirements compared to organizations using traditional development methodologies. These documented outcomes provide strong empirical support for the selection of the Agile model as the guiding development methodology for the proposed Condominium Booking and Operations Management System.

References

Imkumnert, W. (2002). Assumption University: Service condominium information system for condominium rental business

https://scholar.google.com/scholar?q=related:bDJn-Z3LSsAJ:scholar.google.com/&scioq=condominium+Booking+and+operation+manay&hl=en&as_sdt=0,5#d=gs_qabs&t=1774511369898&u=%23p%3DbDJn-Z3LSsAJ

Halkiopoulos, C., Antonopoulou, H., & Giotopoulos, K. (2020). Journal of Tourism, Heritage & Services Marketing (JTHSM) 6 (1), 9-16: Online reservation systems in e-Business: Analyzing decision making in e-Tourism https://scholar.google.com/scholar?hl=en&as_sdt=0%2C5&scioq=condominium+Booking+and+operation+manay&q=halkiopoulos+%282020%29+online+booking&oq=halkiopoulos+%282020%29+online++book#d=gs_qabs&t=1774511670536&u=%23p%3DvCXEbkGU348J

Varutbangkul (2000). Assumption University: Condominium management information system https://www.library.au.edu/

Gallenero, R. C., Flores, J. M., Gundran, R. M., Rosario, J. V., Ubaldo, G. F., Solomo, M. V. S., Fernando-Raguro, M. C., & Lagman, A. C. (2022). Occupy - Real-Estate Management System with Sales Analysis and Customer Segmentation. *2022 IEEE 14th International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management (HNICEM)*, 1-5. https://doi.org/10.1109/HNICEM57413.2022.10109621

Kuo, Y., Chou, J. (2012). Expert Systems with Applications 39 (5), 5418-5425, 2012: Enhancement of condominium

management based on the effect of quality attributes on satisfaction improvement

https://www.sciencedirect.com/science/article/pii/S0957417411015867

Bardi, J. A. (2011). Hotel front office management (5th ed.). John Wiley & Sons.

Brigham, E. F., & Houston, J. F. (2019). Fundamentals of financial management (15th ed.). Cengage Learning.

Clauretie, T. M., & Sirmans, G. S. (2010). Real estate finance: Theory and practice (6th ed.). Cengage Learning.

Cobanoglu, C., Berezina, K., Kasavana, M. L., & Erdem, M. (2013). The impact of technology amenities on hotel guest overall satisfaction. Journal of Quality Assurance in Hospitality & Tourism, 14(3), 272–285.

Date, C. J. (2004). An introduction to database systems (8th ed.). Pearson Education.

Department of Information and Communications Technology. (2020). ICT adoption and utilization survey for small and medium enterprises in the Philippines. DICT Publications.

Dhillon, B. S. (2006). Maintainability, maintenance, and reliability for engineers. CRC Press.

Elmasri, R., & Navathe, S. B. (2016). Fundamentals of database systems (7th ed.). Pearson Education.

Gaither, N., & Frazier, G. (2002). Operations management (9th ed.). South-Western College Publishing.

Horngren, C. T., Datar, S. M., & Rajan, M. V. (2015). Cost accounting: A managerial emphasis (15th ed.). Pearson Education.

Kasavana, M. L., & Brooks, R. M. (2009). Managing front office operations (8th ed.). Educational Institute of the American Hotel & Lodging Association.

Kimmel, P. D., Weygandt, J. J., & Kieso, D. E. (2016). Financial accounting: Tools for business decision-making (8th ed.). John Wiley & Sons.

Laudon, K. C., & Laudon, J. P. (2020). Management information systems: Managing the digital firm (16th ed.). Pearson Education.

Mobley, R. K. (2002). An introduction to predictive maintenance (2nd ed.). Butterworth-Heinemann.

O'Brien, J. A., & Marakas, G. M. (2011). Management information systems (10th ed.). McGraw-Hill/Irwin.

O'Connor, P., & Murphy, J. (2008). Hotel yield management practices across multiple electronic distribution channels. Information Technology & Tourism, 10(2), 161–172.

Pinedo, M. L. (2012). Scheduling: Theory, algorithms, and systems (4th ed.). Springer.

Pizam, A., & Ellis, T. (1999). Customer satisfaction and its measurement in hospitality enterprises. International Journal of Contemporary Hospitality Management, 11(7), 326–339.

Pressman, R. S., & Maxim, B. R. (2020). Software engineering: A practitioner's approach (9th ed.). McGraw-Hill Education.

Schwaber, K., & Sutherland, J. (2017). The Scrum guide: The definitive guide to Scrum, the rules of the game. Scrum.org.

Siguaw, J. A., & Enz, C. A. (1999). Best practices in information technology. Cornell Hotel and Restaurant Administration Quarterly, 40(5), 58–71.

Silberschatz, A., Korth, H. F., & Sudarshan, S. (2019). Database system concepts (7th ed.). McGraw-Hill Education.

Sommerville, I. (2016). Software engineering (10th ed.). Pearson Education.

Tesone, D. V. (2012). Hospitality information systems and e-commerce. John Wiley & Sons.

Turban, E., Volonino, L., & Wood, G. R. (2015). Information technology for management: Digital strategies for insight, action, and sustainable performance (10th ed.). John Wiley & Sons.

Turner, C. (2012). Financial management in the hospitality industry. Goodfellow Publishers.

Warren, C. S., & Jones, J. P. (2010). Corporate financial accounting (11th ed.). Cengage Learning.

Wireman, T. (2004). Developing performance indicators for managing maintenance (2nd ed.). Industrial Press.

2.4 Synthesis of the Reviewed Literature

The body of literature reviewed in the preceding sections of this chapter collectively established a comprehensive and well-grounded theoretical and empirical foundation for the development of the Condominium Booking and Operations Management System proposed in this study. Across the various domains examined—including property management systems, booking and reservation technologies, payment and revenue tracking, maintenance management, database architecture, web-based system design, and the application of the Agile development methodology—a consistent and converging set of insights emerged. These insights affirmed that the operational challenges experienced by Gales Urban Haven under One Spatial Iloilo were not unique to the organization but were, rather, characteristic of a broader class of problems that had been extensively documented in the academic and professional literature on small- to medium-scale property management operations that continued to rely on manual or fragmented record-keeping systems. The synthesis of these reviewed works revealed that the transition from manual, paper-based, or loosely organized digital processes to a centralized, automated, and integrated information system was not merely a technological upgrade but a fundamental transformation in the way a property management business operated, managed its resources, served its tenants, and positioned itself for sustainable growth. The literature consistently demonstrated that organizations that had undertaken this transition realized measurable improvements in operational efficiency, financial accuracy, data accessibility, staff productivity, and overall service quality. The proposed

system for Gales Urban Haven was firmly grounded in these documented outcomes, and its design, features, and development approach were shaped directly by the collective knowledge and evidence presented in the reviewed literature.

Convergence of Key Themes Across the Literature

A careful examination of the reviewed literature revealed that despite the diversity of contexts, geographic settings, and technological environments in which the individual studies were conducted, a set of dominant and recurrent themes emerged with striking consistency. These themes not only validated the problem statement articulated in Chapter One of this study but also provided direct and actionable guidance for the scope, design, and functionality of the proposed system. The following subsections discussed these converging themes in detail and explained how each one informed the development of the Condominium Booking and Operations Management System. Furthermore, the convergence of these themes across multiple studies and literature sources strengthened the theoretical and empirical foundation of the proposed system, demonstrating that the identified operational challenges faced by Gales Urban Haven were not isolated concerns but rather widely recognized issues that had been consistently addressed through the adoption of web-based information systems. The recurring patterns observed across the reviewed literature also served as a guiding framework in determining the most essential features and functionalities that the proposed system needed to

incorporate in order to effectively respond to the operational demands of a condominium rental business. Additionally, the synthesis of these converging themes highlighted the critical role that automation, centralized data management, and real-time monitoring played in modernizing property management operations, further justifying the relevance and necessity of the proposed system. The alignment between the findings of the reviewed literature and the operational context of Gales Urban Haven also reinforced the appropriateness of the chosen system design and development approach, ensuring that the proposed solution was grounded in established knowledge and proven practices. Moreover, the identification of these key themes allowed the researchers to draw meaningful connections between existing studies and the specific objectives of this research, providing a coherent and well-supported academic basis for the development and implementation of the Condominium Booking and Operations Management System.

Ultimately, the convergence of key themes across the literature affirmed that the proposed system represented a timely, relevant, and well-informed response to the operational inefficiencies of Gales Urban Haven, contributing not only to the improvement of its business processes but also to the broader academic discourse on the application of information systems in property management and hospitality operations.

The Pervasive Inadequacy of Manual Property Management Processes

Perhaps the most consistently documented finding across the reviewed literature was the fundamental inadequacy of manual, paper-based, and non-integrated approaches to property management. Studies examining condominium associations, residential rental properties, commercial real estate portfolios, and mixed-use developments all converged on the conclusion that organizations relying on such approaches routinely suffered from a predictable constellation of operational failures. These included delayed and error-prone tenant billing, inconsistent maintenance request tracking, inability to generate accurate real-time financial reports, poor visibility into unit occupancy and availability, and fragmented communication between property managers, maintenance staff, and tenants.

The literature made clear that these failures were not isolated incidents but were the natural and expected consequences of systems that were designed for lower volumes of transactions and simpler operational environments. As the scale and complexity of property portfolios grew—in terms of the number of units managed, the diversity of tenant needs, the variety of services offered, and the stringency of regulatory compliance requirements—the limitations of manual systems became increasingly acute. For Gales Urban Haven, which operated in a growing urban condominium market in Iloilo City, these dynamics were particularly relevant and underscored the urgency and necessity of the proposed system.

The Demonstrated Value of Centralized and Integrated Information Systems

In direct response to the documented inadequacies of manual approaches, the literature overwhelmingly endorsed the adoption of centralized, integrated property management information systems as the most effective and sustainable solution. The reviewed studies documented concrete and quantifiable benefits realized by organizations that had implemented such systems, including reductions in administrative workload, improvements in billing accuracy and collection rates, faster resolution of maintenance issues, and enhanced tenant satisfaction scores.

Centralization was identified as a particularly critical success factor. When all operational data—tenant records, unit availability, booking schedules, payment histories, maintenance requests, and financial summaries—resided in a single, authoritative system of record, the risks of data duplication, inconsistency, and loss were dramatically reduced. Property managers were able to access up-to-date information at any time, from any location, enabling more responsive and informed decision-making. The proposed Condominium Booking and Operations Management System embodied this principle of centralization as one of its foundational design objectives.

The Strategic Importance of Booking and Reservation Management

The literature reviewed on booking and reservation systems established that the management of unit availability and the processing of booking transactions represented some of the most operationally critical and error-prone functions in property management. Scheduling conflicts, double bookings, and the failure to maintain accurate availability calendars were identified as frequent sources of tenant dissatisfaction, revenue loss, and reputational damage for property management organizations.

The reviewed studies demonstrated that automated booking systems equipped with real-time availability tracking, conflict detection, and instant confirmation capabilities effectively eliminated or significantly reduced these problems. The literature further established that the integration of booking management with tenant records and payment processing created a seamless operational workflow that reduced the manual effort required to complete a booking transaction from end to end. These documented outcomes directly informed the booking management module that was developed as a core component of the proposed system for Gales Urban Haven.

The Critical Role of Accurate Financial and Payment Tracking

Across the financial management domain, the reviewed literature identified inaccurate and incomplete payment tracking as one of the most consequential operational deficiencies in property management organizations. The inability to accurately record, reconcile, and report on rent collection, association dues, security deposits, utility charges, and penalty fees created compounding problems that affected cash flow management, financial planning, and regulatory compliance. The literature documented that automated payment tracking systems, particularly those that generated digital receipts, sent payment reminders, flagged overdue accounts, and produced summary financial reports, consistently outperformed manual equivalents in terms of collection rates, accuracy, and audit readiness. The proposed system incorporated a comprehensive payment and revenue tracking module informed by these findings, ensuring that all financial transactions conducted through Gales Urban Haven were recorded with precision, transparency, and full auditability.

As the system became fully operational, the payment tracking module served as a critical instrument in strengthening the financial governance and accountability practices of the organization, providing administrators with a reliable and centralized platform for monitoring all incoming and outgoing financial transactions in real time. The structured financial data generated by the system also enabled management to produce accurate periodic financial statements, facilitating more informed budgeting decisions

and long-term financial planning that supported the sustainable growth of Gales Urban Haven. The payment tracking module also incorporated advanced analytics capabilities that allowed administrators to identify payment trends, forecast revenue streams, and detect anomalies or irregularities in financial records before they escalated into more serious concerns. The system further supported the generation of tenant-specific payment histories, which served as valuable documentation in cases of financial disputes, lease renewals, or regulatory audits conducted by relevant government authorities. The integration of the payment tracking module with other system components, such as booking management and tenant records, created a fully interconnected financial ecosystem that eliminated data redundancy and ensured consistency across all financial and operational records maintained by the organization. The adoption of a digitized and automated payment tracking approach also significantly reduced the administrative burden placed on staff, freeing valuable time and resources that were redirected toward higher-value management activities and tenant relationship building. Ultimately, the financial and payment tracking capabilities embedded within the system laid a strong and enduring foundation for the continued improvement of fiscal responsibility, operational transparency, and financial performance at Gales Urban Haven.

Maintenance Management as a Driver of Tenant Retention

The reviewed literature on maintenance management revealed a strong and statistically significant relationship between

the responsiveness of a property's maintenance operations and the satisfaction and retention rates of its tenants. Properties that implemented structured, technology-supported maintenance request workflows—including formal request submission channels, automated routing to appropriate staff, status tracking, and completion confirmation—were shown to resolve issues faster, reduce the number of repeat complaints for the same issue, and generate higher tenant satisfaction ratings than properties relying on informal or verbal maintenance reporting.

These findings had direct implications for the design of the maintenance management module included in the proposed system. The module was designed to provide tenants with a formal and accessible channel for submitting maintenance requests and tracking their progress, while simultaneously providing property managers and maintenance personnel with a structured workflow for prioritizing, assigning, and documenting the resolution of reported issues.

Database Architecture as the Backbone of System Reliability

The literature reviewed on database systems and architecture established that the choice of database technology and schema design had profound implications for the reliability, performance, scalability, and data integrity of any information system. The reviewed studies consistently recommended the use of relational database management systems (RDBMS) for applications that managed structured, interdependent data entities with complex

relationships—a description that accurately characterized the data model of a property management system.

Key principles drawn from the database literature—including normalization to eliminate data redundancy, the enforcement of referential integrity through foreign key constraints, the use of indexing to optimize query performance, and the implementation of transaction management to prevent data corruption—were all applied in the design of the database layer of the proposed system. These design decisions ensured that the system maintained the integrity and consistency of all stored data even under concurrent access from multiple users.

Web-Based Accessibility as an Enabler of Operational Continuity

The reviewed literature on web-based information systems established that the accessibility and platform independence afforded by browser-based applications represented a significant operational advantage over desktop-installed or server-bound legacy systems. Web-based systems allowed authorized users to access the system from any device with an internet connection, eliminating the need for dedicated hardware installations, simplifying software updates and maintenance, and enabling remote management of property operations.

For a property management context, this accessibility was particularly valuable. Property managers were able to respond to booking inquiries, review payment statuses, and

monitor maintenance requests from outside the office. Tenants were able to submit requests, view their payment history, and communicate with management without having to visit a physical office. These capabilities directly improved the operational responsiveness and service quality of Gales Urban Haven, and the proposed system was designed as a web-based application to realize these benefits fully.

Agile Methodology as the Appropriate Development Framework

The literature reviewed on software development methodologies provided strong justification for the selection of the Agile methodology as the development framework for the proposed system. The Agile approach, characterized by iterative development cycles, continuous stakeholder collaboration, adaptive planning, and incremental delivery of working software, was shown to be particularly well-suited for projects in which the full scope of requirements could not be exhaustively defined at the outset—a condition that accurately described the development context of the proposed system.

The reviewed studies documented that Agile projects consistently achieved higher rates of stakeholder satisfaction, more timely delivery, and better alignment between the delivered system and actual user needs compared to projects managed under the traditional Waterfall methodology. By adopting Agile, the development team was positioned to respond constructively to feedback from the management and staff of Gales Urban Haven throughout the development process, ensuring that the final system

accurately reflected the operational realities and priorities of the organization.

Research Gaps Addressed by the Proposed System

While the reviewed literature provided robust evidence for the value of property management information systems in general, it also revealed a number of research and implementation gaps that the proposed system was positioned to address. These gaps related to the specific context of small- to medium-scale condominium operations in emerging urban markets in the Philippines, the integration of multiple management functions within a single unified system, and the accessibility of such systems to organizations with limited technical infrastructure.

Research Gap Identified in the Literature	**How the Proposed System Will Address It**
Limited studies on integrated property management systems for small-scale condominium operations in Philippine urban markets	The proposed system will be designed specifically for the operational context of Gales Urban Haven, a small-scale condominium in Iloilo City, and will be calibrated to the resource constraints and regulatory environment of similar organizations in the Philippines.

Absence of unified systems that will integrate booking, payment, and maintenance management in a single platform	The proposed system will consolidate booking management, payment tracking, maintenance request handling, and administrative reporting into one cohesive platform, eliminating the fragmentation documented in the literature.
Underrepresentation of web-based, role-specific systems accessible to non-technical property management staff	The system will feature a role-based access control architecture and an intuitive user interface designed for non-technical users, ensuring that property managers, front desk staff, and maintenance personnel will be able to use the system effectively without specialized training.
Lack of documented frameworks for transitioning manual records to digital systems in small condominium operations	The development process will include a structured data migration and onboarding phase informed by best practices documented in the literature, ensuring a smooth and low-disruption transition for Gales Urban Haven.
Insufficient attention to tenant-facing features in	The proposed system will incorporate tenant-facing

property management system research	functionalities, including booking history visibility and maintenance request status tracking, directly addressing the tenant experience dimension that is underexplored in the existing literature.

Theoretical Frameworks Informing the Proposed System

Beyond the empirical findings reported in individual studies, the reviewed literature also surfaced several theoretical frameworks and conceptual models that provided a principled basis for the design decisions incorporated in the proposed system. These frameworks guided not only what the system did but also how it was structured, how its components related to one another, and how it was evaluated upon completion. The theoretical foundations drawn from the reviewed literature ensured that the proposed system was not merely a technical solution to immediate operational problems but rather a thoughtfully designed platform grounded in established principles of information systems development, property management theory, and organizational efficiency. Each framework identified through the review process contributed a distinct perspective that informed specific aspects of the system architecture, user interface design, data management approach, and overall functional scope of the proposed solution.

The application of these conceptual models also provided a systematic basis for justifying the design choices made throughout the development process, ensuring that every

major decision was supported by both theoretical reasoning and empirical evidence drawn from existing literature. Furthermore, the integration of multiple theoretical frameworks allowed the development team to approach the system design from a multidimensional perspective, accounting for the technical, organizational, and human factors that collectively determined the effectiveness and sustainability of the proposed system in a real-world property management environment.

As the system progressed through its development and evaluation phases, these theoretical frameworks continued to serve as reference points against which the system's performance, usability, and alignment with organizational goals were measured and assessed. Future researchers and developers who built upon the findings of this study could also benefit from the theoretical groundwork established herein, as it provided a coherent and well-documented intellectual foundation for subsequent enhancements, adaptations, and expansions of the proposed system. Ultimately, the deliberate incorporation of theoretical frameworks into the system design process elevated the academic rigor and practical relevance of this capstone project, demonstrating that the proposed solution was not only technically functional but also conceptually sound and aligned with the broader body of knowledge in the fields of information technology and property management.

Systems Theory and the Integrated Operations Model

The concept of the organization as an integrated system—in which all operational processes were interdependent and the performance of the whole was a function of the coherence of its parts—provided the theoretical foundation for the integrated design of the proposed system. Rather than developing separate standalone tools for booking, payment, and maintenance, the proposed system was designed as an interconnected platform in which data generated by one module was immediately available to and usable by all other modules. This integrative approach, grounded in systems theory, eliminated the data silos that were identified as a primary source of inefficiency in the reviewed literature.

The application of systems theory to the design of the proposed platform ensured that each module operated not as an isolated component but as a contributing element of a larger, cohesive operational framework that served the holistic information and management needs of Gales Urban Haven. By treating the organization as a unified system of interdependent processes, the proposed system was capable of reflecting the true complexity of condominium property management operations, where a single transaction or event in one area of the business inevitably produced ripple effects across multiple other functional domains. The interconnected architecture informed by systems theory also enabled the proposed system to maintain a single, consistent, and continuously updated data environment, ensuring that all modules operated on the basis of accurate and current information rather than fragmented or duplicated records maintained across separate platforms. As

the system was adopted and utilized, the systemic coherence built into its design contributed to measurable improvements in the speed, accuracy, and reliability of administrative decision-making, as administrators were able to access a comprehensive and integrated view of all operational activities from a single centralized interface. The systems theory foundation embedded in the architecture also provided a scalable and flexible structural framework that accommodated the addition of new modules and functionalities without compromising the integrity or coherence of the overall platform. The insights derived from systems theory also informed the evaluation criteria used to assess the success of the proposed system, with particular emphasis on the degree to which integration across modules reduced redundancy, improved data consistency, and supported more efficient and informed organizational management. Ultimately, the deliberate grounding of the proposed system in systems theory not only strengthened its technical design but also ensured that it remained aligned with the operational realities and strategic goals of Gales Urban Haven.

The Technology Acceptance Model (TAM)

The Technology Acceptance Model, which posited that the adoption and effective use of an information system were determined primarily by users' perceptions of its usefulness and ease of use, informed the user experience design strategy of the proposed system. In line with the TAM framework, the system was designed with a strong emphasis on interface clarity, logical workflow navigation,

and minimal cognitive load for end users. The user interface was iteratively refined through stakeholder feedback sessions conducted during the Agile development process, ensuring that the delivered system met the usability standards necessary for adoption by all categories of users at Gales Urban Haven. Furthermore, the principles of the Technology Acceptance Model guided the researchers in evaluating and measuring the degree to which the proposed system was perceived as both useful and easy to use by its intended users, providing a structured basis for assessing user satisfaction and system acceptance upon implementation.

The TAM framework also informed the design of system features that directly addressed the practical needs of administrators and business owners, ensuring that every functional module contributed meaningful value to their daily operational tasks and decision-making processes. Additionally, the emphasis on perceived ease of use drove the development of an intuitive and straightforward user interface that minimized the time and effort required for staff members to learn and navigate the system, thereby reducing resistance to adoption and encouraging consistent system utilization. The system also incorporated clear instructions, tooltips, and navigational cues that assisted users in understanding and maximizing the full range of system features, further reinforcing their confidence and comfort in using the platform.

Moreover, the application of the TAM framework ensured that the proposed system was not only technically functional but was also embraced and regularly utilized by its end users,

as a system that was perceived as useful and easy to use was more likely to deliver sustained operational benefits to Gales Urban Haven. Ultimately, by anchoring the design and development process within the theoretical principles of the Technology Acceptance Model, the researchers ensured that the Condominium Booking and Operations Management System was a practical, user-centered solution that effectively supported the operational goals and digital transformation of Gales Urban Haven under One Spatial Iloilo.

The CRUD Operational Model

The foundational database operations model—Create, Read, Update, Delete (CRUD)—served as the structural template for all data management functionalities incorporated in the proposed system. Every entity managed by the system, whether tenant records, booking entries, payment transactions, or maintenance requests, was governed by a complete set of CRUD operations, ensuring that authorized users were able to create new records, retrieve existing ones, modify information as circumstances changed, and remove outdated entries in a controlled and auditable manner. Furthermore, the implementation of the CRUD operational model provided a consistent and standardized approach to data management across all system modules, ensuring that every functional component of the proposed system interacted with the centralized database in a uniform, reliable, and predictable manner.

The structured application of CRUD operations also contributed to the overall integrity and accuracy of the system's data, as each operation was subject to validation rules and access controls that prevented unauthorized or erroneous modifications to critical business records. Additionally, the CRUD model served as the foundation for the system's audit trail functionality, as every create, update, and delete action performed within the system was recorded and timestamped, allowing administrators to trace the history of any record and maintain full accountability over all data management activities.

The read operations embedded within the CRUD framework also supported the system's reporting and data retrieval functions, enabling administrators and business owners to access accurate and up-to-date information quickly and efficiently whenever it was needed for operational or decision-making purposes. Moreover, the adoption of the CRUD operational model simplified the development and maintenance of the proposed system's database layer, as the well-established and widely recognized nature of the framework provided the development team with a clear and structured blueprint for implementing all data management functionalities. Ultimately, the CRUD operational model ensured that the Condominium Booking and Operations Management System maintained a high standard of data organization, accuracy, and security, providing Gales Urban Haven with a robust and dependable platform for managing all aspects of its condominium rental operations.

Direct Influence of the Literature on the Proposed System's Design

The reviewed literature not only validated the need for the proposed system but also exerted a direct and traceable influence on its specific design features, functional modules, and technical architecture. The mapping of key literature-derived insights to the corresponding design decisions reflected the features and functionalities incorporated into the Condominium Booking and Operations Management System.

Literature Domain	Key Insight	Design Decision in Proposed System
Property Management Systems	Centralized systems reduce errors and improve operational efficiency	A unified platform will be developed to consolidate all management functions into a single system of record
Booking & Reservation	Real-time availability tracking prevents double-bookings and scheduling conflicts	An automated availability calendar with conflict detection will be implemented in the booking module

Payment & Revenue Tracking	Automated billing and receipt generation improve collection rates and accuracy	A payment tracking module with digital receipt generation, overdue flagging, and revenue summaries will be built
Maintenance Management	Structured request workflows improve resolution times and tenant satisfaction	A ticketing-style maintenance request module with status tracking and staff assignment will be included
Database Architecture	Relational databases with normalized schemas ensure data integrity and query performance	A relational database will be designed with normalized tables, referential integrity constraints, and indexed keys
Web-Based System Design	Browser-based access enables remote management and eliminates hardware dependencies	The system will be developed as a responsive web application accessible from any

		internet-connected device
Agile Methodology	Iterative development with stakeholder feedback improves alignment with user needs	The Agile methodology with sprint-based iterations and regular review sessions will govern the development process
User Interface Design	Intuitive interfaces with role-based access improve adoption by non-technical users	Role-based views will be designed for administrators, front desk staff, and maintenance personnel separately

Validation of the Proposed System's Relevance and Necessity

The weight of evidence accumulated across the reviewed literature collectively validated the relevance, necessity, and appropriateness of the Condominium Booking and Operations Management System proposed in this study. This validation operated at multiple levels. At the problem level, the literature confirmed that the operational challenges documented at Gales Urban Haven were

well-recognized phenomena with known root causes and known solutions. At the solution level, the literature established that technology-enabled, integrated property management systems represented the most effective and sustainable response to these challenges. At the design level, the literature provided specific, evidence-based guidance on the features, architecture, and development approach that maximized the probability of the system achieving its intended outcomes.

Furthermore, the literature affirmed that the investment in such a system yielded benefits that extended well beyond mere operational convenience. Organizations that successfully implemented integrated property management systems were positioned to offer a higher standard of service to tenants, operate with greater financial discipline and transparency, respond more effectively to the demands of regulatory oversight, and scale their operations with confidence as their property portfolios grew. For Gales Urban Haven, these outcomes translated directly into enhanced competitiveness within the Iloilo City condominium market, improved tenant retention, and a stronger operational foundation for sustainable long-term growth.

Dimension	**Documented Outcome in Literature**	**Expected Outcome for Gales Urban Haven**

Operational Efficiency	Reduction in administrative hours spent on manual record-keeping	Staff will spend less time on paperwork and more time on tenant-facing service
Financial Accuracy	Higher billing accuracy and improved payment collection rates	All payment transactions will be recorded, tracked, and reconciled with full digital audit trails
Maintenance Responsiveness	Faster resolution of tenant-reported maintenance issues	Maintenance requests will be formally logged, assigned, and tracked to completion
Data Accessibility	Real-time access to operational data from any location	Managers will access live data on bookings, payments, and maintenance from any device
Tenant Satisfaction	Higher tenant retention in	Tenants will experience faster service responses

	properties using integrated systems	and greater transparency in all dealings
Decision Support	Availability of accurate reports enables data-driven management decisions	Management will have access to financial summaries and occupancy reports to guide planning

Conclusion

In conclusion, the synthesis of the reviewed literature left no ambiguity as to the theoretical grounding, practical necessity, and design direction of the proposed Condominium Booking and Operations Management System. The literature collectively established that the problem the system sought to solve was real, well-documented, and consequential; that the solution approach adopted in this study was evidence-based, appropriate, and aligned with best practices in the field; and that the specific design features incorporated in the proposed system reflected a careful and deliberate reading of the documented experiences of organizations that had navigated similar operational challenges.

The proposed system therefore did not emerge from an arbitrary set of design choices but was the direct product of a systematic and rigorous engagement with the existing body of knowledge in property management, information systems, software engineering, and organizational management. This grounding in the literature served as both the justification and the blueprint for the development work that followed in the succeeding chapters of this study, providing a principled and well-supported basis for every functional and architectural decision that the development team made in the course of building the system for Gales Urban Haven under One Spatial Iloilo.

CHAPTER 3

METHODOLOGY

3.1 Research Design

The study utilized a developmental research design, which focused on the systematic planning, design, development, implementation, and evaluation of the proposed application. Developmental research design was appropriate for the study because it allowed the researchers to create a functional and effective system intended to address existing problems encountered in manual condominium and rental management operations. The study concentrated not only on the development of the application itself but also on evaluating its functionality, usability, reliability, and effectiveness in improving current business processes.

The proposed system was developed to address common issues experienced in manual operations, such as double bookings, misplaced records, delayed transactions, inaccurate computations, data redundancy, and inefficient report generation. Traditional methods of managing tenant information, room availability, reservations, billing, and payments often consumed significant time and effort and resulted in human errors that negatively affected customer satisfaction and operational efficiency. Through the developmental research approach, the researchers identified these operational problems, analyzed the needs of the users, and developed an automated solution that improved the overall management process. The study involved several stages, including requirement gathering, system analysis, system design, coding and development, testing,

evaluation, and implementation. During the requirement gathering phase, the researchers collected relevant information regarding the current management practices, challenges, and user needs from the target users or stakeholders. The gathered data served as the foundation for designing the features and functionalities of the proposed application.

In the system design and development phase, the researchers created the structure, interface, database, and functionalities of the application. The system was designed to manage customer information, room reservations, booking schedules, billing transactions, payment monitoring, and report generation efficiently and accurately. The application also incorporated user authentication and security features to ensure the confidentiality and integrity of stored data.

After the development process, the system underwent testing and evaluation to determine whether it met the required standards and objectives of the study. Different testing procedures, such as functionality testing, usability testing, reliability testing, and user acceptance testing, were conducted to identify system errors, evaluate performance, and ensure that the application operated according to user requirements. Feedback from respondents or evaluators was also considered for possible improvements and refinements to the system.

The developmental research design helped ensure that the proposed application was practical, user-friendly, and responsive to the needs of condominium or rental management businesses. Furthermore, this approach supported the study’s goal of providing a modernized and efficient management solution that minimized manual errors, improved operational efficiency,

enhanced customer service, and contributed to the digital transformation of property management operations.

3.2 System Development Methodology

The development of the Condominium Booking and Operations Management System for Gales Urban Haven, operating under the business name One Spatial Iloilo, was guided by the Agile Software Development Model. Agile was a widely adopted and highly regarded iterative approach to software development that emphasized flexibility, collaboration, continuous feedback, and incremental delivery of functional system components.

Unlike traditional linear development models such as the Waterfall Model, which required the complete definition of system requirements before development could begin, the Agile model accommodated changes in requirements and system design throughout the development process, making it particularly well-suited for projects where operational needs evolved or became more clearly defined as development progressed.

The Agile methodology was structured around iterative development cycles, commonly referred to as sprints or phases, wherein a subset of system features was planned, designed, developed, tested, and reviewed before the next cycle began. This iterative structure allowed development teams to identify and address potential issues early, incorporate user and stakeholder feedback at each stage, and ensure that the system being built remained aligned with the actual operational requirements of the target users. For this study, the Agile

model was particularly appropriate given the complexity of integrating multiple operational modules, including booking management, payment tracking, revenue reporting, calendar-based scheduling, and maintenance monitoring, into a single cohesive platform. The development of the proposed system proceeded through five structured and sequential phases under the Agile framework. Each phase built upon the outputs of the preceding phase, ensuring a coherent and progressive development process that consistently moved toward a fully functional, tested, and deployable system. The following sections described each phase in detail, including the specific activities, outputs, and objectives associated with each stage of the development process.

Phase 1: Requirements Analysis

The Requirements Analysis phase constituted the foundational stage of the system development process and was critical to the overall success of the project. During this phase, the development team conducted a thorough and systematic examination of the existing manual operational processes employed by Gales Urban Haven, including the current methods used for recording reservations, tracking payments, monitoring maintenance activities, and managing scheduling. This examination involved direct consultations with property administrators and key personnel, structured interviews, observation of day-to-day operational workflows, and a review of existing records and documentation.

The primary objective of this phase was to gain a comprehensive understanding of the operational environment, identify the specific inefficiencies, recurring problems, and data management gaps that characterized the current manual system, and translate

these findings into clearly defined system requirements. Both functional requirements, which described what the system needed to do, and non-functional requirements, which addressed system performance, security, usability, and reliability, were documented during this phase. The resulting requirements specification document served as the authoritative reference for all subsequent development activities, ensuring that the system to be built was precisely aligned with the operational needs and expectations of its intended users.

Additionally, this phase involved identifying the key stakeholders of the system, defining the user roles and access levels required for the platform, and establishing the technical constraints and environmental conditions within which the system needed to operate. By completing this phase thoroughly and diligently, the development team laid a solid and reliable groundwork for all subsequent phases of the Agile development process.

Phase 2: System Design

The System Design phase involved the translation of the requirements gathered in the preceding phase into a detailed and comprehensive architectural blueprint for the proposed system. During this phase, the development team designed the overall structure of the web-based platform, including the system architecture, database schema, user interface layouts, navigation flows, and module interdependencies. The design was guided by principles of clarity, simplicity, security, and scalability, ensuring that the resulting system was both user-friendly for administrators and technically robust for long-term operational use.

A key focus of this phase was the design of a secure and intuitive web interface that enabled administrators of One Spatial Iloilo to interact with the system efficiently and with minimal training. The interface design prioritized logical information hierarchy, clear labeling of functions, and streamlined navigation pathways so that users could locate and execute system operations with ease. Wireframes and interface mockups were developed and reviewed during this phase to validate design decisions before development commenced.

The database design component of this phase defined the structure of the centralized database, including the tables, fields, relationships, and constraints that governed the storage and retrieval of all operational data. Entity-relationship diagrams and data flow diagrams were constructed to visually represent the data architecture of the system. Security considerations, including user authentication protocols, access control mechanisms, and data encryption strategies, were also incorporated into the system design during this phase to ensure that the platform adequately protected sensitive operational and financial information from unauthorized access or misuse.

Phase 3: System Development

The System Development phase represented the core implementation stage of the project, during which the actual programming and construction of the proposed system took place based on the specifications and designs established in the preceding phases. This phase involved the development of all functional modules of the Condominium Booking and Operations Management System, including the booking management module, payment tracking module, revenue reporting module, calendar-based scheduling

module, and maintenance monitoring module, as well as the centralized database and user authentication system.

Development proceeded iteratively, with each module being built, integrated, and reviewed in successive development cycles consistent with the Agile methodology. This iterative approach allowed the development team to test individual components as they were completed, detect and resolve technical issues promptly, and progressively assemble the complete system from its constituent modules. Throughout this phase, the team adhered to best practices in software engineering, including modular code design, consistent coding standards, thorough documentation of code logic, and regular version control to manage changes and maintain a coherent codebase.

The development phase also encompassed the integration of all modules into a unified and cohesive system, ensuring that data flowed seamlessly between components and that all modules functioned correctly in relation to one another. For instance, reservation data recorded in the booking module automatically updated unit availability in the calendar scheduling module, while payment transactions recorded in the payment tracking module were reflected in real time within the revenue reporting module. This level of integration was essential for ensuring the operational coherence and data consistency of the final system.

Phase 4: Testing

The Testing phase was a critical quality assurance stage that was conducted systematically to verify that the developed system functioned correctly, reliably, and in accordance with the requirements established during the Requirements Analysis phase.

Testing was conducted at multiple levels to ensure comprehensive coverage of all system components and their interactions. Unit testing was performed on individual modules and functions to confirm that each component behaved as expected under normal and boundary conditions. Integration testing was subsequently conducted to verify that all modules interacted correctly and that data was transferred accurately between system components.

System testing was then carried out on the complete, integrated system to evaluate its overall performance, stability, and compliance with the defined functional and non-functional requirements. This included testing for system responsiveness, data accuracy, error handling, security, and user accessibility. Performance testing assessed how the system behaved under varying load conditions to ensure that it remained stable and responsive during actual operational use.

User acceptance testing was also conducted during this phase, wherein selected administrators and end users of Gales Urban Haven interacted with the system and provided structured feedback on its usability, functionality, and overall suitability for the intended operational environment. Any defects, inconsistencies, or areas for improvement identified during the testing phase were documented, prioritized, and addressed by the development team before the system proceeded to the Deployment phase. The testing phase was essential not only for identifying technical issues but also for building confidence among stakeholders that the system was ready for operational deployment.

Phase 5: Deployment

The Deployment phase represented the concluding stage of the Agile development process for this study, during which the fully developed and tested system was presented and made available for operational evaluation. In the context of this research, the system was deployed as a functional prototype that was formally presented to the administrators and key personnel of Gales Urban Haven for evaluation, demonstration, and user acceptance assessment.

The prototype deployment allowed stakeholders to interact with the complete system in a realistic operational setting and assess its effectiveness, usability, and alignment with the business's operational requirements.

During this phase, the development team also ensured that all necessary system configurations were properly established, that user accounts and access credentials were created and assigned appropriately, and that all operational data structures were correctly initialized within the centralized database. End-user orientation and system walkthroughs were conducted to familiarize administrators with the features and functions of the system, enabling them to operate it effectively from the outset.

The feedback and evaluation results gathered during the deployment phase were used to assess the overall success of the system in addressing the operational needs of Gales Urban Haven and to identify any remaining refinements or enhancements that were required prior to full-scale implementation. The deployment

phase thus served not only as the culmination of the development process but also as the beginning of a continuous improvement cycle consistent with the iterative and feedback-driven principles of the Agile development model.

3.3 System Architecture

The Condominium Booking and Operations Management System for Gales Urban Haven, operating under the business name One Spatial Iloilo, was constructed upon a Client-Server Architecture, which was a widely established and industry-standard architectural paradigm for the design and deployment of web-based information systems. The client-server model divided the computational responsibilities of the system into two distinct but closely interconnected tiers: the client tier, which was responsible for presenting information and receiving user input through a graphical web interface, and the server tier, which was responsible for processing requests, executing business logic, managing application workflows, and maintaining the integrity of the centralized data repository. This architectural approach ensured a clear and logical separation of concerns between the presentation layer and the data processing layer, resulting in a system that was maintainable and scalable as the operational demands of the business grew over time.

The adoption of a client-server architecture for this system offered several significant advantages over alternative approaches. First, it enabled multiple authorized users to access and interact with the system simultaneously from different devices and locations through any standard web

browser, without requiring the installation of dedicated software on client machines. Second, it centralized all data processing and storage operations on the server side, ensuring that all users were always working with the same, current version of operational data. Third, it facilitated efficient system maintenance and updates, as modifications to the application logic or database structure only needed to be applied on the server side rather than on each individual client device. These characteristics made the client-server model ideally suited to the operational environment of One Spatial Iloilo, where administrative staff needed to access the system from different workstations within the property management office.

Client Tier (Presentation Layer)

The client tier of the proposed system constituted the front-end presentation layer through which all authorized users, primarily the administrative staff and property managers of Gales Urban Haven, interacted with the system. This layer was rendered and executed within a standard web browser on the client's device and was responsible for displaying all system interfaces, receiving user inputs, and transmitting requests to the server for processing. The client tier was developed using standard web technologies, including HyperText Markup Language (HTML) for content structure, Cascading Style Sheets (CSS) for visual presentation and layout, and JavaScript for dynamic interface behavior and client-side interactivity.

The web-based interface presented through the client tier was designed with a strong emphasis on usability, clarity, and accessibility. Administrators were able to navigate through clearly organized menus and dashboards to access the various

functional modules of the system, including booking management, payment tracking, revenue reporting, calendar-based scheduling, and maintenance monitoring. The interface presented data in structured, readable formats such as tables, forms, summary cards, and interactive calendar views, enabling users to interpret and act upon operational information efficiently and without unnecessary complexity.

The client tier was also designed to be responsive, ensuring that the system interface rendered correctly and remained fully functional across different screen sizes and device types, including desktop computers, laptop computers, and tablet devices commonly used in an office environment. All interactions initiated by the user through the client interface, such as submitting a booking record, recording a payment transaction, or updating a maintenance request, were transmitted as structured requests to the server tier for validation, processing, and storage. The client tier then received the server's response and updated the displayed interface accordingly, providing the user with immediate and accurate feedback on the outcome of their actions.

Server Tier (Application and Logic Layer)

The server tier served as the central processing engine of the proposed system, responsible for receiving and interpreting all requests transmitted from the client tier, executing the corresponding business logic and data operations, and returning appropriate responses to the client. This layer housed the core application logic that governed all system functions, including the validation of booking submissions, the computation of payment balances and revenue figures, the enforcement of scheduling rules and conflict prevention mechanisms, the

management of maintenance records and status updates, and the generation of financial and operational reports.

The server-side application was developed using a robust and widely supported web development framework, ensuring that the system was capable of handling concurrent user requests efficiently and reliably. The server implemented clearly defined application programming interfaces (APIs) that governed the communication between the client tier and the data layer, ensuring that all data transactions were processed in a controlled, consistent, and secure manner. Business rules and validation logic were enforced at the server level to prevent the recording of invalid or conflicting data, such as duplicate bookings, incomplete payment entries, or overlapping maintenance schedules, thereby preserving the integrity and accuracy of all operational records stored within the system.

The server tier was also responsible for enforcing the system's security protocols, including the authentication of user login credentials, the management of role-based access permissions, and the protection of sensitive operational and financial data from unauthorized access or manipulation. By centralizing security enforcement at the server level, the system ensured that all data access and modification activities were governed by consistent and verifiable security policies, regardless of the device or browser used to access the system.

Database Layer (Data Management Layer)

The database layer formed the foundational data management component of the proposed system, providing a secure, structured, and persistent repository for all operational data generated and utilized by the system. The system employed MySQL

as its relational database management system (RDBMS), a widely trusted and extensively used open-source database platform that was recognized for its reliability, performance, data integrity enforcement, and compatibility with web-based application development environments.

MySQL's support for Structured Query Language (SQL), relational data modeling, and transactional data operations made it a highly appropriate choice for managing the complex and interrelated data sets associated with condominium booking and operations management.

The centralized MySQL database stored and organized all data generated by the system across its various functional modules. This included tenant profiles and contact information, unit records and availability status, booking reservations and reservation histories, payment transactions and outstanding balance records, revenue summaries and financial report data, maintenance requests and service completion logs, and user account credentials and access control settings. The database was structured using a normalized relational schema to eliminate data redundancy, maintain referential integrity between related data sets, and ensure that all stored information remained consistent and accurate throughout its lifecycle.

All interactions between the server tier and the database layer were conducted through parameterized queries and prepared statements, which were established best practices for preventing SQL injection attacks and ensuring the security of database operations. The database was also configured with appropriate indexing to optimize query performance and minimize data retrieval times, ensuring that the system responded promptly to user requests even as the volume of stored data grew over time.

Regular database backup procedures were established to safeguard all operational data against accidental loss or system failure, providing administrators with the assurance that critical business records were preserved and recoverable under adverse conditions.

System Data Flow and Component Integration

The three tiers of the client-server architecture operated as an integrated and cohesive system, with data flowing seamlessly and securely between the client, server, and database layers in response to user actions and system events. When an administrator interacted with the web interface through the client tier, for example by submitting a new booking reservation, the request was transmitted to the server tier via the system's defined communication protocols. The server validated the submitted data against the established business rules, checked the availability of the requested unit within the database, recorded the new booking entry, and returned a confirmation response to the client interface, which immediately updated the booking dashboard and calendar view to reflect the newly recorded reservation.

Similarly, when a payment transaction was recorded by an administrator, the server computed the updated outstanding balance, stored the transaction record in the database, and propagated the updated financial figures to the revenue monitoring module and the tenant's payment history in real time. This seamless cross-module data propagation ensured that all sections of the system consistently reflected the current operational state of the business, eliminating the data inconsistencies and information lags that were common in manual record-keeping systems.

The integrated architecture of the proposed system thus provided Gales Urban Haven with a unified, reliable, and responsive operational platform in which all business data was managed centrally, all user interactions were processed consistently, and all system components functioned in coordinated alignment. This architectural foundation supported the system's long-term scalability, enabling additional modules, features, or user roles to be incorporated into the platform without disrupting existing functionality or compromising system performance.

3.4 Tools and Technologies

This section presented a comprehensive overview of the tools, frameworks, libraries, and runtime environments that were employed in the development of the proposed system. The selection of each technology was guided by its maturity, community support, performance characteristics, and alignment with the functional and non-functional requirements of the project. The development team adopted a modern, full-stack JavaScript/TypeScript architecture, ensuring type safety, developer productivity, and long-term maintainability across both the client-side and server-side components of the application.

3.4.1 Frontend Technologies

The frontend layer of the system served as the primary interface through which users interacted with the application. It was responsible for rendering dynamic content, managing user interactions, communicating with backend services, and

presenting data in a structured and accessible manner. The following technologies were utilized to build the frontend:

React 19

The project utilized React 19, the latest major release of the React JavaScript library, as its primary frontend framework. React was an open-source library developed and maintained by Meta (formerly Facebook) and was widely recognized as one of the most popular and battle-tested UI frameworks in the web development industry. React followed a component-based architecture, wherein the user interface was decomposed into reusable, self-contained components that managed their own state and lifecycle.

React 19 introduced several enhancements over its predecessor, including the stable release of the React Compiler, improved support for server components, and a refined concurrent rendering model. The concurrent rendering capabilities of React 19 allowed the application to interrupt, pause, and resume rendering work as needed, thereby keeping the interface responsive even under heavy computational load.

The component-based model of React enabled the development team to maintain a clean separation of concerns, promote code reusability, and facilitate efficient unit testing of individual UI components. React's declarative programming paradigm also simplified reasoning about application state, as developers only needed to describe what the UI should look like for a given state rather than imperatively manipulating the DOM.

Feature	Description	Benefit to the Project
Component-Based Architecture	UI is divided into reusable, encapsulated components	Promotes modularity and code reuse across views
Concurrent Rendering	React will render updates in the background without blocking the UI	Improved responsiveness for complex user interactions
React Compiler (Stable in v19)	Automatic memorization and optimization of components	Reduces boilerplate and improves runtime performance
Server Components Support	Components that will render on the server and stream to the client	Enables faster initial page loads and reduced bundle size
Hooks API	Functional state and lifecycle management via useState, useEffect, etc.	Simplifies state logic without class components

TypeScript

TypeScript was used as the primary programming language for the frontend codebase. TypeScript was a statically typed superset of JavaScript developed and maintained by Microsoft. It compiled down to plain JavaScript and added optional type annotations, interfaces, generics, and advanced type inference capabilities to the language.

The adoption of TypeScript provided the development team with compile-time type checking, which significantly reduced the likelihood of runtime errors caused by type mismatches, undefined values, or incorrect function signatures. TypeScript's type system also enhanced the developer experience by enabling intelligent autocompletion, inline documentation, and refactoring support through modern code editors such as Visual Studio Code.

Furthermore, TypeScript enforced a more rigorous coding discipline across the team, which proved especially beneficial as the project scaled and multiple developers contributed to the same codebase. Shared type definitions and interfaces served as contracts between different modules and layers of the application, ensuring consistent data shapes across the frontend and backend.

Tailwind CSS

For styling and visual design, the project employed Tailwind CSS, a utility-first CSS framework that provided a comprehensive

set of low-level utility classes. Unlike traditional CSS frameworks such as Bootstrap, which offered pre-built components with opinionated styles, Tailwind CSS allowed developers to construct custom designs directly in the HTML markup by composing utility classes.

Tailwind CSS was integrated into the build pipeline and configured to purge unused styles in production, resulting in highly optimized CSS bundles. The framework's responsive design utilities enabled the team to implement a mobile-first layout strategy, ensuring that the application's interface rendered correctly across a wide range of screen sizes and device types.

The use of utility classes eliminated the need to write custom CSS files for most styling requirements, accelerating development speed and reducing the risk of CSS specificity conflicts. Tailwind's design token system, including configurable color palettes, spacing scales, typography scales, and breakpoints, was customized to align with the visual design specifications of the project.

Vite

The frontend application was scaffolded and bundled using Vite, a next-generation build tool and development server created by Evan You, the creator of Vue.js. Vite leveraged native ES Modules in the browser during development, eliminating the need for bundling during the development phase and resulting in significantly faster cold start times and near-instantaneous hot module replacement (HMR).

For production builds, Vite used Rollup as its underlying bundler, producing highly optimized output with automatic code splitting, tree shaking, and asset optimization. The Vite

configuration was tailored to the project's requirements, including TypeScript transpilation via esbuild, path aliasing, environment variable management, and proxy configuration for communicating with the backend API during local development.

Vite's plugin ecosystem was also leveraged to integrate additional build-time capabilities such as SVG imports, image optimization, and Tailwind CSS processing. The fast feedback loop that Vite provided was instrumental in maintaining developer productivity throughout the project's development lifecycle.

Technology	Version / Type	Role in the Project
React	19 (Latest Stable)	UI component framework and rendering engine
TypeScript	5.x	Statically typed programming language for all frontend code
Tailwind CSS	3.x / 4.x	Utility-first CSS framework for responsive styling
Vite	5.x	Frontend build tool, bundler, and development server

3.4.2 Backend Technologies

The backend layer of the system was responsible for handling business logic, processing client requests, managing data persistence, enforcing security policies, and exposing a well-defined API that the frontend consumed. The following technologies were used to build and operate the backend:

Node.js

The backend runtime environment was Node.js, an open-source, cross-platform JavaScript runtime built on Chrome's V8 JavaScript engine. Node.js allowed the development team to write server-side code in JavaScript and TypeScript, enabling a unified language across the entire technology stack. This reduced the cognitive overhead of context switching between programming languages and allowed shared libraries, utility functions, and type definitions to be reused across the frontend and backend.

Node.js operated on an event-driven, non-blocking I/O model, which made it particularly well-suited for handling a large number of concurrent connections with minimal resource consumption. This architectural characteristic was beneficial for the proposed system, as the backend was required to handle multiple simultaneous client requests, interact with external services, and perform database operations in a responsive manner.

The Node.js Package Manager (npm) ecosystem, one of the largest collections of open-source libraries in existence, provided the team with access to a vast array of pre-built modules and utilities, accelerating the implementation of common backend

functionalities such as authentication, data validation, and API communication.

Express

The backend web framework of choice was Express, a minimal and flexible web application framework for Node.js. Express provided a robust set of features for building RESTful APIs, including routing, middleware composition, request parsing, and response formatting. Its minimalist design philosophy afforded the development team full control over the application's architecture without imposing unnecessary abstractions or conventions.

Express's middleware system was utilized to implement cross-cutting concerns such as request logging, authentication and authorization via JSON Web Tokens (JWT), input sanitization and validation, error handling, and Cross-Origin Resource Sharing (CORS) configuration. Each middleware function was composed in a pipeline, allowing for a clean and modular approach to request processing.

The API endpoints exposed by the Express server followed RESTful design principles, using standard HTTP methods (GET, POST, PUT, PATCH, DELETE) and meaningful resource-based URL structures. Response payloads were formatted as JSON to ensure compatibility with the frontend and any potential third-party integrations.

TypeScript (Backend)

Consistent with the frontend, the backend codebase was also written entirely in TypeScript. The use of TypeScript on the server side provided the same type safety and developer experience benefits described in the frontend section, while

additionally enabling the definition of shared types and interfaces that could be referenced by both layers of the application.

Strongly typed request and response schemas were defined using TypeScript interfaces and, where applicable, runtime validation libraries such as Zod. This dual-layer validation strategy—compile-time checking via TypeScript and runtime validation via schema validators—significantly enhanced the reliability and security of the API by ensuring that only well-formed data was processed by the server.

tsx and esbuild

To execute TypeScript source files directly during development without a separate compilation step, the project used tsx (TypeScript Execute), a lightweight Node.js enhancement that allowed TypeScript files to be run natively. tsx used esbuild internally to perform extremely fast transpilation, stripping TypeScript type annotations and converting modern JavaScript syntax into a format that Node.js could execute.

esbuild also served as the primary bundler for the production backend build. Written in Go, esbuild was significantly faster than traditional JavaScript-based bundlers, producing optimized output in a fraction of the time. The production build pipeline transpiled and bundled the TypeScript backend source code into a single JavaScript file that could be deployed to any Node.js-compatible hosting environment.

This toolchain configuration streamlined the development workflow by providing a fast feedback loop during local development through tsx watch mode while simultaneously producing compact, efficient artifacts for production

deployment. The combination of tsx and esbuild eliminated the need for complex Webpack or Babel configurations on the backend, reducing build complexity and maintenance burden.

Technology	Role	Key Benefit
Node.js	Server-side JavaScript runtime environment	Event-driven, non-blocking I/O; unified language stack
Express	Web framework for building RESTful APIs	Minimalist, flexible, rich middleware ecosystem
TypeScript	Statically typed language for backend code	Type safety, shared types with frontend, fewer runtime errors
tsx	TypeScript execution tool for development	Native TypeScript execution without pre-compilation step

esbuild	High-performance transpiler and bundler	Near-instant build times; optimized production artifacts

Rationale for Technology Selection

The technologies described above were not selected arbitrarily; each choice reflected a deliberate evaluation of the project's functional requirements, team expertise, ecosystem maturity, and long-term maintainability considerations. The following subsections presented the key rationale behind the overall technology strategy.

Unified Language Across the Stack

By adopting TypeScript on both the frontend and backend, the development team was able to share code, type definitions, and utility functions across both layers of the application. This isomorphic approach reduced code duplication, minimized inconsistencies between client and server data models, and allowed developers to contribute to all parts of the system without switching mental models or language paradigms.

Performance and Developer Experience

Vite and esbuild collectively ensured that both the development experience and the production build pipeline operated at a high level of performance. Vite's instantaneous hot module replacement reduced the feedback loop during frontend

development, while esbuild's fast build times accelerated the CI/CD pipeline and deployment workflows.

Industry Adoption and Community Support

Each of the selected technologies had broad industry adoption and active open-source communities. React, Node.js, Express, TypeScript, Tailwind CSS, and Vite were widely used technologies in the web development industry. This widespread adoption ensured the availability of comprehensive documentation, third-party tutorials, community forums, and regularly maintained packages, all of which served as valuable resources throughout the project's development lifecycle.

Scalability and Maintainability

The chosen technology stack was well-suited to support the growth of the application over time. The component-based architecture of React made it straightforward to add new UI features without disturbing existing functionality. The modular middleware design of Express allowed new API routes and capabilities to be introduced incrementally. TypeScript's type system acted as a safety net during refactoring, reducing the risk of regressions as the codebase evolved.

Summary of Technologies

The table below provided a consolidated overview of all technologies used in the project, categorized according to their respective roles in the system architecture.

Layer	Technology	Category	Purpose
Frontend	**React 19**	UI Framework	Building and rendering the component-based user interface
Frontend	**TypeScript**	Programming Language	Providing static type safety across all frontend source files
Frontend	**Tailwind CSS**	CSS Framework	Applying utility-first, responsive styling to UI components
Frontend	**Vite**	Build Tool	Serving the dev environment and bundling the app for production

Backend	**Node.js**	Runtime Environment	Executing server-side JavaScript/TypeScript code
Backend	**Express**	Web Framework	Handling HTTP routing, middleware, and RESTful API responses
Backend	**TypeScript**	Programming Language	Providing static type safety across all backend source files
Backend	**tsx**	Dev Execution Tool	Running TypeScript files directly in development without pre-build
Backend	**esbuild**	Bundler / Transpiler	Compiling and bundling TypeScript backend code for production

In summary, the technology stack described in this section provided a solid, modern, and cohesive foundation for the development of the proposed system. The combination of React 19, TypeScript, Tailwind CSS, and Vite on the frontend, together with Node.js, Express, TypeScript, tsx, and esbuild on the backend, enabled the team to deliver a high-quality, performant, and maintainable application that met the objectives outlined in the earlier chapters of this capstone research.

Database

MySQL served as the relational database management system for the Condominium Booking and Operations Management System. MySQL stored all persistent application data, including tenant profiles, booking records, payment transactions, unit availability information, and maintenance activity logs, in a structured relational schema that enforced data integrity through primary keys, foreign key constraints, and normalization principles. The relational model ensured that related records across different tables—such as a booking record linked to a specific tenant and a specific unit—remained consistent and accurate even as data volumes grew.

MySQL's support for complex SQL queries, indexing strategies, and transaction management enabled the system to retrieve and manipulate data efficiently, supporting the real-time reporting and search functionalities that administrators relied upon for day-to-day decision-making. Additionally, MySQL's widespread

adoption, extensive documentation, and compatibility with Node.js database client libraries simplified both development and long-term system maintenance.

Development Environment

Visual Studio Code (VS Code) was used as the integrated development environment (IDE) for all phases of system development. VS Code provided a lightweight yet feature-rich coding environment that supported the entire technology stack employed in this project, including JavaScript, TypeScript, SQL, and configuration file formats such as JSON and YAML.

The extensibility of VS Code through its marketplace of plugins allowed developers to install language-specific linters, formatters, database client extensions, version control integrations, and debugging tools that enhanced productivity and enforced consistent code quality standards throughout the development lifecycle. The built-in Git integration within VS Code facilitated version control management, enabling the development team to track code changes, maintain development branches, and coordinate collaborative contributions systematically.

3.5 Data Gathering Techniques

The development of the Condominium Booking and Operations Management System was informed by a systematic data gathering process designed to establish a thorough understanding of the existing operational environment at Gales Urban Haven (One Spatial Iloilo) and to identify functional requirements grounded in real-world practice. Two primary data gathering techniques

were employed: process analysis and the review of existing platforms. Each technique contributed distinct and complementary insights that guided system design decisions and ensured that the proposed solution was both practically relevant and technically sound.

Process Analysis

Process analysis involved a structured examination of the existing manual workflows and administrative procedures employed by Gales Urban Haven in managing its condominium booking, payment tracking, scheduling, and maintenance operations. This technique entailed direct observation of day-to-day administrative activities, structured interviews with staff members responsible for managing bookings and financial records, and a review of the paper-based and spreadsheet-driven tools in use.

Through process analysis, the researchers mapped the end-to-end flow of each operational process, from the initial receipt of a tenant inquiry through to the recording of a booking confirmation, payment collection, maintenance request submission, and unit handover. By documenting each step of these workflows in detail, the researchers identified specific points of inefficiency, redundancy, and error-proneness that the proposed system was designed to address. Particular attention was given to identifying instances where manual data transfer between records created opportunities for transcription errors, where the absence of a centralized scheduling tool led to booking conflicts, and where the informal nature of maintenance tracking resulted in unresolved or forgotten property concerns.

The findings of the process analysis directly informed the functional specifications of the system's core modules, ensuring that the automated workflows implemented in the system accurately reflected and improved upon the legitimate operational logic embedded within existing processes while eliminating unnecessary manual steps and inefficiencies.

Review of Existing Platforms

A review of existing property management systems and related software platforms was conducted to analyze how comparable solutions addressed the operational challenges identified through process analysis. This review examined established property management information systems used in residential and hospitality contexts, with particular attention to their approaches to booking management, payment tracking, calendar scheduling, maintenance monitoring, and reporting functionalities.

The review evaluated these platforms against the specific operational context of Gales Urban Haven, identifying features and design patterns that represented applicable best practices, as well as limitations or over-complexities that were inappropriate for a small-to-medium condominium rental business. User interface design conventions, data model structures, reporting formats, and security implementation approaches observed across the reviewed platforms informed the design of analogous components within the proposed system.

By grounding the system design in both the specific operational realities of the client organization and the broader landscape

of existing solutions, this data gathering approach ensured that the proposed Condominium Booking and Operations Management System was both contextually appropriate and aligned with proven industry practices.

3.6 Testing Procedures

A comprehensive and multi-layered testing strategy was implemented throughout the development lifecycle of the Condominium Booking and Operations Management System to ensure that all modules functioned correctly in isolation, integrated seamlessly with one another, and satisfied the operational requirements of the end users. Testing was conducted in three sequential phases: unit testing, system testing, and user acceptance testing. Each phase employed distinct methodologies and evaluation criteria appropriate to its scope and objectives.

Unit Testing

Unit testing constituted the first and most granular phase of the testing process, focusing on the verification of individual system modules and functions in isolation from the broader application environment. Each discrete functional unit of the system—including the user authentication and login module, the booking record creation and retrieval functions, the payment recording and calculation routines, the calendar availability update logic, and the maintenance request submission and status update mechanisms—was subjected to targeted test cases designed to confirm that the unit produced the correct output for all valid inputs and handled invalid or boundary-case inputs

gracefully without causing application errors or data corruption.

Unit tests were written and executed by the development team during the implementation phase, following a test-driven approach wherever feasible. Test cases were documented systematically, specifying the input conditions, expected outputs, and actual outputs for each tested function. Any discrepancies between expected and actual outputs were investigated, root causes were identified, and corrective code revisions were applied before the affected module advanced to subsequent testing phases. The thoroughness of unit testing established a stable foundation of individually verified components upon which system-level integration proceeded with confidence.

System Testing

System testing was conducted upon the successful completion of unit testing for all individual modules, examining the behavior of the fully integrated system as a cohesive whole. This phase verified that data flowed correctly across module boundaries—for example, that a booking record created through the booking management module was accurately reflected in the calendar availability view, that payment records linked to a booking were correctly aggregated in the revenue reporting dashboard, and that a maintenance request associated with a specific unit was appropriately flagged in the unit status display accessible to property managers.

System testing also encompassed performance testing to assess the system's responsiveness under realistic concurrent usage conditions, security testing to verify that authentication

controls and role-based access restrictions functioned as intended across all protected endpoints, and regression testing to confirm that code modifications introduced to address identified defects did not inadvertently disrupt previously verified functionalities. All system testing activities were conducted in a dedicated testing environment that closely mirrored the intended production deployment configuration, ensuring that test results were representative of actual operational conditions.

User Acceptance Testing

User acceptance testing (UAT) represented the final phase of the testing process and involved the direct participation of end users from Gales Urban Haven, including administrative staff responsible for booking management and payment recording, and property management personnel responsible for maintenance monitoring and scheduling. UAT evaluated whether the system, as delivered, met the practical operational requirements and usability expectations of its intended users in a realistic working context.

During UAT, participating staff members were guided through a series of realistic operational scenarios corresponding to the primary workflows supported by the system, including recording a new tenant booking, processing a payment and verifying its reflection in the revenue summary, checking unit availability through the integrated calendar, submitting and updating a maintenance request, and generating a financial report for a specified period. Participants were asked to complete these tasks independently and to provide structured feedback on the clarity of the interface, the logical flow of navigation, the

accuracy of system outputs, and any points of confusion or difficulty encountered during task completion.

Feedback gathered during UAT was systematically analyzed and used to inform final refinements to the user interface, workflow design, and system documentation prior to the formal deployment of the system. The successful completion of UAT, evidenced by the ability of end users to complete all primary operational tasks accurately and without technical assistance, confirmed that the system was ready for production use.

3.7 Evaluation Criteria

The performance and effectiveness of the Condominium Booking and Operations Management System were evaluated against a set of clearly defined criteria that collectively reflected the operational needs of Gales Urban Haven and the quality standards expected of a production-grade property management information system. Each criterion was assessed through a combination of quantitative measurements, structured user feedback, and technical verification procedures conducted during the testing and post-deployment evaluation phases. The evaluation criteria were defined as follows.

The evaluation framework was designed to provide a comprehensive and multi-dimensional assessment of the proposed system, ensuring that all critical aspects of its performance, functionality, and usability were thoroughly examined and objectively measured against established benchmarks and quality standards. The combination of quantitative measurements and

structured user feedback ensured that the evaluation process captured both the technical performance characteristics of the system and the practical user experience of its intended end users, providing a balanced and well-rounded assessment of the system's overall effectiveness and operational value.

Additionally, the evaluation criteria were directly aligned with the research objectives established at the outset of the study, ensuring that the assessment process produced meaningful and relevant findings that accurately reflected the degree to which the proposed system successfully addressed the identified operational challenges and inefficiencies of Gales Urban Haven.

The technical verification procedures conducted during the evaluation phase also subjected all core system modules and functionalities to rigorous testing under realistic operational conditions, ensuring that the system demonstrated consistent and reliable performance across all intended use cases and user scenarios. Moreover, the structured user feedback collected during the evaluation phase involved direct participation from administrators, business owners, and other authorized end users of Gales Urban Haven, whose practical insights and firsthand assessments provided the researchers with valuable and actionable information regarding the system's strengths, limitations, and areas for further improvement.

The results generated through the application of the defined evaluation criteria were systematically documented and analyzed, producing a comprehensive evaluation report that served as a transparent and evidence-based record of the system's performance, usability, and overall fitness for purpose as a property management information system. Ultimately, the rigorous and multi-faceted evaluation of the proposed Condominium Booking

and Operations Management System against clearly defined and operationally relevant criteria ensured that the final delivered system was validated as an effective, reliable, and high-quality solution that met the expectations of all stakeholders and delivered meaningful improvements to the operational management of Gales Urban Haven under One Spatial Iloilo.

Functionality

Functionality assessed the degree to which the system accurately and completely performed all specified operational tasks, including the recording, updating, and retrieval of reservation records, the processing and tracking of tenant payment transactions, the generation of revenue and occupancy reports, the management of unit availability through the integrated calendar, and the logging and monitoring of maintenance activities. The system was considered functionally adequate when all defined use cases were executed successfully under both standard and boundary-case conditions without producing data errors, unhandled exceptions, or system failures.

Functionality was verified through the systematic execution of test cases during unit and system testing, with pass rates documented and analyzed to ensure comprehensive coverage of the system's operational scope. The functionality assessment also examined the seamless interaction and data exchange between the seven core system modules, ensuring that the integrated operation of the booking, payment, maintenance, calendar, and reporting functionalities produced consistent and accurate results across all interconnected system components without data loss, synchronization errors, or functional conflicts.

Each defined use case was subjected to a structured testing protocol that simulated realistic operational scenarios drawn from the actual day-to-day workflows of Gales Urban Haven, ensuring that the functionality assessment accurately reflected the system's performance under the specific conditions and data volumes that administrators and staff members encountered during regular system use. Additionally, the functionality evaluation included the testing of edge cases and boundary conditions that represented the extreme limits of the system's operational parameters, verifying that the proposed system handled exceptional inputs, unusual data combinations, and high-volume processing demands without compromising the accuracy, consistency, or reliability of its outputs.

Usability

Usability evaluated the ease with which administrative staff and property management personnel learned to operate the system, navigated between its modules, and completed their designated operational tasks without requiring extensive technical training or external assistance.

A usable system presented information in a clear, logically organized, and visually consistent manner, provided informative feedback in response to user actions, and minimized the number of steps required to accomplish common tasks such as creating a booking, recording a payment, or updating a maintenance request status. Usability was assessed primarily through structured observation and feedback collection during user acceptance testing, supplemented by a post-UAT satisfaction survey administered to participating staff members. The findings were used to identify interface elements or navigation flows that required simplification or redesign prior to deployment.

Accuracy

Accuracy measured the correctness and reliability of all data computations and record-keeping functions performed by the system, with particular emphasis on financial calculations, including payment totals, outstanding balance computations, monthly revenue aggregations, and occupancy rate statistics. The system was evaluated for accuracy by comparing its computed outputs against manually verified benchmark values derived from known input datasets, confirming that computed figures were mathematically correct and consistently reproduced across repeated executions.

Accuracy also encompassed the integrity of relational data linkages within the database, ensuring that booking records were correctly associated with the appropriate tenant profiles and unit records, and that payment entries were accurately linked to their corresponding bookings.

Security

Security evaluated the effectiveness of the system's mechanisms for protecting sensitive operational and financial data from unauthorized access, modification, or disclosure.

The security evaluation examined the robustness of the JWT-based authentication implementation, verifying that unauthenticated requests to protected API endpoints were consistently rejected and that tokens could not be forged or reused after expiration. Role-based access control was tested to confirm that users assigned to different roles, such as administrative staff and system administrators, were appropriately restricted from

accessing or modifying data outside the scope of their designated permissions.

Additional security considerations, including the use of parameterized database queries to prevent SQL injection attacks, the enforcement of HTTPS communication protocols, and the secure storage of user credentials through industry-standard hashing algorithms, were also verified as part of the security evaluation.

Efficiency

Efficiency assessed the extent to which the system reduced the time, manual effort, and likelihood of error associated with the administrative processes it automated compared to the manual methods previously employed by Gales Urban Haven. Efficiency gains were evaluated by measuring the time required for staff to complete representative operational tasks using the system and comparing these measurements against baseline estimates of the time required to perform equivalent tasks manually.

Specific efficiency indicators included the reduction in time required to record and confirm a booking, the elimination of manual payment reconciliation steps, the prevention of double bookings through automated availability checking, and the reduction in maintenance request resolution time attributable to systematic tracking and assignment.

The overall reduction in manual workload attributable to the system's automation features served as the primary indicator of operational efficiency improvement and informed the assessment of the system's value contribution to the organization.

3.8 Supporting Diagrams

The following diagrams provided a comprehensive visual representation of the system architecture, data relationships, and user interactions that characterized the proposed Condominium Booking and Operations Management System. These supporting diagrams served as foundational references throughout the research project, guiding system design, database construction, and functional implementation, and ensuring that all components of the proposed system were developed in alignment with a coherent, well-documented, and thoroughly validated structural model.

The supporting diagrams presented in this section enabled all members of the development team to maintain a shared and consistent understanding of the system's structural design and functional logic, thereby facilitating more effective collaboration, communication, and technical decision-making throughout every phase of the development process. Each diagram was carefully and methodically constructed to accurately reflect the specific relationships, workflows, and data flows that governed the operation of the proposed system, ensuring that all visual representations functioned as reliable, precise, and professionally structured guides for every stage of system development and implementation.

Furthermore, the supporting diagrams provided academic evaluators, institutional stakeholders, and future researchers with a clear and comprehensive visual reference that substantially aided in the understanding, review, and critical

assessment of the proposed system's architecture, data management approach, and overall functional design.

The diagrams also served as permanent and indispensable documentation artifacts that were maintained alongside the system's technical specifications, providing an accessible and enduring record of the structural and functional decisions made during the design and development phases of the research. All supporting diagrams were developed in accordance with established diagramming standards, notational conventions, and academic documentation practices, ensuring that every visual representation was professionally structured, academically appropriate, and readily interpretable by all intended audiences—including technical developers, business stakeholders, and academic reviewers.

The inclusion of multiple diagram types—including system architecture, data entity relationships, and user interaction workflows—ensured that every critical dimension of the proposed system's design was thoroughly documented and visually communicated. No significant aspect of the system's structure or operational functionality was left without a corresponding, clearly defined, and well-annotated visual reference. Ultimately, the supporting diagrams presented in this section played an essential role in ensuring the coherence, technical accuracy, and overall completeness of the proposed Condominium Booking and Operations Management System's design documentation, establishing a strong, well-organized, and visually grounded foundation that guided the successful development, implementation, and evaluation of the system throughout the research project.

Data Flow Diagram (DFD — Level 0)

The Data Flow Diagram at Level 0 presented a high-level contextual overview of the entire proposed system, depicting how data flowed between the primary external entity—the Administrator—the central processing system, and the MySQL relational database that served as the system's persistent data store. As the principal actor interfacing with the system, the Administrator supplied a broad range of operational inputs, including booking records, payment transaction data, maintenance service requests, and unit scheduling information. Upon receipt of these inputs, the system processed each data type in accordance with the applicable business logic and validation rules defined within its operational framework, subsequently transmitting the resulting processed data to the MySQL database for structured and persistent storage.

In return, the system continuously retrieved stored records from the database in order to generate operational reports, populate administrative dashboards, and present comprehensive operational summaries back to the Administrator for review and decision-making. This bidirectional flow of data between the system and its underlying database ensured that all information was consistently maintained, accurately reflected, and accessible to authorized users in real time. The DFD Level 0 thus served as the primary contextual model for understanding the system's fundamental data transformation processes, establishing the boundary between the system and its external environment while clarifying the nature and direction of all high-level data exchanges that occurred during system operation.

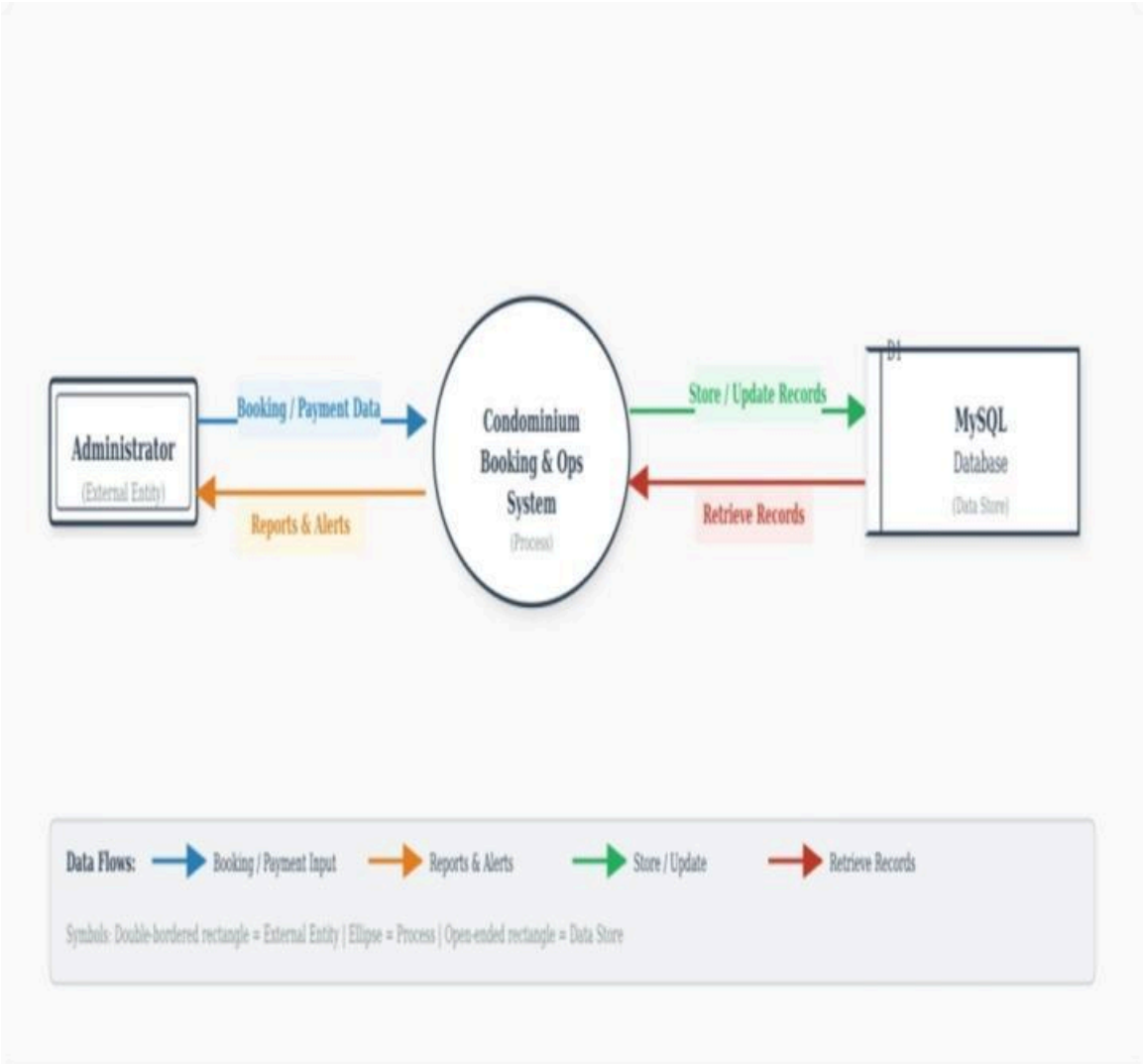


Figure 3.1. Data Flow Diagram (DFD Level 0) — Condominium Booking and Operations Management System

Entity Relationship Diagram (ERD)

The Entity Relationship Diagram illustrated the complete logical structure of the proposed system's relational database, formally defining all entities, their respective attributes, and the relationships that connected them within the MySQL database schema. The database was organized around five core entities—Users, Units, Bookings, Payments, and Maintenance—each

of which encapsulated a distinct and critical domain of the system's operational data.

The Users entity stored the credentials and profile information of all registered system users, including their assigned role designations that governed access permissions, functional privileges, and data visibility throughout all modules of the system. The Units entity maintained comprehensive records of all condominium units managed under the property, capturing essential attributes such as unit type, floor level designation, monthly rental rate, and current availability status, all of which were updated dynamically as the system processed bookings and tenant changes.

The Bookings entity represented the central transactional record of the entire system, establishing a direct relational link between tenants and their assigned condominium units through foreign key references to both the Users and Units tables. Each booking record captured the agreed check-in and check-out dates, the current booking status, the total computed rental amount, and any supplementary notes relevant to the tenancy arrangement. The Payments entity maintained a detailed and auditable financial ledger of all monetary transactions associated with each active booking, systematically recording payment amounts, transaction dates, and accepted payment methods through a well-defined foreign key relationship to the Bookings table, thereby ensuring full traceability of all financial activities within the system.

The Maintenance entity tracked and documented all property maintenance activities linked to specific condominium units, recording detailed issue descriptions, assigned priority levels, current resolution status, and timestamps for both the initial

reporting event and the final resolution. This supported both operational monitoring and historical performance analysis. The one-to-many relational associations formally defined across all five entities enforced referential integrity throughout the database schema and enabled comprehensive, cross-module reporting and data querying capabilities that supported the full range of administrative functions provided by the proposed system.

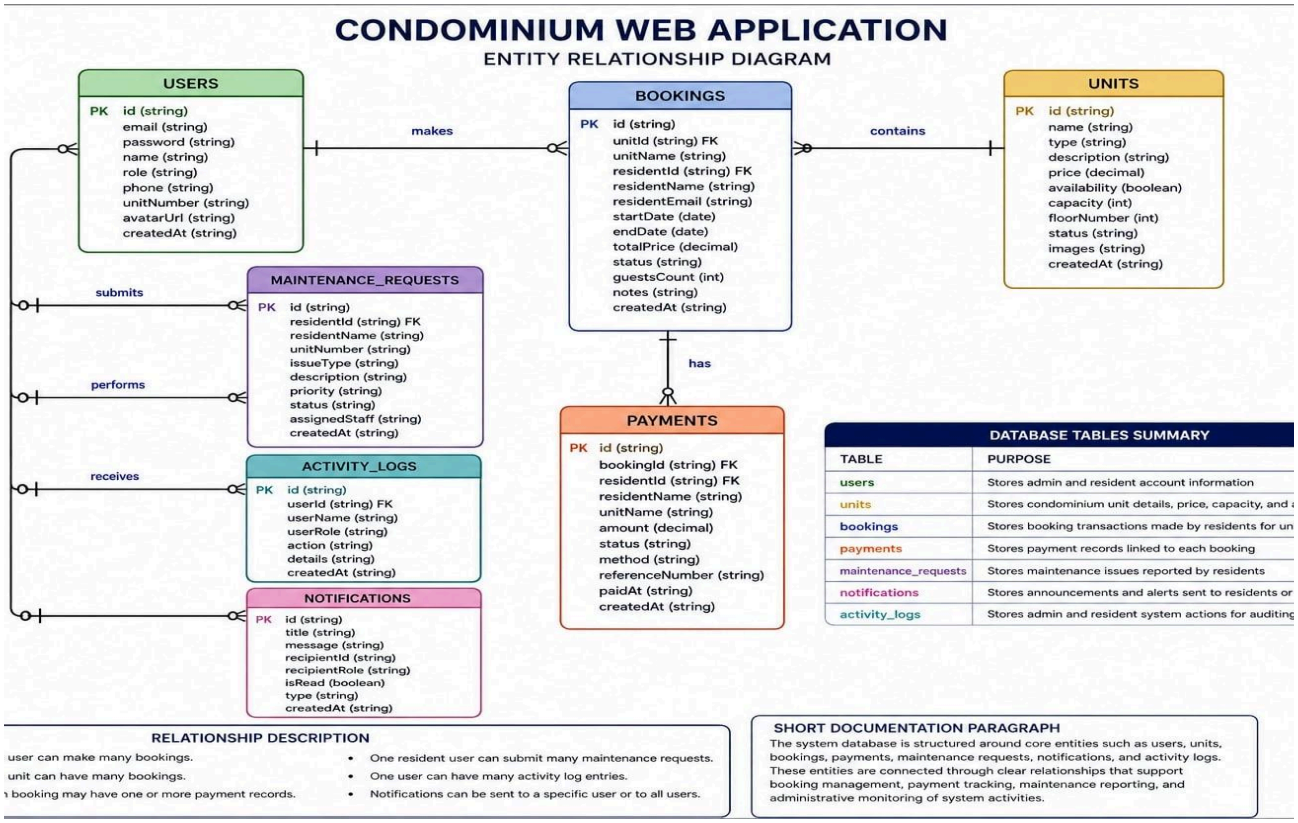


Figure 3.2. Entity Relationship Diagram (ERD) — Condominium Booking and Operations Management System

Chapter 4

Condominium Booking and Operation Management System

4.2 Overview of the Developed System

The system developed for this study, One Spatial Iloilo, was a web-based condominium management platform built to bring the booking, billing, and administrative processes of the One Spatial Iloilo residence under a single online system. Before the platform was built, tasks such as reserving a unit, tracking payments, filing a maintenance concern, or checking occupancy figures had been handled manually or through disconnected tools — a setup that made records harder to consolidate and slowed down both residents and staff. Reservation slips got misplaced, payment records lived in separate ledgers or spreadsheets, and a maintenance request often meant a phone call or a note left at the front desk, with no guarantee it reached the right person on time. Administrators, meanwhile, had to manually cross-check these scattered records just to get a clear picture of who owed what, which units were booked, or how many concerns were still unresolved.

The developed system addressed these gaps by giving residents and administrators separate, purpose-built portals that connected to the same underlying database, so that an action as simple as a booking made by a resident was immediately reflected in the admin's records without any manual re-entry. This meant that the moment a resident confirmed a reservation or settled a bill, the administrator saw it too — no waiting for someone to update a spreadsheet or relay the information by hand.

The public-facing side of the platform consisted of a landing page that introduced the property and its amenities, and a login page through which returning users signed in and new users registered. From there, the system branched into two portals depending on the account type.

The Resident Portal served as the interface used by unit owners and tenants. Once logged in, a resident was brought to a personal dashboard summarizing active bookings, total payments made, and any outstanding balance — giving them, at a glance, a sense of where they stood without having to dig through paperwork or ask the front office. From this dashboard, residents could view and manage their bookings, review their payment history, update their profile information, respond to notifications, and raise support or maintenance concerns whenever an issue came up. Every action a resident took, whether confirming a booking or settling a bill, was logged and made visible to the administrator in real time, closing the gap between what residents did and what the admin office actually knew.

The Admin Portal, by contrast, is built for the property's management staff. It opens to a dashboard that summarizes the state of the entire property at a glance, including total bookings, occupied and available units, and total revenue for the period, alongside a running list of recent bookings and an occupancy-rate figure. From this central point, administrators can move into more focused modules for the following purposes:

Booking Management, for reviewing, confirming, or cancelling reservations and monitoring their status from pending to completed;

Unit Management, for maintaining the inventory of units together with their type, floor, rate, and current availability;

Payment Tracking, for recording transactions and flagging paid, pending, or overdue accounts;

Analytics Dashboard, for viewing revenue trends, booking volume, occupancy rate, and the units booked most often over a selected period;

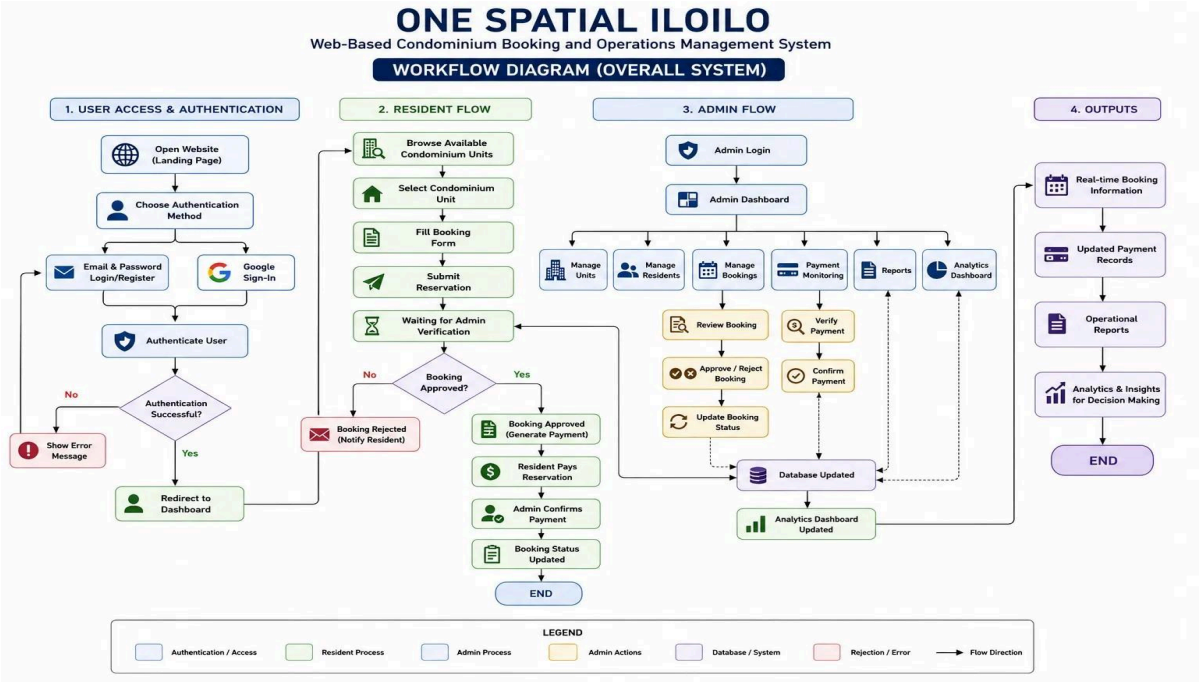
Resident Records, for keeping an updated list of residents and their unit assignments;

Maintenance Requests, for logging, assigning, and tracking the resolution of reported unit and facility issues;

Notifications and Settings, for issuing system-wide alerts and managing account and security preferences.

Taken together, these modules were designed so that a single booking, payment, or maintenance concern is only ever entered once, after which it becomes visible, in the appropriate form, to both the resident who raised it and the administrator responsible for acting on it. This is the core problem the system was built to solve: replacing scattered, manual record-keeping with one consistent source of information that both residents and management can rely on.

Figure 4.1 illustrates the overall workflow of the developed system.



The workflow connects residents and administrators through a centralized system, making condominium booking, payment verification, record management, reporting, and analytics more organized and efficient.

4.3 System Development Methodology

The researchers adopted a structured, phase-by-phase approach to building the system, moving from requirements gathering to system planning, system design, software development, testing, deployment, and finally maintenance. Each phase was substantially completed before the next began, which kept the scope manageable and made it possible to validate decisions, such as the layout of the dashboards or the fields in a database table, before they were carried forward into later stages of

development. The sections that follow describe what was done at each phase.

Requirements Gathering and Analysis

This phase focused on understanding what the system needed to do before any design or coding work began. The researchers gathered information through interviews and informal discussions with prospective users of the system, namely residents of One Spatial Iloilo and members of the property's administrative staff, supplemented by a review of the existing manual processes used for bookings, billing, and maintenance requests. From these sources, the researchers identified two broad sets of requirements. Functional requirements covered the specific tasks the system had to support, such as account registration and login, unit booking, payment recording, maintenance request logging, and the generation of occupancy and revenue reports. Non-functional requirements covered the qualities the system had to exhibit regardless of feature, including ease of use for residents who may not be technically inclined, reasonable response times when loading dashboards or tables, and safeguards for financial and personal data. These requirements were consolidated into a single reference document and used as the basis for all subsequent planning and design decisions.

System Planning

With the requirements defined, the researchers next planned how the project would be carried out. This involved breaking the system down into manageable modules, roughly corresponding to the dashboards and functions identified during requirements

gathering, and sequencing them according to dependency, so that, for example, the account and login system was planned ahead of any module that assumed a logged-in user. The researchers also selected the technologies to be used for the front end, back end, and database at this stage, weighed against the team's existing skills and the project timeline, and drew up a working schedule with target dates for each phase. Potential risks, such as delays in gathering accurate resident data or the possibility of scope creep as new feature requests emerged, were also identified and discussed at this point so that they could be watched for as development progressed.

System Design

The system design phase translated the requirements into concrete blueprints for both the interface and the underlying data structure. On the interface side, the researchers produced wireframes and, subsequently, high-fidelity mockups for each screen, including the landing page, login page, resident and admin dashboards, and the individual modules for bookings, units, payments, analytics, profiles, maintenance, notifications, and settings, ensuring a consistent visual language and navigation pattern across the resident and admin portals. On the data side, the researchers designed an entity-relationship model covering users, units, bookings, payments, and maintenance requests, and mapped out how these entities related to one another, for instance how a booking record ties a resident to a specific unit and a corresponding payment. The overall system architecture, discussed in Section 4.3.8, was also finalized during this phase, establishing how the presentation, application, and data layers would communicate with one another.

Software Development

Development proceeded module by module, generally in the order established during system planning. The user interface was built first for a given module, followed by the server-side logic needed to handle its data, and finally the connections to the database that allowed the module to read and write real records. Core account and authentication features were implemented first, since nearly every other module in the system depended on knowing who the logged-in user was and whether that user was a resident or an administrator. From there, the researchers built out the resident-facing modules, namely the dashboard, bookings, payments, profile, notifications, and support features, followed by the corresponding administrative modules for booking management, unit management, payment tracking, resident records, maintenance requests, and analytics. Code was version-controlled throughout, and each module was reviewed for consistency with the interface designs and the data model established in the previous phase before being considered complete.

Testing

Once a module was functionally complete, it was subjected to testing before being marked ready for release. Unit testing was carried out on individual functions, such as the calculation of a resident's outstanding balance or the filtering of bookings by status, to confirm that each behaved correctly in isolation. Integration testing followed, checking that modules worked correctly together, for example that a confirmed booking on the resident side immediately produced a corresponding entry in the admin's booking management table. System testing was then

performed on the platform as a whole, working through complete user scenarios, such as a resident registering, booking a unit, and paying the reservation fee, or an administrator confirming a booking and updating a unit's status, from start to finish. Finally, a small group of prospective residents and administrative staff were asked to use the system directly and provide feedback, which surfaced usability concerns that were not always apparent from developer testing alone, such as unclear labels or steps that took more clicks than necessary. Issues identified at any of these stages were logged, corrected, and re-tested before the affected module was considered stable.

Deployment

After testing confirmed that the system was stable, the researchers moved it from the local development environment to a hosting environment where it could be accessed over the internet. This involved configuring the production database, migrating the finalized schema and any seed data, and setting up the necessary server and hosting infrastructure for the front end and back end. Before the system was opened to actual residents and staff, the researchers conducted a final round of checks in the live environment to confirm that the deployed version behaved the same way as the tested version, since differences in server configuration can sometimes introduce problems that do not appear during local testing. Once these checks were satisfactory, user accounts were created for the participating residents and administrative staff, and the system was formally turned over for use.

Maintenance

The final phase covers the ongoing upkeep of the system following deployment. This includes monitoring the platform for errors or performance issues that only surface under real usage, applying corrective fixes as they are identified, and updating dependencies or security configurations as needed to keep the system reliable over time. It also includes accommodating reasonable adjustments requested by residents and administrative staff once they have had the opportunity to use the system in their day-to-day routine, such as refinements to a report layout or additional filters on a table. The researchers regard maintenance not as a one-time cleanup after launch but as a continuing responsibility for as long as the system remains in active use.

Agile Development Diagram

The project can also be represented using an Agile development workflow, where requirements are handled in iterative sprints and improvements are made based on feedback.

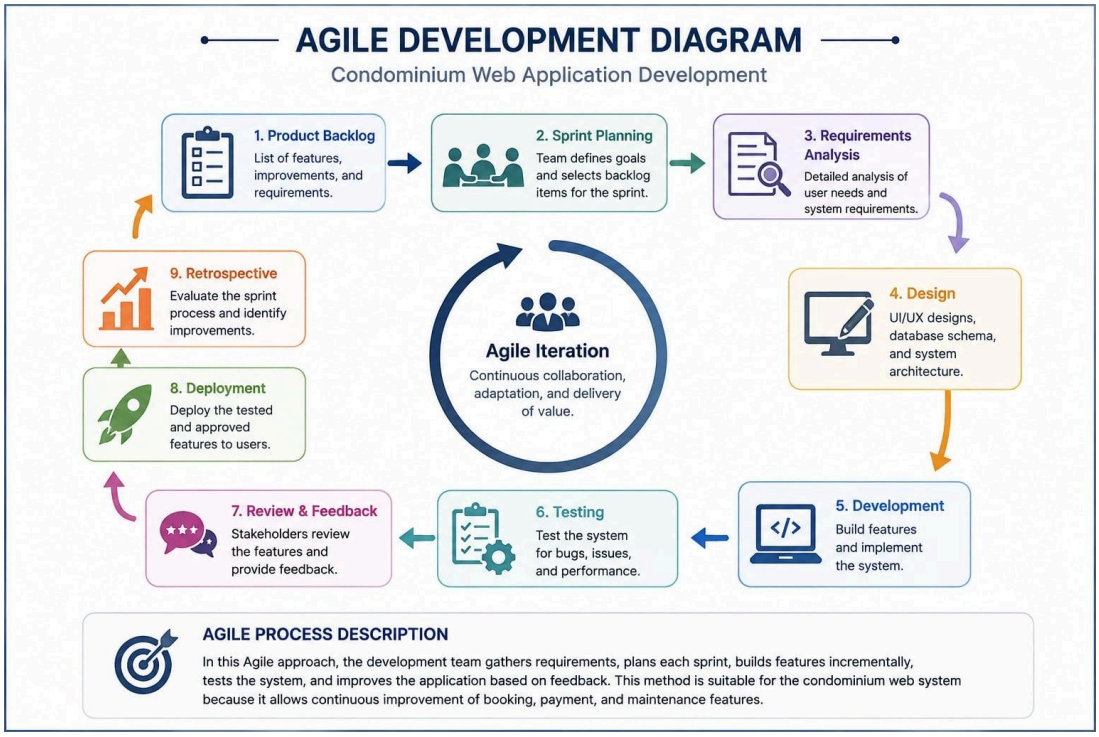


Figure 4.2 presents the Agile Software Development Life Cycle adopted in the study.

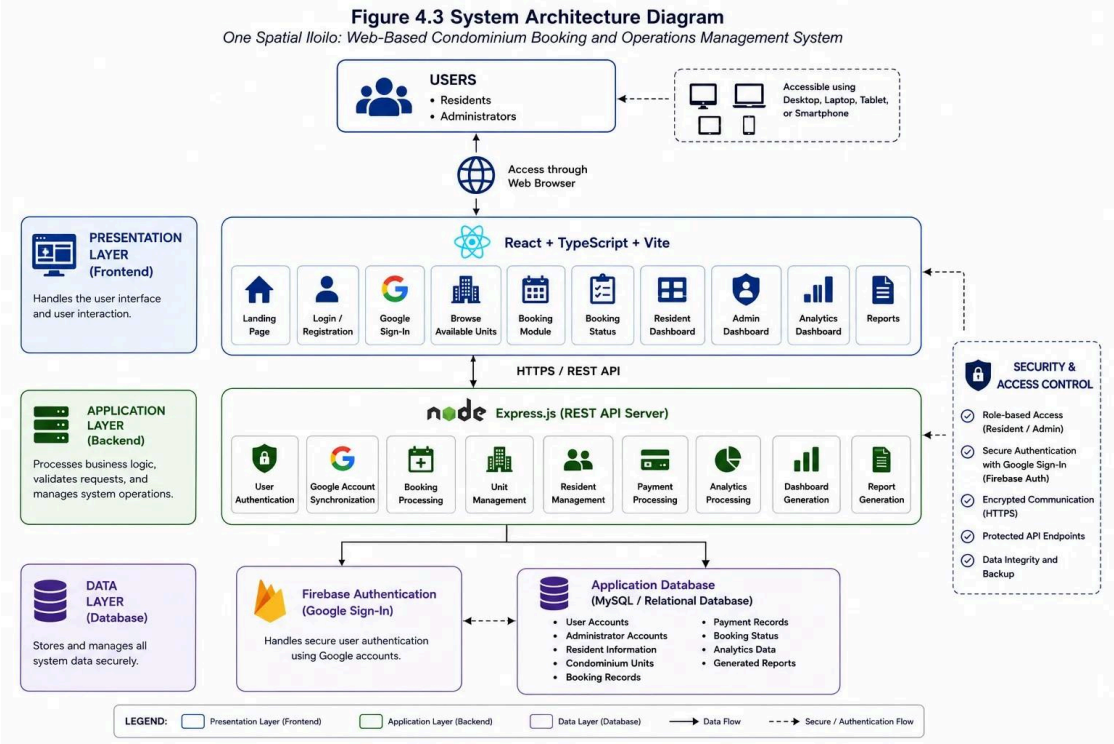
Agile Process Description

In this Agile approach, the development team gathers requirements, plans each sprint, builds features incrementally, tests the system, and improves the application based on feedback. This method is suitable for the condominium web system because it allows continuous improvement of booking, payment, and maintenance features.

4.4 System Architecture

The system was designed around a three-layer architecture, separating the concerns of presentation, application logic, and data storage. This separation keeps each layer independently

maintainable: changes to how information is displayed do not require changes to how it is processed, and changes to how data is processed do not require changes to how it is stored.



Presentation Layer

This is the layer the user directly interacts with, comprising the landing page, login page, and the resident and admin dashboards along with their respective modules. It is responsible for rendering information retrieved from the layers beneath it in a readable, navigable form, and for capturing user input, such as a new booking request or an updated profile field, before passing that input down to the application layer for processing. The presentation layer was built with

responsiveness in mind so that residents and administrators can access the system consistently across different screen sizes.

Major interface components include:

- Landing Page
- Login Page
- Resident Dashboard
- Admin Dashboard
- Booking management interface
- Unit management interface
- Payment Tracking interface
- Analytics Dashboard
- Resident Profile Interface
- Maintenance Requests interface
- Notification Interface

Application Layer

Positioned between the presentation and data layers, this layer contains the system's business logic. It validates incoming requests, such as confirming that a unit is available before allowing a booking to proceed, enforces rules like restricting certain actions to administrator accounts, computes derived values such as outstanding balances and occupancy rates, and coordinates the exchange of data between the interface and the database. Authentication and session handling, which determine whether a logged-in user is treated as a resident or an administrator, also reside in this layer.

Its primary responsibilities include:

- Authentication and session management
- Booking logic
- Payment Processing logic
- Maintenance request workflow
- Notification engine
- Analytics and reporting engine
- Profile Account management logic

Data Layer

This is the layer responsible for the persistent storage and retrieval of the system's records, including user accounts, unit listings, bookings, payments, and maintenance requests. It is organized around the entity-relationship model produced during the system design phase, with the relationships between residents, units, bookings, and payments enforced at the database level to preserve data integrity. Every figure shown in the presentation layer, whether a resident's payment history or an administrator's revenue chart, ultimately traces back to records maintained in this layer.

In practice, a single user action illustrates how the three layers work together: when a resident submits a booking through the presentation layer, the application layer checks the unit's availability and computes the applicable fees, and the data layer records the new booking and updates the unit's status, after which the presentation layer reflects the confirmed booking back to both the resident's and the administrator's dashboards. This layered structure is what allows the system, described in Section 4.2, to keep resident- and admin-facing

information synchronized without requiring any manual reconciliation between the two portals.

Stored information includes:

- User account records
- Unit records
- Booking Records
- Payment records
- Maintenance Request Records
- Notification records

4.5 Hardware and Software Requirements

This section presents the minimum and recommended hardware and software requirements for developing, deploying, and accessing the One Spatial Iloilo: Web-Based Condominium Booking and Operations Management System. The hardware specifications ensure that administrators and residents can access and use the system efficiently through a web browser, supporting essential functions such as booking, payment monitoring, report generation, and analytics visualization. Meanwhile, the software requirements identify the tools and technologies used for system development and execution, including frontend and backend development tools, database management, authentication, version control, package management, and application testing. These requirements help ensure reliable system performance, compatibility, security, and smooth operation.

4.5.1 Hardware Requirements

Table 4.1 Presents the minimum and recommended hardware specifications for running the One Spatial Iloilo: Web-Based Condominium Booking and Operations Management System. These specifications ensure that both administrators and residents can access the system efficiently through a web browser and experience smooth system performance during booking, payment monitoring, report generation, and analytics visualization.

Table 4.1 Hardware Requirements

Component	**Minimum Specification**	**Recommended Specification**
Processor	Intel Core i3 (8th Generation) or AMD Ryzen 3	Intel Core i5 (10th Generation) or higher / AMD Ryzen 5 or higher
Memory (RAM)	4 GB	8 GB or higher
Storage	256 GB SSD	512 GB SSD or higher
Monitor	1366 x 768 Resolution	Full HD (1920 × 1080) Display
Internet Connection	10 Mbps	50 Mbps or higher
Mobile Device	Android 10 or iOS 14	Android 13 or iOS 17 or later

The recommended hardware configuration provides sufficient processing power to ensure smooth execution of the web

application, including condominium booking, user authentication, payment monitoring, report generation, and analytics visualization. It also enhances the overall user experience by reducing loading times and improving system responsiveness across different devices.

4.5.2 Software Requirements

Table 4.2 presents the software requirements used in the development and execution of the One Spatial Iloilo: Web-Based Condominium Booking and Operations Management System. These software tools were selected to support frontend and backend development, authentication, version control, package management, and application testing.

Table 4.2 Software Requirements

Component	**Purpose**
Windows 10/ Windows 11	Operating System
Visual Studio Code	Source Code Editor
Node.js	JavaScript runtime environment
npm	Dependency and package management
React	Front-end Development Library
Typescript	Front-end programming language

Vite	Front-end build tool and Development server
Express.js	Backend development framework
Firebase Authentication	Google sign in authentication
JSON database	Local data storage
Google Chrome	Web browser for testing and system access
Git	Version control
Github	Source code repository

The selection of these software tools was based on their reliability, compatibility, community support, and suitability for modern web application development. React, TypeScript, and Vite were utilized to develop a responsive user interface, while Express.js and Node.js handled the backend operations and business logic. Firebase Authentication was implemented to provide secure Google Sign-In functionality, and the JSON database (db.json) was used to store application data during development. Git and GitHub were used for version control and collaborative development throughout the project.

4.6 Database Design

The database for this capstone project was designed using a relational structure composed of seven interconnected tables, each representing a core entity within the property/unit rental and management system. The design followed normalization principles to reduce data redundancy and maintain data integrity across the system. At the center of the design was the **users**

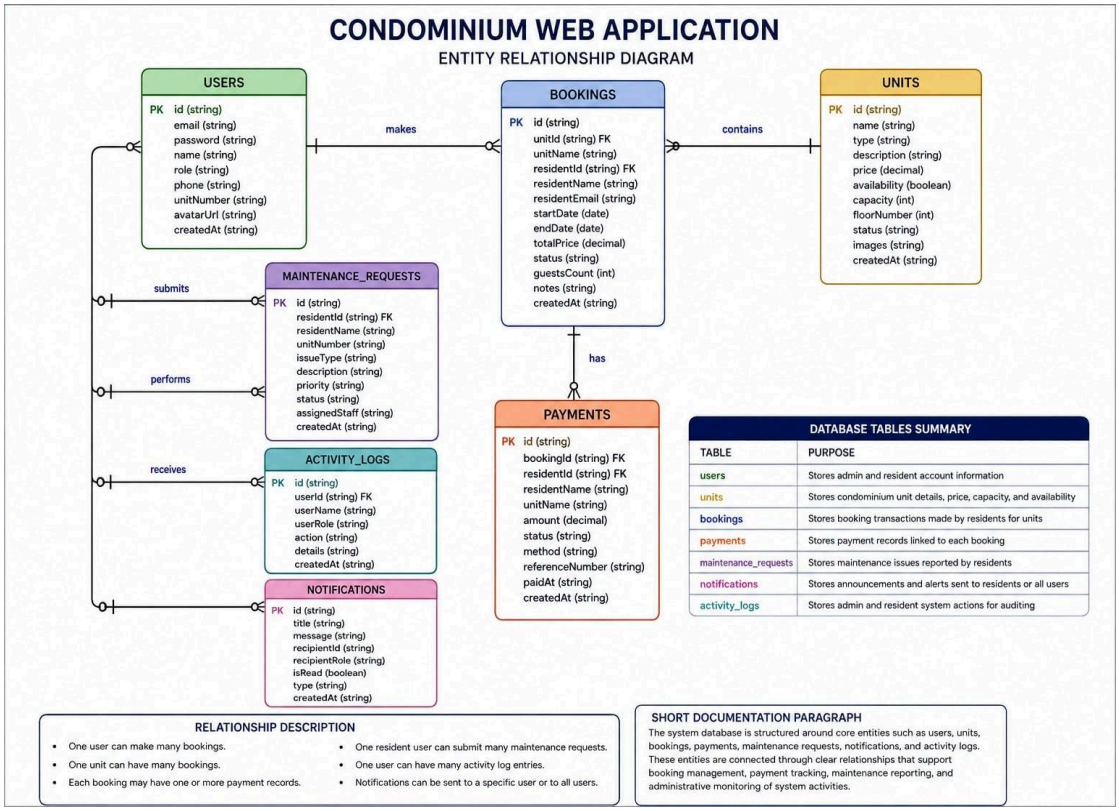
table, which stored account and profile information for both administrators and residents, and served as a reference point for several other tables through foreign key relationships. The units table was designed independently to hold property-related details such as pricing, capacity, and status, and was linked to the **bookings** table through a foreign key (unitId), establishing a one-to-many relationship where a single unit could be associated with multiple bookings over time.

The “bookings’’ table, in turn, was connected to the users table through the residentId field, allowing each booking to be traced back to the resident who made it. To support financial tracking, the “payments” table was linked to the “bookings” table via bookingId, ensuring that every payment transaction could be associated with its corresponding reservation, while also maintaining a reference to the resident who made the payment. This relationship allowed the system to accurately monitor payment statuses and amounts relative to specific bookings.

For maintenance operations, the “maintenance_requests” table was designed to reference residents through the residentId field, allowing issues to be logged, tracked, and resolved on a per-user basis. Meanwhile, the “notifications” table was structured to support both individual and role-based messaging, using recipientId and recipientRole fields to determine whether a notification was intended for a specific user or a broader group, such as all residents or all administrators.

Finally, the "activity_logs" table was designed to capture system-wide user actions, linking back to the "users" table through userId while also storing denormalized fields such as userName and userRole for quick reference and easier auditing without requiring additional table joins. Overall, the database design emphasized clear relationships between entities, consistent use of primary and foreign keys, and a balance between normalization and practical query efficiency, resulting in a structure that could reliably support the system's core functionalities.

Figure 4.4 presents the Entity Relationship Diagram (ERD) of the system.



The primary entities of the database include:

- Users
- Bookings
- Units
- Payments
- maintenance _requests
- Notifications
- Activity _logs

Users Table

This table stores account credentials and user information.

Field	**Description**
user_id	Primary Key
Email	User email
password	User password
name	Full name
role	Admin/residents
phone	Contact number
Unit_number	Resident unit number
AvatarUrl	Profile image URL
CreatAt	Date Created

Units Table

Stores condominium unit information and availability.

Field	Description
id	Primary key
name	Unit name
type	Unit type
description	Unit description
price	Rental price
availability	Availability status
capacity	Maximum guest capacity
floorNumber	Floor number
status	Available, Booked, or Maintenance
images	Unit image URLs
createdAt	Date created

Booking table

Stores resident booking and reservation details.

Field	Description
id	Primary key
unitId	Foreign key to units
unitName	Unit name
residentId	Foreign key to users
residentName	Resident name

residentEmail	Resident email
startDate	Booking start date
endDate	Booking end date
totalPrice	Total booking price
status.	Booking status
guestsCount	Number of guests
notes	Additional notes
createdAt	Date created

Payments Table

Stores payment and transaction details.

Field	**Description**
id	Primary key
bookingId	Foreign key to bookings
residentId	Foreign key to users
residentName	Resident name
unitName	Unit name
amount	Payment amount
status	Payment status
method	Payment method
referenceNumber	Payment reference number
paidAt	Payment date
createdAt	Date created

maintenance_requests table

Stores resident maintenance requests and their status.

Field	Description
id	Primary key
residentId	Foreign key to users
residentName	Resident name
unitNumber	Unit number
issueType	Type of issue
description	Issue description
priority	Priority level
status	Maintenance status
assignedStaff	Assigned staff
createdAt	Date created

notifications table

Stores notifications sent to users.

Field	Description
id	Primary key
title	Notification title
message	Notification content
recipientId	Recipient user ID or all
recipientRole	Recipient role

isRead	Read status
type	Notification type
createdAt	Date created

activity_logs table

Stores records of user activities in the system.

Field	**Description**
id	Primary key
userId	Foreign key to users
userName	User name
userRole	Admin or resident
action	Performed action
details	Action details
createdAt	Date created

The system database is structured around core entities such as users, units, bookings, payments, maintenance requests, notifications, and activity logs. These entities are connected through clear relationships that support booking management, payment tracking, maintenance reporting, and administrative monitoring of the system.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presented the final phase of the study, which focused on the development and evaluation of the Condominium Booking and Operations Management System. It brought together the results gathered from the preceding chapters, particularly the findings drawn from system testing, user evaluation, and data analysis, in order to determine whether the objectives set at the beginning of the study were achieved. The researchers designed the system to address the recurring problems encountered by condominium administrators and residents in managing amenity reservations, tracking payments, and coordinating maintenance requests, which were traditionally handled through manual logbooks, phone calls, and disconnected spreadsheets.

The discussion that followed synthesized the overall outcomes of the study by presenting a summary of the findings, the conclusions derived from those findings, and a set of recommendations intended to guide future improvement and further research. The chapter also examined the broader impact that the system created for its intended users and closed with a concise summary of the entire study.

5.2 Summary of Findings

The findings of the study were drawn from the results of the system evaluation, which involved condominium administrators, front desk personnel, and unit owners who tested the Condominium Booking and Operations Management System over the course of the

testing period. The evaluation made use of a combination of functionality testing, usability assessment, and standardized instruments such as the ISO 25010 software quality model to measure how well the system performed against its intended objectives.

The results showed that the system was able to automate the booking of shared amenities such as function halls, swimming pools, and fitness areas, which significantly reduced the scheduling conflicts that were previously common under the manual reservation process.

Respondents observed that the real-time calendar and automatic conflict-checking feature prevented double bookings that used to occur when multiple residents submitted requests for the same time slot.

The findings also revealed that the integration of an online payment and billing module improved the accuracy and timeliness of dues collection. Administrators reported that generating financial reports took considerably less time compared to the manual consolidation of receipts and ledgers that the association previously relied upon. In terms of operations management, the maintenance request and tracking feature allowed residents to log concerns directly through the system and allowed the maintenance staff to update the status of each request, which improved transparency between the association and the residents.

Usability testing indicated that the majority of respondents found the system's interface easy to navigate, even for users

who described themselves as having limited technical background. The system likewise obtained favorable ratings in terms of reliability and response time during simulated peak usage, which suggested that it could accommodate the volume of transactions typically generated within a mid-sized residential condominium. A smaller number of respondents, however, pointed out occasional delays in notification delivery and requested clearer guidance for first-time users, which were noted as areas for further refinement.

5.3 Conclusion

Based on the findings gathered throughout the study, the researchers concluded that the Condominium Booking and Operations Management System successfully met the objectives that were originally set for its development. The system provided an organized and centralized platform that replaced the fragmented, manual processes previously used for amenity booking, billing, and maintenance coordination within the condominium setting.

The study further concluded that automation played a critical role in minimizing human error and administrative workload, allowing condominium staff to redirect their time and effort toward more substantive tasks rather than repetitive clerical work. The positive feedback

obtained from both administrators and residents affirmed that a digital, integrated system was not only feasible but also a practical improvement over the traditional methods of condominium operations management.

Overall, the researchers concluded that the system, in its current form, was capable of streamlining condominium operations, improving communication between administration and residents, and providing a foundation that could be expanded further to accommodate the evolving needs of residential communities.

5.4 Recommendations

In light of the findings and conclusions presented, the researchers formulated the following recommendations to strengthen the system and to guide those who wished to pursue related studies in the future.

System Improvements

- The researchers recommended the refinement of the notification module to ensure that booking confirmations, payment reminders, and maintenance updates were delivered to users without delay, since occasional lags were observed during the evaluation period.
- It was recommended that a more detailed onboarding guide or in-app tutorial be incorporated to assist first-time users, particularly older residents who may be less familiar with digital platforms.
- The researchers suggested the addition of a dedicated mobile application, in addition to the existing web-based platform, to give residents greater convenience in accessing booking and payment features.
- It was further recommended that the system's reporting module be expanded to generate more comprehensive financial

and occupancy analytics that administrators could use for long-term planning and budgeting.

- The researchers also recommended strengthening the system's data backup and security protocols to safeguard sensitive resident and financial information against potential breaches or data loss.

Future Research Directions

- Future researchers were encouraged to explore the integration of artificial intelligence or predictive analytics to forecast amenity demand and to recommend optimal maintenance schedules based on historical usage data.
- It was recommended that future studies examine the applicability of the system across larger, multi-tower condominium developments or gated communities with more complex organizational structures.
- Future researchers may also consider incorporating biometric or facial recognition technology for gate access and visitor management to further extend the scope of operations covered by the system.
- A longitudinal study that measured the system's performance and financial impact over an extended period, such as one to two years of actual deployment, was recommended to validate the long-term benefits observed during this study.

Implementation Strategies

- The researchers recommended that condominium associations interested in adopting the system begin with a pilot

implementation involving a limited number of units or a single amenity before proceeding to a full-scale rollout.

- It was recommended that a structured training program be conducted for administrative staff and homeowners' association officers prior to system deployment to ensure smooth adoption and to minimize resistance to change.
- The researchers suggested that a technical support or help-desk arrangement be established during the initial months of implementation to promptly address user concerns and system issues.
- It was likewise recommended that the system be introduced alongside the existing manual process for a brief transition period, allowing residents and staff to gradually adjust before the manual system was fully phased out.

5.5 Impact of the Study

The study carried significant implications for the residents, administrators, and management personnel of the condominium where it was conducted. For residents, the system provided a more convenient and transparent means of reserving amenities and settling dues, which reduced the frustration commonly associated with manual booking procedures. For administrators and front desk personnel, the system lightened the administrative burden by automating routine tasks such as scheduling, billing, and maintenance tracking, allowing them to focus on higher-value responsibilities.

Beyond its immediate users, the study contributed to the growing body of knowledge on the application of information technology in property and facilities management, particularly within the

context of residential condominiums in the local setting. It demonstrated that even small to mid-sized property management operations could benefit from digital transformation without requiring extensive financial investment or highly specialized technical expertise.

The study likewise served as a reference for future information technology practitioners, students, and researchers who wished to design similar systems for other types of residential or commercial property management, thereby extending the practical and academic value of the research beyond its immediate scope.

5.6 Summary

This chapter presented the conclusion and recommendations drawn from the development and evaluation of the Condominium Booking and Operations Management System. It began with an introduction that framed the purpose of the chapter, followed by a summary of the findings obtained from system testing and user evaluation. The chapter then presented the conclusions reached by the researchers, affirming that the system successfully addressed the problems it was designed to solve. Recommendations were subsequently offered in three areas, namely system improvements, future research directions, and implementation strategies, all of which were intended to enhance the system further and to guide future studies in the same field. The chapter also discussed the impact of the study on its immediate users as well as its broader contribution to property management practice and research. Taken together, the discussions in this chapter affirmed that the Condominium Booking and Operations Management System achieved its intended purpose and offered a meaningful improvement over traditional, manual condominium operations.